NOTES ON

Gravity and Cosmology

version 1.0

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For those who come after

PREFACE

> *In a sense, you might say I'm getting quite fond of writing these kind of notes.*

Not long ago, it occurred to me how cool it is when someone unexpectedly releases a very detailed and all-comprehensive version of their notes, especially when dealing with a course that has a handful of topics often interacting together in unpredictable, yet fascinating, ways.

These notes will be mainly based on *my* own notes of the lectures by Professor Omar Zanusso for the DG, Hamiltonian, and BHs parts, and on the notes of the lectures by Professors Giovanni Marozzi and Daniele Gaggero for the Cosmology part.

These notes refer to the lectures of the academic year 2025-2026, so future versions of these lectures might differ slightly.

At any rate, since I take little to no pride in my messy notes, I'll be often using some of the references you can find on the course catalogue page or that have been cited during class. Either way, I'll do my best keeping track of them all in the bibliography of this humble collection.

You can report errors (whatever their nature might be) and suggestions for additions at g.pannocchia3@studenti.unipi.it or through whatever convoluted way (conventional or not) you prefer[1].

You can find all the notes I've redacted here.

Without further ado, we'd better not lose much more time on a preface and get started with it.

> *There was Eru, the One, who in Arda is called Ilúvatar; and he made first the Ainur* [...] *But for a long while they sang only each alone, or but few together, while the rest hearkened; for each comprehended only that part of the mind of Ilúvatar from which he came, and in the understanding of their brethren they grew but slowly.*
> *Yet ever as they listened they came to deeper understanding, and increased in unison and harmony.*

Ainulindalë, "The music of the Ainur",
Silmarillion, J. R. R. Tolkien

1 I'd like, however, not to see my house stormed by homing pidgeons.

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1 BASIS OF DIFFERENTIAL GEOMETRY

Coming soon [5].

2 | HAMILTONIAN APPROACH

Coming soon.

3 Black Holes

Coming soon.

4 COSMOLOGY

PRE-RELATIVISTIC COSMOLOGY AND ITS PROBLEMS

Long before GR as we know it today was a thing, Cosmology was already an open field of study. Of course, in this first age of Cosmology, studies were based on Newtonian mechanics and the Newtonian theory of gravity.

Newtonian Cosmology assumed an infinite, homogeneous, and static Universe, which was also eternal and responded to Euclidean geometry. Not surprisingly, under this set of un-reasonable hypothesis[1] (as an old professor of mine would say), problems began to sprout overnight.

Consider for example the so called Olber's paradox (1826). Under the assumptions of Newtonian Cosmology, Olber's paradox implicates an always-bright sky, while we have empirical evidence it isn't. In fact, the total luminosity of a set of stars with solar luminosity $L_\odot$ within a sphere of radius R

$$\phi_s = \int_0^R \frac{L_\odot}{4\pi r^2} n_s 4\pi r^2 \,\mathrm{d}r$$

will tend to infinity once $R \to \infty$. That is because, under the hypothesis of homogeneity, the surface density of stars n_s cannot depened on the spatial coordinates.

Olber then tried to solve his own paradox, claiming that it was solved allowing the presence of dust in the ISM (read: Interstellar Medium). But, if you think about it, it really doesn't solve anything: Sooner or later, the dust itself will get into thermodynamical equilibrium with the radiation, and at that point dust will begin to emit on its own.

A second solution was given in 1872 from Zolbnev, who assumed that the Universe is *closed* with a *positive curvature*. The Universe would then be finite, and the assumption of homogeneity wouldn't let the integral above diverge.

However, solved a problem, there's always another one[2] creeping around the corner. In this particular case it's the problem of gravitational stability. In fact, it's evident that local instabilities in the gravitational potential must have occurred at some point in time through fluctuations of density, leading to collapses and star formation epochs. As I said, the problem is mainly local, and instabilities are generally balanced on the larger scales.

A cosmologist called Bentley then showed that perturbations on spatial scale r have, under the assumption of homogeneity, an associated potential $\propto r^2$ and an associated acceleration $\propto r$, which leads to unstable equilibrium.

This problem is however partially mitigated by an argument brought by Seelinger (1895), who assumes gravity to be *shielded* on the large scales

$$\phi = G \int \frac{\rho}{r} \,\mathrm{d}^3 x \to \phi = G \int \frac{\rho}{r} e^{-\lambda r} \,\mathrm{d}^3 x$$

1 The only reasonable assumption was for the Universe to be homogeneous and, only at times, static. The rest of the hypothesis were arguable at best.

2 "Always two there are."

where the exponential term is an absorption/shielding factor. This correction will thus bring about a modification of Poisson's equation

$$\nabla^2\phi = 4\pi G\rho \rightarrow \nabla^2\phi - \lambda\phi = 4\pi G\rho$$

Much expectedly, quite a lot changed since 1917, when relativistic cosmology was officially born [8]. Einstein's GR was based on three main principles:

1. (Weak) Equivalence Principle: Inertial acceleration and gravitational acceleration are proportional to each other. Hence all free-falling systems are locally equivalent to an inertial system;

2. General Covariance Principle: When written in covariant form, the laws of Physics are the same in all reference frames;

3. Mach's Principle[3]: The momentum of all bodies is determined by all the masses in the Universe. Alternatively, the only things with physical sense are those that are measurable.

Only a year prior to Einstein's 1917 article, Schwarzschild had already found a solution to Einstein's celebrated equation under the assumption of a single spherical body in an otherwise empty space. Such solution, that is nowadays known as *Schwarzschild's metric*, has to the perk to degenerate into the standard spacetime's Minkowskian metric at infinite distance from the object. This, of course, felt like a big problem for Mach's philosophical principle.

Einstein, perhaps conditioned from the beliefs of his time, twisted his theory so to produce a closed, static Universe. To do so, he only had to add a single term to his equation, the Cosmological Constant

$$G_{\mu\nu} - \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu} \tag{4.1}$$

which is actually perfectly valid, even from a theoretical point of view[4]. That said, the interpretation of the Λ term was somewhat...elusive.

In 1918, Erwin Schrödinger suggested that the Cosmological Constant was equivalent to the energy-momentum tensor of a fluid with negative pressure, and could thus be included in the right-hand side term of the equation.

In 1917, Vesto Slipher observed a discrepancy in the observed emission lines of some galaxies (with respect to the expected frequencies). He, however, failed to correlate such discrepancy to the distance of the aforementioned galaxies.

Only when Hubble measured their distance [11] through the lightcurve of Cepheid variables did cosmologists start to realize that the Universe was expanding rather than being static; the expansion law of the Universe still carries Hubble's name, who was the first to suggest such correlation.

Another viable model to describe the Universe is the so-called Einstein-de Sitter Universe. This model essentially assumes the Universe to be akin to a non-relativistic fluid for the sake of computing the resulting metric and the properties stemming from it. This model was however abandoned because it produced errors in the estimation of the age of the Universe[5].

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3 This one is a bit "philosophical", but it's interesteing to at least mention it.

4 I assume the four readers of these notes to be at least familiar with the terms and symbols used. If not, I suggest reading [5] or any graduate/postgraduate textbook about GR.

5 It turned out that some white dwarves, low mass stars that had reached the end of their evolution cycle, would be older than the Universe itself.

Thermodynamical Considerations on the Big Bang

The Laws of Thermodynamics speak rather clearly: Energy is (locally) conserved, and Entropy either remains constant or grows. But, if the Universe were already in a state of maximum entropy, there would be no thermodynamic free energy available to drive further macroscopic change, and it would meet its end in a rather ominous-sounding way: Thermal Death[6].

If, on the other hand, entropy is allowed to grow, then it is possible for the system to switch to more probable microscopic configurations. So, starting from a low-entropy state, the Universe would then evolve through thermodynamical means into the messy Universe we see today.

4.1 MAXIMALLY SYMMETRIC UNIVERSES

Note: Parts of this section will be eventually removed and integrated into Chapter 1 when I'll have it sorted out. So, for the moment, you'll have to take it at face value, which, admittedly, feels a tad random.

4.1.1 Isometries and Killing Vectors

Consider a metric $g_{\mu\nu}$ and a given map of coordinates x_μ (which roughly translates into choosing a reference frame). The metric tensor will thus transform according to the standard tensor tranformation rule

$$g'_{\mu\nu}(x') = \frac{\partial x^\rho}{\partial x'^\mu}\frac{\partial x^\sigma}{\partial x'^\nu} g_{\rho\sigma}(x) \tag{4.2}$$

We'll say that a particular transformation $x \to x'$ is an *isometry* if

$$g'_{\rho\sigma}(y) = g_{\rho\sigma}(y) \tag{4.3}$$

If we consider an infinitesimal transformation of the likes

$$x'^\mu = x^\mu + \epsilon\xi^\mu \tag{4.4}$$

with $|\epsilon| \ll 1$, we can substitute directly into the transformation equation for the metric. Under the request of the transformation being an isometry, we'll obtain

$$\partial_\rho \xi^\mu g_{\mu\sigma} + \partial_\sigma \xi^\nu g_{\rho\nu} + \xi^\mu \partial_\mu g_{\rho\sigma} = 0 \tag{4.5}$$

We can then introduce Christoffel's symbols to cast the equation in an equivalent form

$$\partial_\rho \xi_\sigma + \partial_\sigma \xi_\rho - 2\xi_\mu \Gamma_\rho{}^\mu{}_\sigma = 0 \tag{4.6}$$

The vector ξ^μ will be a *Killing vector* if it satisfies Killing's equation

$$\nabla_\rho \xi_\sigma + \nabla_\sigma \xi_\rho = 0 \tag{4.7}$$

If a given spacetime manifold has n spatial dimensions, then the space can have *at most* $n(n+1)/2$ Killing vectors. Such space is called *maximally symmetric*.

6 Other less ominous versions were proposed, such as the Big Chill and Big Freeze. You can check here for more.

4.1.2 The Bare Minimum to Begin

Contemporary cosmological models are based on the idea that the Universe is pretty much the same everywhere, a stance sometimes known as *Copernican Principle* or *Cosmological principle*. If we take it at face value, such a claim seems crazy, if at all logical: There's little resemblance, say, between the center of the Sun (which is kinda hot) and the desolate cold of the ISM. The Cosmological Principle starts to make a little more sense if we take it to apply only to the very largest scales, where local variations in density are averaged over and magically disappear.

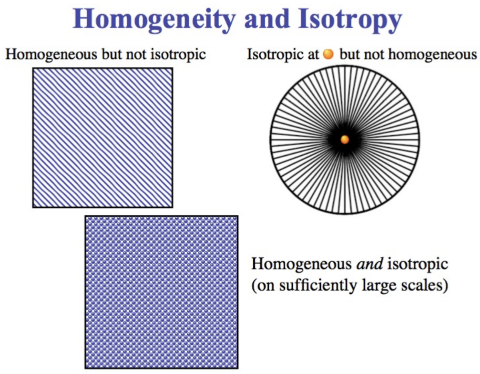

Figure 4.1: The Cosmological Principle automatically translates in two very specific requests: Homogeneity and Isotropy.

The Cosmological principle is related to two more mathematically precise properties that we can request our spacetime manifold to have: *Isotropy* and *Homogeneity*, which are given a pictorial representation in Fig.4.1.

Homogeneity

> Consider a family of hypersurfaces defined by some parameter t which allows us to foliate the manifold. Given any two points p and q in a manifold M, there is an isometry that takes p into q. See [5].

The assumption of Homogeneity essentially requires that the metric is the same throughout the manifold.

Isotropy

> A manifold M is isotropic around a point p if, for any two vectors V and W in the tangent space of M to the point p T_pM, there is an isometry (symmetry) of M such that the *pushforward* of W under the isometry is parallel with V (not pushed forward). See [5].
> Alternatively, we can consider the set of geodesics that defines a congruence[a] tangent to a vector u^μ, orthogonal to the fixed-time hypersurfaces. The manifold is said isotropic if there exists an isometry of the metric that transforms the generators of the hypersurfaces in other generators.
>
> a You can also see [5], Appendix F for more.

Isotropy, instead, states that space looks exactly the same around a given point, no matter in what direction you're looking.

It turns out that a manifold that is simultaneously homogeneous and isotropic is also maximally symmetric! The transformations associated to isotropy are the rotations

$$x'^{\mu} = R^{\mu}{}_{\nu} x^{\nu} \tag{4.8}$$

which generate $n(n-1)/2$ Killing vectors. On the other hand, the transformations associated to homogeneity are the translations

$$x'^{\mu} = x^{\mu} + \xi^{\mu} \tag{4.9}$$

which generate n Killing vectors. The total number of Killing vector is, as expected, exactly $n(n+1)/2$. A spacetime which has a maximally symmetric spatial section must have a special metric that reflects such properties. Let's start from a general metric

$$\mathrm{d}s^2 = g_{\mu\nu}\,\mathrm{d}x^{\mu}\,\mathrm{d}x^{\nu} \tag{4.10}$$

Clearly, off-diagonal terms like g_{0i} must be vanishing $g_{0i} = 0$, or the request of an isotropic manifold wouldn't be met. This already leaves us with a metric with the following expression

Note that we'll be using a metric with Pseudo-Riemannian signature $(-,+,+,+)$, which might be different from that used in other courses.

$$\mathrm{d}s^2 = -(u_{\mu}\,\mathrm{d}x^{\mu})^2 + g_{ij}(t)\,\mathrm{d}x^i\,\mathrm{d}x^j \tag{4.11}$$

where the first term of the right-hand side represents the projection of the displacement along the tangent vector u^{μ}, which is the "generator" of the congruence and, by virtue of its own definition, perpendicular to the spatial hypersurfaces.

Let's focus on the spatial metric g_{ij}. Thanks to isotropy, the temporal dependence of the metric has to factor out, so that

$$g_{ij}(t) = a^2(t)\tilde{g}_{ij} \tag{4.12}$$

where we've introduced the *scale factor* $a(t)$.

Taking $u_{\mu} = (-1, 0, 0, 0)$, our metric turns out to be

$$\mathrm{d}s^2 = -\,\mathrm{d}t^2 + a^2(t)\tilde{g}_{ij}\,\mathrm{d}x^i\,\mathrm{d}x^j \tag{4.13}$$

Let's now consider a n-dimensional geometry which we embed into a $n+1$ dimensional space.

3-Dimensional Space

So let's, for example, write down the metric for a 2-dimensional sphere embedded in a 3-dimensional space. The 3-dimensional Euclidean metric is

$$\mathrm{d}l^2 = \mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2 \tag{4.14}$$

where we can define the radius of a given sphere a as

$$a^2 = x_1^2 + x_2^2 + x_3^2 \tag{4.15}$$

which we can implicitly differentiate to obtain an expression for, say, dx_3, which we'll plug into (4.14)

$$dx_3 = -\frac{x_1\,dx_1 + x_2\,dx_2}{\pm(a^2 - x_1^2 - x_2^2)^{1/2}} \tag{4.16}$$

so that the line element induced on the 2-sphere is

$$dl^2 = dx_1^2 + dx_2^2 + \frac{(x_1\,dx_1 + x_2\,dx_2)^2}{a^2 - x_1^2 - x_2^2} \tag{4.17}$$

We can define a vector $\vec{x} = (x_1, x_2)$, so that the line element can be expressed in a more compact fashion

$$dl^2 = d\vec{x}^2 + \frac{(\vec{x}\cdot d\vec{x})^2}{a^2 - \vec{x}^2} \tag{4.18}$$

Flat Space

The case of a flat space with vanishing curvature can be dealt with starting from the metric induced on the sphere, simply imposing $a \to \infty$. Not surprisingly, the resulting metric will be

$$dl^2 = d\vec{x}^2$$

Similarly, we can also calculate the induced metric on a 2-dimension hyperboloid

$$a^2 = x_1^2 + x_2^2 - x_3^2 \tag{4.19}$$

The passages are essentially identical to those for the sphere, and the resulting metric is

$$dl^2 = d\vec{x}^2 - \frac{(\vec{x}\cdot d\vec{x})^2}{a^2 + \vec{x}^2} \tag{4.20}$$

For our future discussions it will be useful to define a scaled-down x vector $\tilde{x} = \vec{x}/a$, so that the expression for the line element can be conveniently expressed as

$$dl^2 = a^2\left(d\tilde{x}^2 + k\frac{(\tilde{x}\cdot d\tilde{x})^2}{1 - k\tilde{x}^2}\right) \tag{4.21}$$

where we've introduced the parameter $k = 0, \pm 1$, which defines the topological properties of the space we're considering.

Sometimes (read: almost always) it is useful to express the metric through the polar coordinates of the 2-dimensional sphere. With such purpose in mind, we're defining x_3 as the "height" of a certain circular section, r' the radius of such section, and θ the azimuthal angle with respect to axis parallel to x_1. In formulas

$$x_3^2 = a^2 - r'^2$$
$$x_1 = r'\cos\theta$$
$$x_2 = r'\sin\theta$$

With this new set of coordinates, the line element of the 2-dimensional sphere becomes

$$dl^2 = dr'^2 + r'^2\,d\theta^2 + \frac{(r'\,dr')^2}{a^2 - r'^2} = a^2\left(\frac{dr^2}{1 - r^2} + r^2\,d\theta^2\right) \tag{4.22}$$

where we've also introduced the adimensional radius $r = r'/a$, which lives in the interval $r \in [0, 1]$. We can repeat the calculations for the hyperboloid and the flat space in a similar fashion. The final result for the three scenarios we've considered can be once again expressed in a single expression thanks to the k parameter

$$dl^2 = a^2 \left(\frac{dr^2}{1 - kr^2} + r^2 \, d\theta^2 \right) \tag{4.23}$$

4-Dimensional Space

Perhaps it might have occurred to you that we do not, in fact, live in a 3-dimensional space, but rather in a 4-dimensional one with 3-dimensional spatial sections. The purpose of this section is essentially to replicate the calculations we've already went through for the 3-dimensional space but with one more additional dimension to address.

For this purpose, it comes somewhat in handy recalling (4.21), which allows us to immediately find the expression for the line element induced on a 3-dimensional surface embedded in a 4-dimensional space. All we have to do is replacing the metric of the 1-sphere S_1 ($d\theta^2$), with that of the 2-sphere S_2, the solid angle $d\Omega^2_{(2)}$

$$dl^2 = a^2 \left(d\tilde{x}^2 + k \frac{(\tilde{x} \cdot d\tilde{x})^2}{1 - k\tilde{x}^2} \right) = a^2 \left(\frac{dr^2}{1 - kr^2} + r^2 \, d\Omega^2_{(2)} \right) \tag{4.24}$$

For example, the metric induced on the sphere ($k = 1$) can be obtained defining the spherical coordinates (χ, θ, ϕ) and a radial distance a, so that

$$x_1^2 + x_2^2 + x_3^2 + x_4^2 = a^2 \tag{4.25}$$

and the line element is

$$dl^2 = dx_1^2 + dx_2^2 + dx_3^2 + dx_4^2 \tag{4.26}$$

Performing the due substitutions you'd obtain

$$dl^2 = a^2 (d\chi^2 + \sin^2 \chi \, d\Omega^2_{(2)}) \tag{4.27}$$

The metric induced on the hyperboloid is obtained replacing $\sin \chi$ with $\sinh \chi$. Note that the new radial coordinate χ is adimensional, in the range $[0, \pi]$ for the sphere, and defined through

$$d\chi = \frac{dr}{\sqrt{1 - kr^2}} \tag{4.28}$$

In the hyperboloid case, the radial coordinate χ will now be in the range $[0, \infty)$, and the associated Cartesian coordinates are

$$\begin{aligned}
x_1 &= a \sinh \chi \sin \theta \cos \phi \\
x_2 &= a \sinh \chi \sin \theta \sin \phi \\
x_3 &= a \sin \chi \cos \theta \\
x_4 &= a \cos \chi
\end{aligned}$$

We can go through several boring substitutions to find the induced line element

$$\mathrm{d}l^2 = a^2(\,\mathrm{d}\chi^2 + \sinh^2\chi\,\mathrm{d}\Omega^2_{(2)}) \tag{4.29}$$

and the compact expression for the three surfaces we're considering is

$$\boxed{\mathrm{d}l^2 = a^2\left(\mathrm{d}\chi^2 + \frac{1}{k}\sin^2(\sqrt{k}\chi)\,\mathrm{d}\Omega^2_{(2)}\right) \quad k = 0, \pm 1} \tag{4.30}$$

where we've made use of the fact that the line element can be conventiently expressed as

$$\mathrm{d}l^2 = a^2(\,\mathrm{d}\chi^2 + f_k^2(\chi)\,\mathrm{d}\Omega^2_{(2)}) \tag{4.31}$$

with $f_k(\chi)$ the solution of (4.28), and thus given by

$$f_k(\chi) = \begin{cases} k^{-1/2}\sin(\sqrt{k}\chi) & k = 1 \\ \chi & k = 0 \\ (-k)^{-1/2}\sinh(\sqrt{-k}\chi) & k = -1 \end{cases} \tag{4.32}$$

which can be further compressed into

$$f_k(\chi) = (k)^{-1/2}\sin(\sqrt{k}\chi) \tag{4.33}$$

which exploits the known limit for $k = 0$ for the sine and the definition of the hyperbolic sine for $k = -1$.

4.2 FRIEDMANN-ROBERTSON-WALKER METRICS

At this point we have all the elements needed to reintroduce time into the picture

$$\mathrm{d}s^2 = -\,\mathrm{d}t^2 + a^2(t)\tilde{g}_{ij}\,\mathrm{d}x^i\,\mathrm{d}x^j = -\,\mathrm{d}t^2 + \,\mathrm{d}l^2 \tag{4.34}$$

If the Universe is spatially homogeneous and isotropic at all times, and if distances are allowed to expand or contract as a function of time only[7], there's only one metric possible and is the FLRW metric:

$$\boxed{\mathrm{d}s^2 = -\mathrm{d}t^2 + a(t)^2\left(\frac{\mathrm{d}r^2}{1-kr^2} + r^2\mathrm{d}\Omega^2_{(2)}\right)} \tag{4.35}$$

Since $\mathrm{d}s^2 = g_{\mu\nu}\,\mathrm{d}x^\mu\,\mathrm{d}x^\nu$, the elements of the metric tensor can be read directly from (4.35)

$$\begin{aligned} g_{00} &= -1 \\ g_{rr} &= \frac{a^2}{1-kr^2} \\ g_{\theta\theta} &= r^2a^2 \\ g_{\phi\phi} &= a^2r^2\sin^2\theta \end{aligned}$$

7 This is an inevitable consequence of the request of having a metric that is isotropic and homogeneous. If all points didn't evolve at the same rate in time, it would be impossible for the Universe to be isotropic.

while g_{0i} and off-diagonal terms of g_{ij} are vanishing.

From the coefficient of the metric, we can also compute Christoffel's symbols

$$\Gamma_{\mu\ \nu}^{\ \alpha} = \frac{1}{2} g^{\lambda\alpha} (\partial_\mu g_{\lambda\nu} + \partial_\nu g_{\mu\lambda} - \partial_\lambda g_{\mu\nu}) \tag{4.36}$$

which have been reported in Appendix C.

Sometimes it is useful to perform a slight manipulation of the FLRW metric (4.35) and define a conformal time

$$a(\eta)\mathrm{d}\eta = \mathrm{d}t \tag{4.37}$$

so that the metric is, guess what, conformal.

Conformal Metric and Transformations

A conformal transformation is essentially a local change in scale. Since distances are measured by the metric, such transformations are implemented by multiplying the metric by a spacetime-dependent (nonvanishing) function

$$\tilde{g}_{\mu\nu} = \omega^2(x) g_{\mu\nu}$$

for some nonvanishing function $\omega(x)$. Notably *null curves are invariant under conformal transformations.*. See [5], Appendix G for more.

The scale factor a is therefore factorized and (4.35) becomes

$$\mathrm{d}s^2 = a(\eta)^2 \left(-\mathrm{d}\eta^2 + \frac{\mathrm{d}r^2}{1-kr^2} + r^2 \mathrm{d}\Omega^2_{(2)} \right) \tag{4.38}$$

Note that the quantity within parentheses is independent of the conformal time. On top of that, if $k = 0$ (flat space), the FLRW metric is conformal to a Minkowski metric, and the metric is said "conformally flat".

4.2.1 Geodesics

Consider a particle initially at rest in a FLRW metric (4.35). The geodesics equation for such a particle will be

$$\frac{\mathrm{d}^2 x^i}{\mathrm{d}\tau^2} = -\Gamma_{0\ 0}^{\ i} \frac{\mathrm{d}x^0}{\mathrm{d}\tau} \frac{\mathrm{d}x^0}{\mathrm{d}\tau} \tag{4.39}$$

since $\mathrm{d}x^i/\mathrm{d}\tau = 0$, where τ is the particle's proper time. However, since $\Gamma_{0\ 0}^{\ i} = 0$, the particle isn't accelerated and its radial coordinate remains constant!

Note that what remains constant is actually the particle's *comoving coordinate*, while its *proper distance* as measured from some origin of the coordinate system changes because of the time-evolution of the scale factor a.

Once a system of axis is fixed so that a given point $P_1 = (r_1, 0, 0)$, we can define its proper distance as

$$d(r,t) = \int_0^{r_1} \mathrm{d}l(r) \tag{4.40}$$

where $\mathrm{d}l(r)$ is the line element induced on the manifold, in our case

$$\mathrm{d}l^2 = a(t)^2 \left(\frac{\mathrm{d}r^2}{1-kr^2} + r^2 \mathrm{d}\Omega^2_{(2)} \right) \tag{4.41}$$

Ignoring the angular dependence, i.e. a point object for which $\mathrm{d}\Omega = 0$, the proper distance at some given time t is

$$d(r,t) = \int_0^{r_1} \frac{a(t)\,\mathrm{d}r}{(1-kr^2)^{1/2}} \tag{4.42}$$

which is solved by

$$d(r,t) = a(t) \begin{cases} \arcsin(r), & k=1 \\ r, & k=0 \\ \operatorname{arcsinh}(r), & k=-1 \end{cases} \tag{4.43}$$

Please note that unlike it's done by other textbooks, like [5], here the scale factor a has the dimension of a length rather than being adimensional. You can think of it as the result of the product between an adimensional scale factor (which is the one really changing) and the curvature radius of the Universe at some time.

Let us now consider a free particle with non-vanishing initial momentum. The general geodesics equation is

$$\frac{\mathrm{d}^2 x^\sigma}{\mathrm{d}\tau^2} = -\Gamma_{\mu\ \nu}^{\ \sigma} \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau} \frac{\mathrm{d}x^\nu}{\mathrm{d}\tau} \tag{4.44}$$

It's useful to remember the definition of the 4-momentum $p^\mu = mu^\mu = m\mathrm{d}x^\mu/\mathrm{d}\tau$ (valid for timelike paths, at least), from which we know that

$$|\vec{p}|^2 = m^2 g_{ij} \frac{\mathrm{d}x^i}{\mathrm{d}\tau} \frac{\mathrm{d}x^j}{\mathrm{d}\tau}$$

If we consider a system of coordinates such that at the instant considered the particle is arbitrarily close to the center of the coordinate system, the metric can be approximated as $\tilde{g}_{ij} = \delta_{ij} + o(x^2)$, where we're essentially neglecting the scalar product contribution to (4.21).

Then the Christoffel symbol $\Gamma_{j\ l}^{\ i} \propto \partial_l g^i_{\ j} \propto o(x)$ is negligible, since the space is always locally flat for each point we choose arbitrarily close to the origin. The geodesics equation can be thus expanded

$$\begin{aligned} \frac{\mathrm{d}^2 x^i}{\mathrm{d}\tau^2} = -\Gamma_{\mu\ \nu}^{\ i} \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau} \frac{\mathrm{d}x^\nu}{\mathrm{d}\tau} &= -2\Gamma_{0\ j}^{\ i} \frac{\mathrm{d}x^j}{\mathrm{d}\tau} \frac{\mathrm{d}x^0}{\mathrm{d}\tau} \\ &= -2\frac{\dot{a}(t)}{a(t)} \delta^i_{\ j} \frac{\mathrm{d}x^j}{\mathrm{d}\tau} \frac{\mathrm{d}x^0}{\mathrm{d}\tau} \end{aligned}$$

At this point we can multiply both terms by $\mathrm{d}\tau/\mathrm{d}t$

$$\frac{\mathrm{d}^2 x^i}{\mathrm{d}\tau^2} \cdot \frac{\mathrm{d}\tau}{\mathrm{d}t} = -2\frac{\dot{a}(t)}{a(t)} \delta^i_{\ j} \frac{\mathrm{d}x^j}{\mathrm{d}\tau} \frac{\mathrm{d}x^0}{\mathrm{d}\tau} \cdot \frac{\mathrm{d}\tau}{\mathrm{d}t} \tag{4.45}$$

and exploit the identity

$$\frac{\mathrm{d}x^0}{\mathrm{d}\tau} \cdot \frac{\mathrm{d}\tau}{\mathrm{d}t} = 1 \tag{4.46}$$

resulting in a first-order differential equation for $\mathrm{d}x^i/\mathrm{d}\tau$

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}x^i}{\mathrm{d}\tau}\right) = -2\frac{\dot{a}(t)}{a(t)}\frac{\mathrm{d}x^i}{\mathrm{d}\tau} \tag{4.47}$$

This equation yields as a solution $\mathrm{d}x^i/\mathrm{d}\tau \propto a^{-2}$, which means that $|\vec{p}|^2 \propto m^2a^2 \cdot a^{-4} \propto a^{-2}$. This means that a massive particle "slows down" with respect to the comoving coordinates as the Universe expands!

If we instead consider a null geodesic (massless particle), we find out that

$$|\vec{p}| = p = \frac{\hbar}{\lambda} \implies \lambda \propto a(t) \tag{4.48}$$

which means that the wavelength of a photon changes depending on the time at which it was emitted, resulting in an effect commonly known as *cosmological redshift*.

4.2.2 A short detour on redshift

The purpose of this section is to show that we can obtain the same results that we've obtained in the last section using a more elegant approach that makes use of the symmetry properties of GR and the FLRW metric. Incidentally, this will also allow us to introduce a most useful quantity in Cosmology, which is the redshift z.

Consider the four-velocity of comoving observers $U^\mu = (1, 0, 0, 0)$, then the metric admits a Killing tensor of the form

$$K_{\mu\nu} = a^2(g_{\mu\nu} + U_\mu U_\nu) \tag{4.49}$$

that you can check[8] that it satisfies $\nabla_{(\sigma}K_{\mu\nu)} = 0$ where ∇_σ is the covariant derivative. As stems from the definition of Killing vector (the generalization to a Killing tensor is straightforward), the quantity

$$K^2 = K_{\mu\nu}V^\mu V^\nu = a^2\left[V_\mu V^\mu + (U_\mu V^\mu)^2\right] \tag{4.50}$$

is conserved along the geodesics. Here $V^\mu = \mathrm{d}x^\mu/\mathrm{d}\lambda$. For massive particles, since $V_\mu V^\mu = -1$ (note the convention on the signature of the metric), we also have

$$(V^0)^2 = 1 + |\vec{V}|^2 \quad |\vec{V}|^2 = g_{ij}V^iV^j$$

Recalling the formula for the four-velocity of the observer, we have $U_\mu V^\mu = -V^0$, so (4.50) implies

$$|\vec{V}| = \frac{K}{a}$$

Notice how this means that the particle "slows down" with respect to the comoving coordinates as the Universe expands. If we instead consider null geodesics (that is what we're actually interested in)

$$V_\mu V^\mu = 0 \implies U_\mu V^\mu = \frac{K}{a}$$

8 Perhaps it was trivial, but it took me a long while to figure out and convince myself of the proof, which you'll find in Appendix D.

but for null-like particles, like photons, we can write the four-velocity as $V^\mu = (\omega, \vec{k})$. So the frequency of the photon as measured by a comoving observer is $\omega = -U_\mu V^\mu$ (remember that $c = 1$). The frequency emitted will still be given by the same functional expression but calculated in the local inertial frame of the emitter, so $\omega_{em} = -K/a_{em}$. We then find out that

$$\frac{\omega_{obs}}{\omega_{em}} = \frac{a_{em}}{a_{obs}}$$

To make things easier we then introduce a quantity called *redshift* defined as

$$z_{em} = \frac{\lambda_{obs} - \lambda_{em}}{\lambda_{em}} \tag{4.51}$$

so that, if the observation takes place today ($a_{obs} = a_0$), we can just write

$$\boxed{a_{em} = \frac{a_0}{1 + z_{em}}} \tag{4.52}$$

and

$$\boxed{\frac{\nu_{obs}}{\nu_{em}} = \frac{1}{1 + z_{em}}} \tag{4.53}$$

Let's give an example. If we observe, say, a galaxy with a redshift $z = 2$, we are observing it as it was when the Universe had a scale factor $a(t_e) = a_0/3$.

The redshift we observe for a distant object thus depends only on the relative scale factors at the time of emission and the time of observation, it doesn't depend on how the transition between $a(t_e)$ and $a(t_0)$ was made.

Calculating the redshift through spectroscopic observations, it was possible to correlate the recession velocity of objects in the Universe to their proper distance, obtaining the so called *Hubble's law*

$$v = cz = H_0 d_P \tag{4.54}$$

Recent measurements [16] managed to estimate the present-day Hubble constant H_0 with great precision

$$H_0 = \frac{\dot{a}(t_0)}{a_0} = 72.53 \pm 0.99\,\mathrm{km\,s^{-1}\,Mpc^{-1}} \tag{4.55}$$

from which we get an order-of-magnitude estimate of the age of the Universe

$$H_0^{-1} = 9.78 h^{-1}\,\mathrm{Gyr} \tag{4.56}$$

using $H_0 = 100h\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. We can also calculate the so-called Hubble radius

$$R_H = H_0^{-1} c = 3h^{-1}\,\mathrm{Gpc} \tag{4.57}$$

The variable h keeps track of the dispersion on the value of H_0 produced by the different measurement technique that was implemented.

Please note that Hubble's law is only valid on relatively large scales, where the peculiar velocity of objects become much smaller than the expansion velocity at a given distance.

On a concluding note, please note that although it's a useful meter of comparison to understand what we're talking about, cosmological redshift has *nothing to do* with a Doppler shift.

In fact, even if the Universe underwent an age in which $\dot{a} = 0$ (so $H_0 = 0$), you wouldn't be able to interpret the shift in frequency as the effect of a Doppler shift ($v = 0$ after all), yet you'd still observe the shift, since $a(t) \neq 0$.

4.2.3 Hubble-Le Maître's Law

Let's get back to the definition of redshift starting from the scale factor a which we can Taylor expand for emission times t "close" to the observation time

$$a(t) = a(t_0) + \left(\frac{\partial a}{\partial t}\right)_{t_0} (t - t_0) + o\left((t - t_0)^2\right) \tag{4.58}$$

so that

$$a(t) = a(t_0)\left[1 + H_0(t - t_0) + o\left((t - t_0)^2\right)\right] \tag{4.59}$$

The expansion won't be ill-defined only if $H_0^{-1} \gg (t - t_0)$. We can now plug (4.59) into the definition of redshift (4.52)

$$z \approx H_0(t_0 - t) \implies z = \frac{H_0 d}{c} \tag{4.60}$$

where $d = c(t_0 - t)$ is the distance crossed by photons in the same time interval. Equation (4.60) is the correct way to express Hubble-Le Maître's law for not too large distances. For larger distances we would have to take into account the modification brought about by the second order term in the expansion of a, forcing us to introduce an additional parameter, the *deceleration parameter* q.

4.3 THE ROAD TO FRIEDMANN'S EQUATIONS

Conserved current for a non-collisional fluid

For the time being, let's consider a Universe completely filled with a fluid of cosmological relevance with negligible collision rate. The number of particles that make the fluid is conserved, and since the fluid must obey the Cosmological Principle, collective motion of the fluid can't have a privileged direction, hence $\langle v^i \rangle = 0$. We can thus define a 4-current vector J^μ

Please note that we're working, as usual, with units in which $c = 1$.

$$J^\mu = n u^\mu = n\frac{dx^\mu}{d\tau} \quad u^\mu = (u^0, \gamma\vec{v}) \tag{4.61}$$

where n is the number density of particles. To see how the expansion of the Universe affects the number density n we have to first compute

$$\nabla_\mu J^\mu = \partial_\mu J^\mu + \Gamma_\mu{}^\mu{}_\nu J^\nu \tag{4.62}$$

But since the metric is the FLRW, the covariant divergence of J^μ is vanishing. In fact, since J^i is zero on average, the covariant divergence is reduced to simply

The demonstration also follows from recalling the equation for the covariant divergent of a vector

$$\nabla_\mu J^\mu = \frac{1}{\sqrt{-g}}\partial_\mu\left(\sqrt{-g}J^\mu\right)$$

$$\partial_0 J^0 + \Gamma_i{}^i{}_0 J^0 = 0 \quad \Gamma_i{}^i{}_0 = \frac{\dot a}{a}\delta^i{}_i = 3\frac{\dot a}{a} \tag{4.63}$$

which means

$$\frac{dn}{dt} = -3\frac{\dot a}{a} \implies n(t) \propto a^{-3} \tag{4.64}$$

4.3.1 Energy-Momentum Tensor and Its Conservation

For the sake of simplicity, let's consider first the case of a flat-space. The energy-momentum tensor can be introduced from the minimal-action principle, imposing that the variation of a properly defined action functional is vanishing *on shell*[9]

$$\delta S = \delta \left[\int \mathrm{d}t\, \mathrm{d}V \mathcal{L} \left(q^{(a)}, \frac{\partial q^{(a)}}{\partial x^\mu} \right) \right] = 0 \tag{4.65}$$

where $\mathcal{L}$ is the lagrangian density associated with the field(s) we're considering. Computing the variation we obtain Euler-Lagrange equations

$$\partial^\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial^\mu q)} \right) - \frac{\partial \mathcal{L}}{\partial q} = 0 \tag{4.66}$$

Exploiting Noether's theorem and assuming translational invariance for the lagrangian density, we could show that there exists a conserved quantity representing the 4-momentum flux crossing the fixed-time hypersurfaces, whose normal vector is tangent to the flux itself

$$\partial_\nu \left[\frac{\partial \mathcal{L}}{\partial(\partial_\nu q)} \partial_\mu q - \delta^\nu_{\ \mu} \mathcal{L} \right] = \partial_\nu T^\nu_{\ \mu} = 0 \tag{4.67}$$

The 4-momentum is thus simply defined as the flux of the energy-momentum tensor through the hypersurfaces

$$p^\mu = \int T^{\mu\nu}\, \mathrm{d}S_\nu \quad \mathrm{d}S_\nu = \mathrm{d}V \hat{n}_n \tag{4.68}$$

where $\hat{n}_\nu = (1, 0, 0, 0)$ is the unit vector orthogonal to the fixed-time hypersurfaces. For reasons that I'm sure won't surprise you, T^{00} is the energy density, while T^{i0} is the momentum density. Moreover, you can easily check that

$$\int_V \partial_\alpha T^{i\alpha}\, \mathrm{d}V = 0 \implies \partial_t \int_V T^{i0}\, \mathrm{d}V + \int_V \partial_j T^{ij}\, \mathrm{d}V = 0 \tag{4.69}$$

Applying the divergence theorem to the second integral yields

$$\partial_t \int_V T^{i0}\, \mathrm{d}V + \int_{\partial V} T^{ij}\, \mathrm{d}S_j = 0 \tag{4.70}$$

which means that forces must be evenly balanced by the pressure arising in the fluid.

When a fluid is isotropic and homogeneous in a comoving reference frame, the fluid is said to be *perfect*. Through long, and boring, calculations (which essentially consist in transforming T^{00} and T^{ij} out of the comoving frame), you can show that a perfectly valid expression for the energy-momentum tensor of a perfect fluid is

Note that the components of u^μ are fixed by requiring $u_\mu u^\mu = -1$

$$T^{\mu\nu} = (p + \rho) u^\mu u^\nu + p g^{\mu\nu} \tag{4.71}$$

which we can also present in the form of a (1,1) tensor

$$T^\mu_\nu = g_{\alpha\nu} T^{\mu\alpha} = (p + \rho) u^\mu u_\nu + p(g^{\mu\alpha} g_{\nu\alpha}) \tag{4.72}$$

9 i.e. when calculated on the equations of motion.

from which we can immediately compute the trace

$$T^{\mu}_{\mu} = -(p+\rho) + 4p = 3p - \rho \tag{4.73}$$

What we'd like to show now is what the conservation of the energy-momentum tensor (4.71) we've defined implies if the metric is no longer that of the flat space but rather the FLRW (4.35). To do so, we have to calculate

$$\nabla_\nu T^{\mu\nu} = \nabla_\nu \left[(p+\rho)u^{\mu\nu}\right] + \nabla_\nu \left[p g^{\mu\nu}\right] = 0 \tag{4.74}$$

After exploiting metric compatibility, and expanding the covariant derivative, we can rewrite the previous expression as

$$\partial_\nu(p+\rho)u^\mu u^\nu + (\partial_\nu p)g^{\mu\nu} + (p+\rho)\left[\Gamma_{\nu\ \lambda}^{\ \mu} u^\lambda u^\nu + \Gamma_{\nu\ \lambda}^{\ \nu} u^\mu u^\lambda\right] = 0 \tag{4.75}$$

At this point we can notice that, since the fluid is comoving, $u^\mu = (1,0,0,0)$, and thus

$$\Gamma_{\nu\ \lambda}^{\ \mu} u^\lambda u^\nu = \Gamma_{0\ 0}^{\ \mu} u^0 u^0 \quad \Gamma_{\nu\ \lambda}^{\ \nu} u^\mu u^\lambda = \Gamma_{\nu\ 0}^{\ \nu} u^\mu u^0 \tag{4.76}$$

But, given the FLRW metric, $\Gamma_{0\ 0}^{\ \mu} = 0$ for all values of μ, while the second term can be simplified as

$$\Gamma_{\nu\ 0}^{\ \nu} u^\mu u^0 = \frac{1}{2} g^{\nu i} \partial_0 g_{\nu i} u^\mu \tag{4.77}$$

Putting everything back together

$$\partial_\nu(p+\rho)u^\mu u^\nu + (\partial_\nu p)g^{\mu\nu} + (p+\rho)\frac{1}{2} g^{\nu i} \partial_0 g_{\nu i} u^\mu = 0 \tag{4.78}$$

Let's focus on the purely spatial component $\mu = j$, which ultimately simplifies to just $\partial_i p g^{ij}$ being vanishing essentially because of the Cosmological Principle (homogeneous pressure). The only non-trivial component is the temporal one $\mu = 0$, which yields

$$\partial_0(p+\rho) - \partial_0 p + \frac{p+\rho}{2} g^{ij} \partial_0 g_{ij} = 0 \tag{4.79}$$

The last derivative can be hand-wavingly calculated expanding $g_{ij} = a^2 \tilde{g}_{ij}$, where $\tilde{g}_{ij}$ is independent of time. This way, we obtain the so-called *fluid equation*

$$\boxed{\partial_0 \rho + 3(p+\rho)H(t) = 0} \tag{4.80}$$

It might be worth pointing out that ρ is here an energy density rather than a matter density.

4.3.2 Ricci Tensor in a FLRW Metric

Honestly, I'm not in the mood to calculate all non-trivial components of the Riemann tensor

$$R^{\mu}_{\ \nu\alpha\beta} = \partial_\alpha \Gamma_{\nu\ \beta}^{\ \alpha} - \partial_\beta \Gamma_{\nu\ \mu}^{\ \alpha} + \Gamma_{\sigma\ \alpha}^{\ \mu} \Gamma_{\nu\ \beta}^{\ \sigma} - \Gamma_{\sigma\ \beta}^{\ \mu} \Gamma_{\nu\ \alpha}^{\ \sigma} \tag{4.81}$$

to then calculate Ricci's $R_{\nu\beta} = R^{\alpha}_{\ \nu\alpha\beta}$. You can find all non-trivial components of the Ricci tensor in Appendix C.

Without doing all the troublesome calculations, I'll just say that in principle the calculation is made a little easier if we consider small deviations from flat-space for the time independent metric $\tilde{g}_{ij} = g_{ij}/a^2$, which helps significantly to decrease the effort.

It is worth mentioning, however, that we might as well expect the components R_{0i} to be vanishing because of the request of homogeneity and isotropy. An elegant way of proving it without losing your wits in the process might be considering the non-canonical version of Einstein's equation

$$R_{\mu\nu} = 8\pi G \left(T_{\mu\nu} - \frac{1}{2} T g_{\mu\nu} \right) \tag{4.82}$$

where T is the trace of the energy-momentum tensor. From here is trivial to prove our previous claim if we assume $T_{\mu\nu}$ to be the energy momentum tensor of a perfect fluid as seen by a comoving observer.

For future reference, I'm going to write the non-trivial components of the Ricci tensor down below

$$R_{00} = -\frac{3\ddot{a}}{a} \qquad R_{ij} = \frac{1}{a^2}(2k + a\ddot{a} + 2\dot{a}^2) g_{ij} \tag{4.83}$$

4.3.3 Friedmann's Equation

Consider the non-canonical form of Einstein's equation (4.82). The energy momentum tensor is (4.71), and its trace is $T = 3p - \rho$, so it's immediate to solve Einstein's equation.

The temporal component yields

$$\boxed{\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p)} \tag{4.84}$$

Since we have three independent variables and only two viable equations[10], we're in the need of introducing and equation of state (EoS) that relates the pressure to the energy density. Cosmology usually deals with dilute gases, for which the equation of state is simple.

For substances of cosmological importance

$$p = w\rho \tag{4.85}$$

where w is a dimensionless number. The behaviors of our favorite sources are summarized in the following table

	w_i
matter	0
radiation	$\frac{1}{3}$
curvature	$-\frac{1}{3}$
vacuum	-1

10 The equations are actually 3: Two Friedmann equations and the equation coming from the conservation of the temporal component of the energy-momentum tensor. The three of them are not independent, since they're related through Bianchi's identity.

With our newly gained EoS in mind, we can rewrite Friedmann equation as

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\rho(1+3w) \tag{4.86}$$

which tells us that the Universe is accelerating if and only if

$$\frac{\ddot{a}}{a} > 0 \iff w < -\frac{1}{3} \tag{4.87}$$

The spatial component of (4.82), on the other hand, yields

$$\frac{2k}{a^2} + \frac{\ddot{a}}{a} + 2\left(\frac{\dot{a}}{a}\right)^2 = -4\pi G(p-\rho) \tag{4.88}$$

where the common factor $\tilde{g}_{ij}$ was suppressed. It is customary however to subtract the 1st Friedmann equation (4.84) from the equation we've just derived, obtaining the 2nd Friedmann equation

$$\boxed{\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}} \tag{4.89}$$

Suppose that the Universe contains N different components, with the i-th component having an energy density ρ_i and an EoS parameter w_i. As long as there is no interaction between the different components, the fluid equation (4.80) must hold for each component separately. So

$$\frac{\dot{\rho}_i}{\rho_i} = -3(1+w_i)\frac{\dot{a}(t)}{a(t)} \tag{4.90}$$

which can be immediately integrated. Behold

$$\boxed{\rho_i(a) = \rho_{i,0}\left(\frac{a(t)}{a(t_0)}\right)^{-3(1+w_i)}} \tag{4.91}$$

Equation (4.91) implies that for (non-relativistic) matter $\rho \propto a^{-3}$, while for radiation (or relativistic matter) $\rho \propto a^{-4}$. This fact has a simple explanation: The energy density of non-relativistic matter decreases as a increases because of the expanding volume of the Universe, while relativistic matter also gets redshifted. This has also another striking consequence: There had been times when different components were equivalent!

$$\frac{\rho_\Lambda(a)}{\rho_m(a)} = 1 = \frac{\rho_{\Lambda,0}}{\rho_{m,0}}a^3 \implies a_{\Lambda,m} = \left(\frac{\rho_{m,0}}{\rho_{\Lambda,0}}\right)^{1/3} \approx 0.766 \implies z_{\Lambda,m} = 0.31$$
$$\frac{\rho_m(a)}{\rho_r(a)} = 1 = \frac{\rho_{m,0}}{\rho_{r,0}}a \implies a_{r,m} = \frac{\rho_{r,0}}{\rho_{m,0}} \approx 2.9\cdot 10^{-4} \implies z_{r,m} = 3400$$

If the Universe contains different components with different values of w, then in the limit $a \to 0$, the component with the largest value of w is dominant.

Now that we have a relation between ρ and a we can set to solve the Friedmann equation (4.89) in the special case of a spatially flat ($k=0$) and single-component Universe. Plugging in (4.91), we find

$$\dot{a}^2 = \frac{8\pi G\rho_0}{3}\frac{a^{-(1+3w)}}{a_0^{-(1+3w)}} \tag{4.92}$$

Also note that $\dot{a} \propto (\rho a^2)^{1/2}$ which is also proportional to $\propto a^{-(1+3w)/2}$. Hence, a fluid with $w = -1/3$ has constant expansion rate $\dot{a}$.

which can be solved explicitly

$$\boxed{a(t) = a_0 \left(\frac{t}{t_0}\right)^{\frac{2}{3+3w}} \qquad t_0 = \frac{2}{3H_0(1+w)} \qquad w \neq -1} \tag{4.93}$$

For $w = -1$, the equation is solved by

$$\boxed{a(t) = a_0 e^{\pm H_0(t-t_0)}} \tag{4.94}$$

A detour on conformal time

Let's now write Friedmann's equations using the conformal time $\mathrm{d}\eta = \mathrm{d}t/a$. Through a basic application of the chain rule

$$\frac{\mathrm{d}a}{\mathrm{d}t} = \frac{\mathrm{d}a}{\mathrm{d}\eta}\frac{\mathrm{d}\eta}{\mathrm{d}t} = a'\frac{1}{a} \tag{4.95}$$

where with the superscript we're indicating derivatives with respect to the conformal time. In short

$$a' = \dot{a}a \quad a'' = a(a\ddot{a} + \dot{a}a') \quad \mathcal{H} = \frac{a'}{a} = \dot{a} \quad \mathcal{H}' = a\ddot{a} \tag{4.96}$$

With these new variables, Friedmann's equations become

$$\mathcal{H}^2 = \frac{8\pi G}{3}a^2\rho - k \tag{4.97}$$

$$\mathcal{H}' = -\frac{4\pi G}{3}a^2(\rho + 3p) \tag{4.98}$$

while the conformal fluid equation is

$$\rho' + 3\mathcal{H}(p + \rho) = 0 \tag{4.99}$$

Assuming $w \neq 1$, we can integrate the definition of conformal time (4.37) to obtain the scale factor as a function of the conformal time

$$\eta - \eta_0 = \int_{t_0}^{t} a^{-1}\,\mathrm{d}t \propto \int_{t_0}^{t} t^{-2/3(1+w)}\,\mathrm{d}t \tag{4.100}$$

which yields

$$a(\eta) \propto \eta^{2/1+3w} \quad (w \neq -1) \tag{4.101}$$

If, on the other hand, $w = -1$, the integral is trivial and we find that

$$a(\eta) \propto -\frac{1}{\eta - \eta_0} \quad (\eta - \eta_0 < 0) \tag{4.102}$$

4.3.4 Einstein Universe

Consider a stationary Universe for which Einstein's equation is modified to include the addition of Cosmological Constant Λ

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu} \tag{4.103}$$

and consider the modified energy-momentum tensor $\tilde{T}_{\mu\nu}$ defined so to include the contribution of the stray Cosmological Constant

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi G\tilde{T}_{\mu\nu} = 8\pi G\left(T_{\mu\nu} - \frac{\Lambda}{8\pi G}g_{\mu\nu}\right) \tag{4.104}$$

Moreover we can compute the new energy density

$$\tilde{T}_{00} = \rho + \frac{\Lambda}{8\pi G} \tag{4.105}$$

and define an energy density for vacuum

$$\rho_\Lambda = \frac{\Lambda}{8\pi G} \tag{4.106}$$

Similarly, we can also calculate the new pressure exerted by the cosmological fluid

$$\tilde{p} = \tilde{T}_{ii} = \left(p - \frac{\Lambda}{8\pi G}\right)g_{ii} = p - \frac{\Lambda}{8\pi G} \tag{4.107}$$

and define the pressure for vacuum, which turns out to be

$$p_\Lambda = -\frac{\Lambda}{8\pi G} = -\rho_\Lambda \tag{4.108}$$

So the vacuum, or Cosmological Constant, would behave like a fluid with $w = -1$. We can see what that implies if we plug this last result into Friedmann's equations

$$\begin{aligned} 3\ddot{a} &= -4\pi Ga(\tilde{\rho} + 3\tilde{p}) \\ a\ddot{a} + 2\dot{a}^2 + 2k &= 4\pi Ga^2(\tilde{\rho} - \tilde{p}) \end{aligned}$$

which under Einstein's assumption of a static Universe ($\dot{a} = 0$ and $\ddot{a} = 0$) implies

$$\tilde{\rho} = -3\tilde{p} \quad 2k = 4\pi Ga^2(\tilde{\rho} - \tilde{p}) \tag{4.109}$$

and so

$$\tilde{\rho} = \frac{3k}{8\pi Ga^2} \quad \tilde{p} = -\frac{k}{8\pi Ga^2} \tag{4.110}$$

The conditions for a static Universe thus require it to be a *closed Universe* ($k = 1$). If we assume non-relativistic matter to be the only component of $T_{\mu\nu}$, then

$$\tilde{p} = -\frac{\Lambda}{8\pi G} \implies \tilde{p} = -\frac{\Lambda}{8\pi G} = -\frac{k}{8\pi Ga^2} \tag{4.111}$$

and thus

$$\Lambda = \frac{k}{a^2} \tag{4.112}$$

This means that the physical dimension of the Universe would be fixed and equal to $a = \Lambda^{-1/2}$ ($k = 1$). This already conflicts with present observations (since $H_0 \neq 0$), but to push worse to worst, Einstein's Universe would also be unstable.

Consider in fact $\tilde{T}_{\mu\nu} = T_{\mu\nu} + T^\Lambda_{\mu\nu}$, so that

You have to simply substitute (4.106) and the value we've found for $\tilde{\rho}$, using $\Lambda = a^{-1}$.

$$\tilde{\rho} = \rho_M + \rho_\Lambda \implies \rho_M = \frac{2\Lambda}{8\pi G} \tag{4.113}$$

If we add a positive perturbation to the scale factor $a = a_E + \Delta a$, then the matter energy density decreases

$$\rho_M = \frac{3k}{8\pi G a^2} - \frac{\Lambda}{8\pi G} \tag{4.114}$$

so that we could write in principle $\rho_M \to \rho_M + \Delta\rho_M$ with $\Delta\rho_M < 0$. But if we plug this into Friedmann's equation

$$\ddot{a} = -4\pi G a(\tilde{\rho}_0 + 3\tilde{p}_0 + \Delta\rho_M) = -4\pi G a \Delta\rho_M = 4\pi G a|\Delta\rho_M| > 0 \tag{4.115}$$

meaning that such a Universe is unstable!

Einstein-de Sitter Universe

Let's try this one more time. Recall Friedmann's second equation (4.89) and consider a flat Universe with only the Cosmological Constant in it

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{\Lambda}{3} \implies a(t) \propto \exp\left(\sqrt{\frac{\Lambda}{3}}t\right) \tag{4.116}$$

and so Hubble's constant $H = \sqrt{\Lambda/3}$. This implies

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho_\Lambda + 3p_\Lambda) = \frac{8\pi G}{3}\rho_\Lambda = \frac{\Lambda}{3} \tag{4.117}$$

4.3.5 Critical Density

Let's write again Friedmann's second equation

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2} \tag{4.118}$$

and define the *critical energy density* as the energy density the Universe should have to be flat

At present time, the critical density $\rho_C(t_0)$ is evaluated at $1.878 \cdot 10^{-29} h^2\, g\, cm^{-3}$ or, equivalently, at $1.05 \cdot 10^4 h^2\, eV\, cm^{-3}$.

$$\rho_C = \frac{3H^2}{8\pi G} \tag{4.119}$$

This quantity, that will generally change with time, is called the critical density because the Friedmann equation (4.89) can be written as

$$\Omega - 1 = \frac{k}{H^2 a^2} \qquad \Omega = \frac{\rho}{\rho_c} \tag{4.120}$$

that univocally determinates the sign of k depending on whether Ω is greater then, equal to, or less than one.

$$\begin{aligned}
\rho < \rho_c &\iff \Omega < 1 \iff k < 0 \iff \text{open}\\
\rho = \rho_c &\iff \Omega = 1 \iff k = 0 \iff \text{flat}\\
\rho > \rho_c &\iff \Omega > 1 \iff k > 0 \iff \text{closed}
\end{aligned}$$

The density parameter Ω, then, tells us which of the three FLRW geometries describes our Universe. Since the right-hand side of (4.120) cannot change sign as the Universe expands, neither can the left-hand side. So, Ω will have the same sign *at all times*.

In a realistic case, we should consider a Universe with multiple components, although for the sake of simplicity we choose them not to be interacting. We can then think of writing a total energy density

$$\rho = \sum_i \rho_i = \rho_{DM} + \rho_B + \rho_r + \rho_\Lambda + ... \tag{4.121}$$

or, in terms of the density parameter(s), we could also write

$$\rho = \rho_C \sum_i \Omega_i \tag{4.122}$$

In these variables, the Friedmann equation (4.89) can be written as

$$H^2 = \frac{8\pi G}{3} \sum_{i(c)} \rho_i \tag{4.123}$$

where the notation $\sum_{i(c)}$ indicates that we sum not only over all the actual components of energy density ρ_i, but also on the contribution of spatial curvature, which we define as

$$\Omega_k = -\frac{k}{a^2 H^2} \tag{4.124}$$

Dividing both sides by H^2 yields a shockingly simple equation

$$1 = \sum_{i(c)} \Omega_i = \Omega_k + \sum_i \Omega_i \tag{4.125}$$

thus producing a consistency relation that must hold at *any* time.

Acceleration Parameter

A few pages ago, back when we first proved Hubble-Le Maître's law, we've introduced the *deceleration parameter* q, which we now properly define

$$q = -\frac{\ddot{a}a}{\dot{a}^2} \tag{4.126}$$

I don't think I will surprise any of you saying that if $q < 0$, then the Universe is accelerating, and if it's $q > 0$, then it is decelerating. Not being surprised was what people back then expected when they first defined this quantity. They were so sure of their results that they even named it *deceleration parameter* rather than acceleration parameter. Shortly after naming it, they realized they should have probably changed its name.

Recall (4.84) with a perfect fluid EoS

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\rho(1+3w) \tag{4.127}$$

This means that the "deceleration" parameter can be written as

$$q = \frac{4\pi G}{3H^2}\rho(1+3w) \tag{4.128}$$

or, equivalently, we can plug in the critical density and density parameter to find

$$2q = \Omega(1+3w) \implies 2q = \sum_i \Omega_i(1+3w_i) \tag{4.129}$$

which can be studied for all the various components our Universe could feature

- $w < -1/3 \implies q < 0$ and hence the Universe accelerates;
- $w = 0 \implies q = \Omega_M/2 > 0$ and hence it decelerates. This was actually the underlying assumption in the Einstein-de Sitter Universe model;
- $w = 1/3 \implies q \approx \Omega_r > 0$;
- $w = -1 \implies q = -\Omega_\Lambda < 0$ and hence the Universe accelerates. This is what a de Sitter Universe would predict.

Vacuum Energy

Achtung! I know nothing of quantum field theory nor have any competence or background knowledge to assure you that what you'll read in this section is ontologically correct.

For this, I'm merely reporting what's written in my notes.

It is the purpose of this section to briefly discuss as for why the Cosmological Constant can be considered as a vacuum energy. First of all, if we consider the Lorentz-invariant quantuum void, the energy density associated to the Cosmological Constant should be roughly equal to present day's critical density

Any mismatch in units has to be attributed to the unspoken choice to use units in which $c = 1$, $k_B = 1$, and either h or $\hbar = 1$.

$$\rho_\Lambda \approx \rho_C(t_0) \approx 10^{-44}\,\mathrm{GeV}^4 \tag{4.130}$$

which, I'm told, assumes that

$$\langle T^{\mathrm{vac}}_{\mu\nu} \rangle = -\langle \rho \rangle g_{\mu\nu} \iff g_{\mu\nu} \text{ is a Lorentz invariant}$$

This implies that $\rho_{\mathrm{vac}} = \langle \rho \rangle$ and $p_{\mathrm{vac}} = \langle \rho \rangle = -\rho_{\mathrm{vac}}$. But to go any further than this we have to somehow assume a field theory of some sort

$$\langle \rho \rangle_{\mathrm{vac}} = \int_0^{k_{\max}} \frac{\mathrm{d}^3 k}{(2\pi)^3} \frac{1}{2} \sqrt{k^2 + m^2} \approx \frac{k^4_{\max}}{16\pi^2} \tag{4.131}$$

where $k_{\max}$ is an arbitrarily-high scale at which we can assume our QFT to be valid. For the sake of simplicity we can assume a system composed by a set of harmonic oscillators[11] with Hamiltonian

$$H_k = \omega_k \left(n_k + \frac{1}{2} \right) \tag{4.132}$$

hence

$$\frac{1}{2}\omega_k = \frac{1}{2}\sqrt{k^2 + m^2} \tag{4.133}$$

If we assume that the highest scale possible at which our theory is valid is the Planck scale

$$k_{\max} = M_{Pl} \sim 10^{19}\,\mathrm{GeV} \tag{4.134}$$

then the average energy density of the vacuum would be as high as $\langle \rho \rangle_{\mathrm{vac}} \sim 10^{74}\,\mathrm{GeV}^4$, which, I'm sure you've noticed, is 120 orders of magnitude larger that the expected critical density.

If we lower our upper bound for the allowed scales to the electroweak scale $\sim 200\,\mathrm{GeV}$, the average energy density of the vacuum would be $\langle \rho \rangle_{\mathrm{vac}} \sim 10^7\,\mathrm{GeV}^4$, which is still 51 orders of magnitude too large.

11 This is the only thing I can gather about all this.

Researchers way more qualified than me have come up with several possible explanation for this unsettling discrepancy:

1. A fluid with negative pressure, a.k.a. *dark energy* (also, Cosmological Constant, or Quintessence);
2. A modification of GR on the very largest scales (possibly including massive gravitons);
3. Non-linear effects of GR caused by interactions over large scales, so some sort of backreaction.

4.4 DYNAMIC AGE OF THE UNIVERSE

The actual model for the Universe, sometimes known as Benchmark model, is a ΛCDM model, where the Λ stands for the Cosmological Constant, while CDM stands for *Cold Dark Matter*.

We've already seen that we can write the total energy density as

$$\rho = \rho_m + \rho_r + \rho_\Lambda \tag{4.135}$$

which all scale differently but according to (4.91). This is most useful, because it allows us to write the total energy density as a function of the present-day density parameters and their associated scaling with the scale factor

$$\rho = \frac{3H_0^2}{8\pi G}\left(\Omega_{\Lambda,0} + \Omega_{r,0}\left(\frac{a_0}{a}\right)^4 + \Omega_{m,0}\left(\frac{a_0}{a}\right)^3\right) \tag{4.136}$$

We can plug this into (4.89) to obtain

$$\frac{\dot{a}^2}{a_0^2 H_0^2} + \frac{k}{a_0^2 H_0^2} = \frac{8\pi G}{3H_0^2}\left(\frac{a}{a_0}\right)^2 \rho \tag{4.137}$$

It might be useful to define and introduce the parameter $x = a(t)/a_0 = 1/(1 + z(t))$ which lives in the range $0 \leq x \leq 1$ since $a(t) \leq a_0$ (we're only looking back). With a realistic model, such as the Benchmark, we can safely put $k = 0$ and simplify the equation

$$\frac{\dot{a}^2}{a_0^2}\frac{1}{H_0^2} = \left(\frac{a}{a_0}\right)^2 \left(\Omega_{\Lambda,0} + \Omega_{r,0}\left(\frac{a_0}{a}\right)^4 + \Omega_{m,0}\left(\frac{a_0}{a}\right)^3\right) \tag{4.138}$$

which we can rewrite as

$$\frac{H^2}{H_0^2} \cdot x^2 = x^2 \left[\Omega_{\Lambda,0} + \Omega_{m,0} x^{-3} + \Omega_{r,0} x^{-4}\right] = x^2 E(x)^2 \tag{4.139}$$

and further manipulate

$$\frac{H}{H_0} x = xE(x) \to \frac{1}{a}\dot{a}\frac{1}{H_0} x = xE(x) \to \frac{1}{a_0}\frac{\mathrm{d}a}{\mathrm{d}t}\frac{1}{H_0} = xE(x) \tag{4.140}$$

But since, by virtue of its own definition, we have

$$\mathrm{d}x = \frac{\mathrm{d}a}{a_0} \to \mathrm{d}t = \frac{\mathrm{d}x}{xH_0 E(x)} \tag{4.141}$$

This we can easily integrate from $x = 0$ ($t = 0$, which corresponds to the BBN in the standard model of the Hot Big Bang)

$$\boxed{t(x) = \int_0^x \mathrm{d}t = \int_0^x \frac{\mathrm{d}x'}{x' H_0 E(x')}} \tag{4.142}$$

which represents the *dynamical age of the Universe*. Sometimes it might be useful to express it in terms of the redshift z so that

$$E^2(z) = \Omega_{\Lambda,0} + \Omega_{m,0}(1+z)^3 + \Omega_{r,0}(1+z)^4 \quad \mathrm{d}x = -\frac{\mathrm{d}z}{(1+z)^2} \tag{4.143}$$

The dynamical age in terms of the redshift is then

$$t(z) = \frac{1}{H_0} \int_z^\infty \frac{\mathrm{d}z'}{(1+z')E(z')} \tag{4.144}$$

For the Einstein-de Sitter Universe ($k = 0$, $\Omega_{m,0} = 1$), the dynamical time evaluates to

$$t_0^{E-dS} = \frac{1}{H_0} \int_0^1 \frac{\mathrm{d}x}{x \cdot x^{-3/2}} = \frac{2}{3H_0} \approx 7.0 h^{-1}\,\mathrm{Gyr} \tag{4.145}$$

which is smaller than a Hubble time H_0^{-1}. The dynamical time produced by a E-dS Universe is too short: It's in tension with the estimated age of some structures and/or unstable chemical elements which we can still observe (e.g., radionuclids with decay time of about $10 - 15\,\mathrm{Gyr}$). It is also in high tension with the estimated age of some old white dwarves and with the estimated age we've obtained from the temperature fluctuations of the CMB.

For a realistic ΛCDM model ($k = 0$, $\Omega_{r,0} \sim 0$, and $\Omega_{\Lambda,0} + \Omega_{m,0} \sim 1$), the dynamical time is evaluated to

$$\begin{aligned} t_0 &\approx \frac{1}{H_0} \int_0^1 \frac{\mathrm{d}x}{x(\Omega_{\Lambda,0} + \Omega_{m,0} x^{-3})^{1/2}} \\ &= \frac{1}{H_0 \Omega_{\Lambda,0}^{1/2}} \int_0^1 \frac{\mathrm{d}x}{x\left(1 + \left(\frac{1}{\Omega_{\Lambda,0}} - 1\right) x^{-3}\right)^{1/2}} \end{aligned}$$

which yields

$$t_0 = \frac{2}{3H_0} \frac{1}{\Omega_{\Lambda,0}^{1/2}} \ln\left[\frac{1 + \Omega_{\Lambda,0}^{1/2}}{(1 - \Omega_{\Lambda,0})^{1/2}}\right] \sim 14\,\mathrm{Gyr} \tag{4.146}$$

5 THERMAL HISTORY OF THE UNIVERSE

For a significant part of this chapter, I'll be following [2]. Please note that for some reason I cannot fathom, Baumann is using the metric signature $(+,-,-,-)$, which is the exact opposite of what we've been using in the previous chapters.

The key to understanding the thermal history of the Universe is the comparison between the rate of interactions Γ and the rate of expansion H. When $\Gamma \gg H$, then the time scale of particle interactions is much smaller than the characteristic expansion time scale

$$t_c = \frac{1}{\Gamma} \ll t_H = \frac{1}{H} \tag{5.1}$$

and Local Thermal Equilibrium (LTE) is reached before the effect of the expansion becomes relevant. As the Universe cools, the rate of interactions may decrease faster than the expansion rate. At $t_c \sim t_H$, the particles decouple from the thermal bath. Different particle species may have different interaction rates and so may decouple at different times.

The reactions we'll be interested in looking into are those of the likes

$$\begin{aligned} x + y &\leftrightharpoons x + y \quad \text{(scattering)} \\ y + \bar{y} &\leftrightharpoons \gamma + \gamma \quad \text{(pair production)} \end{aligned}$$

We mark the beginning of the thermal history of the Universe at $(k)T \sim 1\,\text{MeV}$, which roughly corresponds to $1\,\text{s}$ after the Big Bang. So we'll implicitly assume that quarks are already confined within hadrons, which are already slowed to a non-relativistic regime (while photons, electrons, and neutrinos have not).

Baryon to Photon Ratio

For future reference, let's define now the *Baryon to Photon ratio* η. Provided that photons are not created or destroyed, but merely redshifted, $n_\gamma \propto a^{-3}$. Similarly, if baryons are not destroyed during the evolution of the Universe (that is, they are stable and do not annihilate), their number density merely gets diluted $n_{\text{Baryon}} \propto a^{-3}$. Therefore

$$\eta = n_{\text{Baryon}}/n_\gamma$$

is constant and approximately equal to $\eta \approx 6.1 \cdot 10^{-10}$.

5.0.1 Statistical Properties

Before getting knee-deep into calculations, let's take a moment to recall some statistical properties and properties in general which we'll be using extensively in the following.

First of all, recall the Bose-Einstein/Fermi-Dirac distribution function ($c = \hbar = 1$)

$$f(p,T) = \frac{1}{\exp\left(\frac{E(p)-\mu}{kT}\right) \pm 1} \tag{5.2}$$

where the $+$ identifies the Fermi-Dirac distribution, and the $-$ the Bose-Einstein distribution. We'll be always working under conditions for which $\mu \ll kT$, so we can drop it from (5.2). That being the case, we can define the particle numeric density

$$n = \frac{g}{h^3} \int \mathrm{d}^3 p f(p,T) \tag{5.3}$$

and the associated energy density

$$\rho(T) = \frac{g}{h^3} \int \mathrm{d}^3 p E(p) f(p,T) \tag{5.4}$$

We can assume an isotropic distribution for the momentum $\mathrm{d}^3 p = 4\pi p^2 \, \mathrm{d}p$, and define $m = xT$ and $p = \xi T$, so that the integrals in (5.3) and (5.4) can be expressed as

As often happens, what's really important for you to remember it's not the whole expression, but rather the characteristic scaling of quantities. In this particular case, you should commit to memory $n \propto T^3$ and $\rho \propto T^4$.

$$n(T) = \frac{g}{h} T^3 I_\pm \quad \rho(T) = \frac{g}{h} T^4 J_\pm \tag{5.5}$$

where $I_\pm$ and $J_\pm$ are the two integral functions reported below

$$I_\pm = \int_0^\infty \mathrm{d}\xi \frac{\xi^2}{\exp\left(\sqrt{x^2+\xi^2}\right) \pm 1} \qquad J_\pm = \int_0^\infty \mathrm{d}\xi \frac{\xi^2 \sqrt{x^2+\xi^2}}{\exp\left(\sqrt{x^2+\xi^2}\right) \pm 1} \tag{5.6}$$

In the relativistic limit ($x \to 0$, $m \ll (k)T$) the equation for $I_\pm$ can be solved explicitly

$$\begin{aligned} I_\pm(x=0) &= \int_0^\infty \frac{\xi^2}{\exp(\xi) \pm 1} \mathrm{d}\xi \\ &= \sum_{j=1}^{\infty} (\mp 1)^{j-1} \int_0^\infty \xi^2 e^{-j\xi} \, \mathrm{d}\xi = \begin{cases} 2\zeta(3) & (-) \\ \frac{3}{2}\zeta(3) & (+) \end{cases} \end{aligned}$$

where $\zeta(x)$ is the Riemann ζ-function. Hence we get for the particle density

$$\boxed{\frac{\zeta(3)}{\pi^2} g T^3 \begin{cases} 1 & \text{bosons} \\ \frac{3}{4} & \text{fermions} \end{cases}} \tag{5.7}$$

A similar computation for the energy density gives

$$\boxed{\frac{\pi^2}{30} g T^4 \begin{cases} 1 & \text{bosons} \\ \frac{7}{8} & \text{fermions} \end{cases}} \tag{5.8}$$

In the non-relativistic limit, on the other hand, we have $x \gg 1$, and the integral is the same for both bosons and fermions

$$I_\pm(x) = J_\pm \approx \int_0^\infty \frac{\xi^2}{\exp\left(\sqrt{\xi^2+x^2}\right)} \mathrm{d}\xi \tag{5.9}$$

Most of the contribution to the integral comes from $\xi \ll x$. After Taylor-expanding the square root in the exponential to lowest order in ξ, we obtain

$$I_{\pm}(x) = \sqrt{\frac{\pi}{2}} x^{3/2} e^{-x} \tag{5.10}$$

which introduces a Boltmann-like soppression

Note that in the expression for the energy density, the bulk of the contribution comes from the term mn.

$$n(T) = g\left(\frac{mT}{2\pi}\right)^{3/2} e^{-m/T} \quad \rho(T) = mn + \frac{3}{2} nT \tag{5.11}$$

As expected, massive particles are exponentially rare at low temperatures, $(k)T \ll m$.

5.0.2 Effective Number of Relativistic Species

Let T be the temperature of the photon gas. The total radiation density is the sum over the energy densities of all relativistic species

$$\rho_r = \sum_i \frac{g_i}{2\pi^2} T_i^4 J_{\pm}\left(x_i = \frac{m_i}{T_i}\right) := \frac{\pi^2}{30} g_\star T_\gamma^4 \tag{5.12}$$

where $g_\star$ is the *effective number of relativistic degrees of freedom* at the temperature T, assumed identical for all the different relativistic species. The definition for the effective number of relativistic degrees of freedom is the following

$$\boxed{g_\star = \sum_i g_i \left(\frac{T_i}{T_\gamma}\right)^4 \frac{J_{\pm}(x_i)}{J_{-}(0)}} \tag{5.13}$$

It might be useful to differentiate the relativistic species in thermal equilibrium with the photons

When the temperature drops below the mass m_i of a particle species, it becomes non-relativistic and is removed from the sum. Away from mass thresholds, the thermal contribution is independent of temperature.

$$g_\star^{\text{th}} = \sum_{i=b} g_i + \frac{7}{8} \sum_{i=f} g_i \tag{5.14}$$

from the relativistic species that are *not* in thermal equilibrium with the photons

$$g_\star^{\text{dec}} = \sum_{i=b} g_i \left(\frac{T_i}{T_\gamma}\right)^4 + \frac{7}{8} \sum_{i=f} \left(\frac{T_i}{T_\gamma}\right)^4 g_i \tag{5.15}$$

The evolution of $g_\star$ assuming the Standard Model particle content is better represented with a plot, that is often clearer than dozens of lines of calculations. You can find this plot in Fig.5.1.

Counting DoFs

As I mentioned in some earlier chapter, I know pretty much nothing about QFT and stuff. But since I think it is at least interesting to see how the computation is done, I'll rely on someone who knows better than me, namely Mr. Baumann.
At $(k)T \gtrsim 100\,\text{GeV}$, all particles of the Standard Model are relativistic. Adding up their internal degrees of freedom we get

- $g_b = 28$: Photons (2), $W^\pm$ and Z^0 $(3 \cdot 3)$, gluons $(8 \cdot 2)$, and Higgs (1);
- $g_f = 90$: Quarks $(6 \cdot 12)$, charged leptons $(3 \cdot 4)$, and neutrinos $(3 \cdot 2)$.

hence $g_\star = g_b + 7g_f/8 = 106.75$. As the temperature drops, various particle species become non-relativistic and annihilate. To estimate $g_\star$ at a temperature T you simply add up the contributions from all relativistic degrees of freedom and discard the rest.

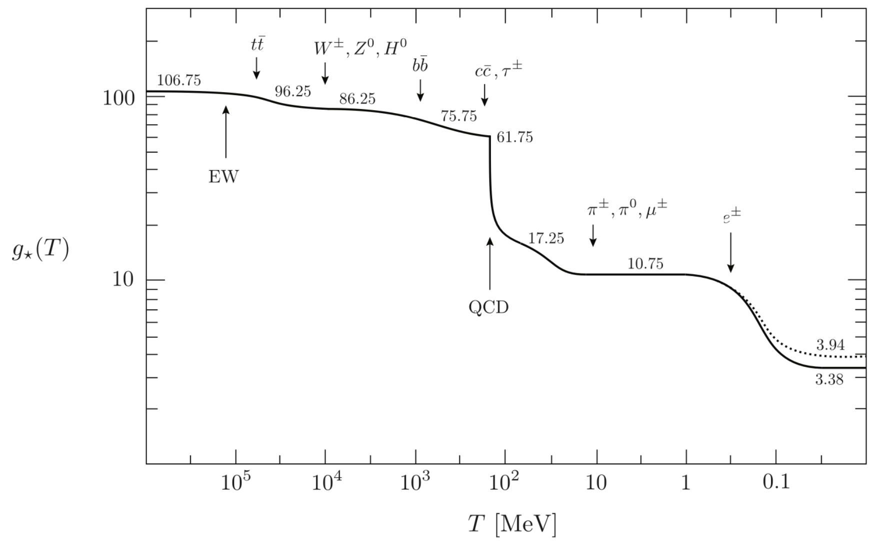

Figure 5.1: Evolution of relativistic degrees of freedom $g_\star(T)$ assuming the Standard Model particle content. The dotted line stands for the number of effective degrees of freedom in entropy $g_{\star S}(T)$. Credits: [2].

5.0.3 Conservation of Entropy

In the field of cosmology, entropy is often more informative than energy. According to the second law of thermodynamics, the total entropy of the Universe only increases or stays constant. Since there are far more photons than baryons in the Universe, the entropy of the Universe is dominated by the entropy of the photon bath (at least as long as the Universe is sufficiently uniform). Any entropy production from non-equilibrium processes is therefore insignificant relative to the total entropy.

To a good approximation we can treat the expansion of the Universe as *adiabatic*, so that the total entropy stays constant even beyond equilibrium. Consider

$$T\,\mathrm{d}S = \mathrm{d}U + p\,\mathrm{d}V \tag{5.16}$$

and define the entropy and energy density respectively as $s = S/V$ and $\rho = U/V$. The equation above thus becomes

$$T\,\mathrm{d}(sV) = \mathrm{d}(\rho V) + p\,\mathrm{d}V \tag{5.17}$$

whence

$$(Ts - \rho - p)\,\mathrm{d}V + V\left(T\frac{\mathrm{d}s}{\mathrm{d}T} - \frac{\mathrm{d}\rho}{\mathrm{d}T}\right)\mathrm{d}T = 0 \tag{5.18}$$

For the identity to be verified it's necessary that

$$s = \frac{p+\rho}{T} \tag{5.19}$$

from which

$$\frac{\mathrm{d}s}{\mathrm{d}T} = \frac{1}{T}\frac{\mathrm{d}\rho}{\mathrm{d}T} \rightarrow \frac{\mathrm{d}s}{\mathrm{d}t} = \frac{1}{T}\frac{\mathrm{d}\rho}{\mathrm{d}t} \tag{5.20}$$

Recall the continuity equation stemming from the FRLW metric (4.80)

$$\dot{\rho} = -3(p+\rho)H = -3HsT \tag{5.21}$$

We're thus able to conclude by simple substitution that

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}(sa^3) = 0} \tag{5.22}$$

It is also possible to show that the entropy density is related to temperature and to the degrees of freedom of the Universal bath by means of

$$s = \frac{2\pi^2}{45} g_{\star S}(T)T^3 \tag{5.23}$$

where we've defined the *effective number of degrees of freedom in entropy*

$$g_{\star S} = \sum_{i=b} g_i \left(\frac{T_i}{T}\right)^3 + \frac{7}{8}\sum_{i=f} g_i \left(\frac{T_i}{T}\right)^3 \tag{5.24}$$

Note that for species in thermal equilibrium $g^{\text{th}}_{\star S}(T) = g^{\text{th}}_{\star}(T)$. However, given that $s_i \propto T_i^3$, only for decoupled species we get (5.24). Hence, $g_{\star S}$ is equal to $g_\star$ only when all the relativistic species are in equilibrium at the same temperature.

Substituting (5.23) into (5.22)

$$\frac{\mathrm{d}}{\mathrm{d}t}(g_{\star S}a^3T^3) = 0 \tag{5.25}$$

which gives us an expression for the Universe's cooling law! In fact, assuming $g_{\star S} \approx$ const. (reasonable for temperatures away from particle mass thresholds), we get

$$\boxed{T \propto \frac{1}{a} \propto 1+z} \tag{5.26}$$

Such scaling relation is surely valid after the last scattering epoch (which we'll be shortly introducing), though it's worth pointing out that this conservation law doesn't include the hyperfine details of irreversible reactions on the smaller scales.

Note that we're implicitly assuming $k \sim 0$ (flat Universe), so that $\Omega_0 = 0$. We're also using the Planck mass $M_{Pl} = \sqrt{\hbar c/8\pi G}$.

Substituting $T \propto g_{\star S}^{-1/3} a^{-1}$ into Friedmann's equation we find

$$H = \frac{\dot{a}}{a} \approx \left(\frac{\rho_r}{3M_{Pl}^2}\right)^{1/2} \approx \frac{\pi}{3}\left(\frac{g_\star}{10}\right)^{1/2} \frac{T^2}{M_{Pl}} \tag{5.27}$$

which for a RD Universe reproduces the usual result $a \propto t^{1/2}$, except that there is a change in the scaling every time $g_{\star S}$ changes. For $T \propto t^{-1/2}$ (as follows from $H = 1/2t$) we can integrate the Friedmann equation and get the temperature as a function of time

$$\left(\frac{T}{\text{MeV}}\right) \approx 1.5 g_\star^{-1/4} \left(\frac{1\,\text{s}}{t}\right)^{1/2} \tag{5.28}$$

5.1 NEUTRINO DECOUPLING

Neutrinos are coupled to the thermal bath via weak interaction processes like

$$\begin{aligned} e^- + \nu_e &\leftrightharpoons e^- + \nu_e \quad \text{(electron scattering)} \\ e^- + e^+ &\leftrightharpoons \nu_e + \bar{\nu}_e \quad \text{(pair production)} \end{aligned}$$

Consider the production rate $\Gamma = n\sigma v \approx H$ (at decoupling). We know that during RD, the Hubble constant was approximately equal to

$$H^2 = \frac{\pi^2}{90} g_\star \frac{T^4}{M_{Pl}^2} \tag{5.29}$$

Moreover, relativistic neutrinos have a particle density given by the fermionic version of (5.7), and their decoupling temperature is significantly lower than the electroweak mass-scale m_ω, which means that the cross-section for the interaction is given by $\sigma = G_F^2 T^2$.

Putting everything together we find out that the interaction rate $\Gamma \propto T^5$, while the Hubble constant $H \propto T^2$, which means that equivalence can be reached! And indeed we can quickly produce a temperature at neutrinos decoupling

Watch out! This is a typical question that is asked during the oral exam!

$$\boxed{T_d \sim \left(\frac{1}{M_{Pl} G_F^2}\right)^{1/3} \sim \text{MeV}} \tag{5.30}$$

After decoupling, neutrinos move freely along geodesics and preserve to an excellent approximation the relativistic Fermi-Dirac distribution (even after they become non-relativistic at later times).

5.2 ELECTRON–POSITRON ANNIHILATION

Shortly after the neutrinos decouple, the temperature drops below the electron mass and electron-positron annihilation occurs (along with Compton scattering that keeps electron coupled to photons)

$$e^-e^+ \leftrightharpoons \gamma\gamma$$
$$e^-\gamma \leftrightharpoons e^-\gamma$$

This, however, is not a decoupling! As temperature decreases, the equilibrium position of the annihilation reaction moves towards the photons, due to the decreasing energy to support pair production. The energy density and entropy of electrons and positrons are transferred to the photons, but not to the decoupled neutrinos. The photons are thus "heated" (the photon temperature does not decrease as much) relative to the neutrinos.

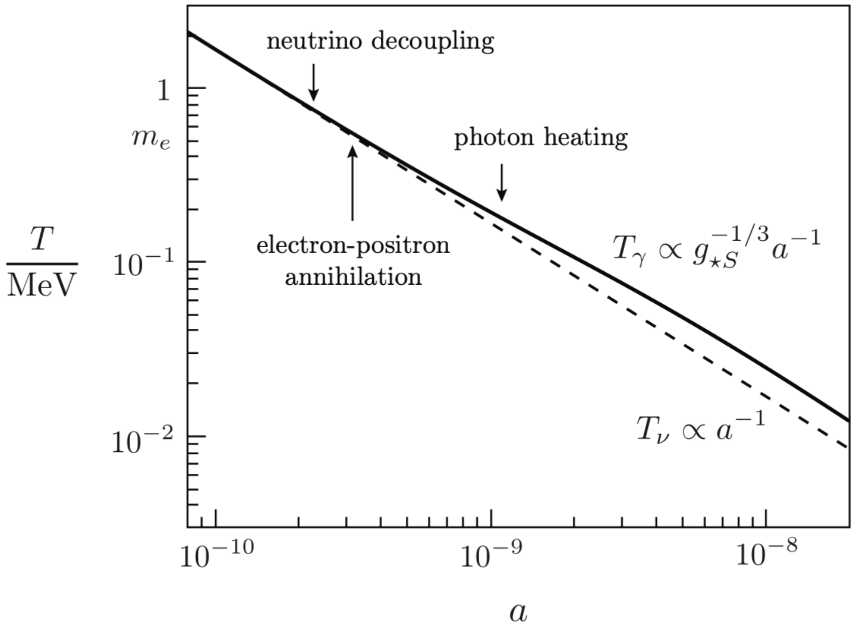

Figure 5.2: Thermal history through electron-positron annihilation. Neutrinos are decoupled and their temperature redshifts simply as $T_\nu \propto a^{-1}$. The energy density of the electron-positron pairs is transferred to the photon gas whose temperature therefore redshifts more slowly, $T_\gamma \propto g_{\star S}^{-1/3} a^{-1}$. Credits: [2].

Let's do a quick entropic balance, assuming $T^I_\gamma = T^I_\nu$, which along with $a^3 T^3_\gamma g_\star = \text{const}$, allows us to compute it separately for both coupled and uncoupled species.

Fix the scale parameter a and assume an adiabatic process, so that if we neglect neutrinos and other decoupled species, we get

$$g^{th}_{\star S} = \begin{cases} 2 + \frac{7}{8}\cdot 4 = \frac{11}{2} & (k)T \gtrsim m \quad (\gamma,\, e^-,\, e^+ \text{with their spins}) \\ 2 & (k)T \lesssim m \quad (\gamma \text{ and their polarizations}) \end{cases} \tag{5.31}$$

This means that the temperature of photons before and after annihilation is in the ratio

$$\frac{T^I_\gamma}{T^F_\gamma} = \left(\frac{4}{11}\right)^{1/3} \approx 0.7 \tag{5.32}$$

This also means that the temperature of neutrinos is slightly lower than the photon temperature after e^+e^- annihilation

$$\boxed{\frac{T_\nu^F}{T_\gamma^F} = \left(\frac{4}{11}\right)^{1/3}} \tag{5.33}$$

This is because, at equilibrium, $g_{\star S}^{th}(aT_\gamma)^3$ remains constant, and we later find that aT_γ increases after electron-positron annihilation, $(k)T < m_e$, by a factor $(11/4)^{1/3}$, while aT_ν remains the same since neutrinos are decoupled.

The present-time temperature of photons is measured from the CMB, $T_\gamma(t_0) = 2.7\,\mathrm{K}$, while the predicted neutrino temperature for a CνB (*cosmic neutrino background*) is $T_\nu(t_0) = 1.95\,\mathrm{K}$ (yet to be confirmed experimentally). The associated particle densities are quickly found

$$n_\gamma \sim g_\gamma T_\gamma^3 \quad n_\nu \sim \frac{3}{4} g_\nu T_\nu^3 \tag{5.34}$$

We can write the latter in terms of the former using the temperature ratio we've previously found

$$n_\nu = \frac{3}{4}\left(\frac{4}{11} n_\gamma\right) \approx 112\,\mathrm{cm}^{-3} \quad \text{(per flavour)} \tag{5.35}$$

At the predicted temperature for the CνB, two out of three neutrino flavours would be non-relativistic. But that's not a big problem since the scaling of density with temperature is already fixed (they've long since decoupled).

Cosmologic Bound to Neutrino Masses

Achtung! Neutrinos, while the Universe expanded, have turned non-relativistic, that's why we're using $\rho \approx mn$.

Consider the density parameter $\Omega_\nu(t_0)$ for neutrinos

$$\Omega_\nu(t_0) = \frac{\rho_\nu(t_0)}{\rho_c} = \frac{\sum_i m_{\nu,i} n_{\nu,i}(t_0)}{\rho_c} \tag{5.36}$$

which yields

$$\boxed{\Omega_\nu(t_0)h^2 \approx \frac{\sum_i m_{\nu,i}}{94\,\mathrm{eV}}} \tag{5.37}$$

where we've used $H_0 = 100h\,\mathrm{km\,s^{-1}\,Mpc}$ and $h = 0.695$. By demanding that neutrinos don't over close the Universe, i.e. $\Omega_\nu < 1$, we can give a cosmological upper bound on the sum of the neutrino masses

$$\sum_i m_{\nu,i} \leq 14 - 15\,\mathrm{eV} \tag{5.38}$$

while from the combined effort of the Planck collaboration and analysis of BAO[1] we get an even stricter upper bound $\sum_i m_{\nu,i} \leq 0.13\,\mathrm{eV}$. On the other hand, we also have a lower bound given by neutrinos oscillations

$$\sum_i m_{\nu,i} \geq 0.06\,\mathrm{eV} \quad \text{(normal ordering)}$$
$$\sum_i m_{\nu,i} \geq 0.1\,\mathrm{eV} \quad \text{(inverted ordering)}$$

1 Baryonic Acoustic Oscillations.

5.3 PHOTON RECOMBINATION AND DECOUPLING

An important event in the history of the early Universe is the formation of the first atoms. At temperatures above $\sim 1\,\text{eV}$, the Universe still consisted of a plasma of free electrons and nuclei. Photons were tightly coupled to the electrons via Compton scattering, which in turn strongly interacted with protons via Coulomb scattering. There was very little neutral Hydrogen. When the temperature became low enough, the electrons and nuclei combined to form neutral atoms (*recombination*), and the density of free electrons fell sharply.

5.3.1 Hydrogen recombination

What we're going to look into now is the recombination reaction of Hydrogen[2]

$$e^- + p^+ \leftrightharpoons \mathrm{H} + \gamma \quad (\chi = 13.6\,\text{eV})$$

Clearly, the electrons participating in such reaction are those that had survived annihilation. We can write a chemical balance for the species involved

$$\frac{n_{\mathrm{H}}}{n_e n_p} = \frac{g_{\mathrm{H}}}{g_e g_p}\left(\frac{m_{\mathrm{H}}}{m_e m_p}\frac{2\pi}{T}\right)^{3/2} \exp\left(\frac{m_p + m_e - m_{\mathrm{H}}}{(k)T}\right) \exp\left(\frac{\mu_p + \mu_e - \mu_{\mathrm{H}}}{(k)T}\right) \tag{5.39}$$

In the prefactor, we can use $m_H \approx m_p$, but in the exponential the small difference between m_H and $m_p + m_e$ is crucial.

but both the ratio of the state degeneracies g and the exponential of the chemical potentials are equal to 1 at equilibrium. If the Universe is globally neutral, then $n_e = n_p$. Also, we can introduce the *free electron fraction*

$$X_e = \frac{n_e}{n_e + n_{\mathrm{H}}} = \frac{n_e}{n_p + n_{\mathrm{H}}} = \frac{n_e}{n_b} = \frac{n_e}{\eta n_\gamma} \tag{5.40}$$

where in the last equality we have introduced the baryon to photon ratio η. To simplify the discussion, let us ignore all nuclei other than protons (over 90% (by number) of the nuclei are protons). The total baryon number density can then be approximated as $n_b \approx n_p + n_{\mathrm{H}} = n_e + n_{\mathrm{H}}$ and hence

$$\frac{1 - X_e}{X_e^2} = \frac{n_b n_{\mathrm{H}}}{n_e^2} \tag{5.41}$$

which, plugged in the equation for chemical balance (5.39), yields the celebrated *Saha equation*

$$\boxed{\left(\frac{1 - X_e}{X_e^2}\right)_{\mathrm{eq}} = \frac{2\eta\zeta(3)}{\pi^2} T^3 \left(\frac{2\pi}{m_e k T}\right)^{3/2} \exp\left(\frac{m_p + m_e - m_{\mathrm{H}}}{(k)T}\right)} \tag{5.42}$$

As a matter of fact, at times Saha's equation is written simply as

$$1 - X_e(T) = f(t) X_e^2(T) \tag{5.43}$$

From here we can choose an arbitrary threshold for when we can consider the Universe "recombined." Some choose the moment at which $X_e \leq 0.5$,

2 Technically speaking, we should also consider the recombination reaction for Helium.

while others choose $X_e \leq 0.1$; you choose, the resulting difference is minimal. Following Baumann, we find a recombination temperature

$$T_{\text{rec}} = 0.3\,\text{eV} \approx 3600\,\text{K} \tag{5.44}$$

Please notice that $T_{\text{rec}} < \chi = 13.6\,\text{eV}$, since there are very many photons for each Hydrogen atom and the high-energy tail of the photon distribution contains photons with energy larger the ionization threshold, allowing Hydrogen ionization.

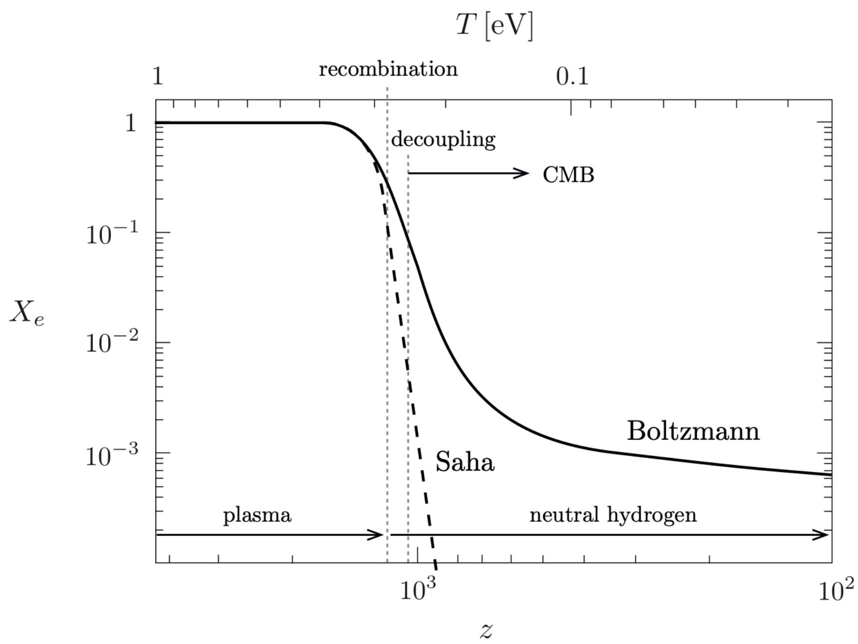

Figure 5.3: Free electron fraction as a function of redshift. The horizontal axis is reversed to replicate the natural flow of time from left to right. Credits: [2].

Using $T_{\text{rec}} = T_0(1 + z_{\text{rec}})$, with $T_0 = 2.7K$, gives the redshift of recombination

$$z_{\text{rec}} \approx 1320 \tag{5.45}$$

Since matter-radiation equality is at $z_{\text{eq}} \approx 3500$, we conclude that recombination occurred in the matter-dominated era.

On an ending note, notice how thanks to Boltmann's contribution in the non-relativistic regime the ionization fraction never reaches zero. The period during which the Universe is mostly neutral is often called *Dark Ages*.

At a much later time ($z \sim 6 - 8$), the Universe will eventually be reionized thanks to the formation of the first massive O-class stars[3].

5.3.2 Photon Decoupling

Photons are most strongly coupled to the primordial plasma through their interaction with electrons

$$e^- + \gamma \leftrightharpoons e^- + \gamma$$

3 You can check my Extragalactic Astrophysics Notes if you're interested.

with an interaction rate given, as usual, by $\Gamma = n_e \sigma c$. Here the interaction cross section is that of Thomson scattering $\sigma_T \approx 2 \cdot 10^{-3}\,\mathrm{MeV}^{-2}$. Photons and electrons decouple roughly when the interaction rate becomes smaller than the expansion rate

$$\Gamma_\gamma(T_{\mathrm{dec}}) \sim H(T_{\mathrm{dec}}) \tag{5.46}$$

Writing

Recall that both recombination and decoupling both occur after matter-radiation equality, hence the Universe is roughly matter dominated.

$$\begin{aligned} \Gamma_\gamma(T_{\mathrm{dec}}) &= n_b X_e(T_{\mathrm{dec}})\sigma_T = \frac{2\zeta(3)}{\pi^2}\eta\sigma_T X_e(T_{\mathrm{dec}})T_{\mathrm{dec}}^3 \\ H(T_{\mathrm{dec}}) &= H_0\sqrt{\Omega_m}\left(\frac{T_{\mathrm{dec}}}{T_0}\right)^{3/2} \end{aligned}$$

we get

$$X_e(T_{\mathrm{dec}})T_{\mathrm{dec}}^{3/2} \sim \frac{\pi^2}{2\zeta(3)}\frac{H_0\sqrt{\Omega_m}}{\eta\sigma_T T_0^{3/2}} \tag{5.47}$$

Thanks to Saha's equation we find

$$T_{\mathrm{dec}} \sim 0.27\,\mathrm{eV} \tag{5.48}$$

Notice that although T_{dec} isn't far from T_{rec}, the ionization fraction decreases significantly between recombination and decoupling

$$X_e(T_{\mathrm{rec}}) \approx 0.1 \to X_e(T_{\mathrm{dec}}) \approx 0.01$$

This shows that a large degree of neutrality is necessary for the Universe to become transparent to photon propagation.

The redshift and time of decoupling are, respectively, $z \approx 1090$, and $t \approx 370,000\,\mathrm{yr}$.

5.4 DARK MATTER

Dark matter is a...thorny subject: At the same time we know little to nothing about it, and yet know plenty. We know that it must be abundant, much more abundant than standard baryonic matter, and must not interact with electromagnetic radiation. That, incidentally, also means that we can't see it, hence the name.

So how do we know that it must exist? To put it simply, we have several astrophysical evidences, gravitational in nature, that suggest the presence of this exotic type of matter.

To name a few, you might know that galactic rotation curves would look rather different if DM halos didn't exist: The first discrepancies between the expected and the observed rotation curves were investigated as early as the 1970s [3], [17] bringing the astrophysicists of the time to speculate the existence of something they couldn't see but that had to have a density $\rho \propto r^{-2}$ so that rotation curves could be as flat as they looked like.

Another evidence came from structure formation theory. If DM didn't exist, there wouldn't have been enough time for standard baryonic matter to collapse and produce structures much denser than the average background of the Universe like galaxies or clusters.

MOND: Modified Newtonian Dynamics

At the time, not everyone was buying the thing about dark matter.

It felt...convenient, too convenient. So of course they produced another extremely convenient theory, which is that of *Modified Newtonian Dynamics*. Essentially MOND's stans claimed (and apparently some still claim) that Newton's celebrated 2nd law of dynamics $\vec{F} = m\vec{a}$ didn't hold equally for all acceleration scales, but rather had to be modified introducing an acceleration-dependent function $\mu(a/a_0)$, with $a_0 \sim 10^{-10}\,\mathrm{m\,s^{-2}}$.

Newton's second law would then look like

$$\vec{F}_N = m\vec{a} \cdot \begin{cases} 1 & a/a_0 \gg 1 \\ a/a_0 & a/a_0 \ll 1 \end{cases} \tag{5.49}$$

Applying this to an object of mass m in circular orbit around a point mass M yields

$$m\left(\frac{a^2}{a_0}\right) = \frac{GMm}{r^2}$$

In the deep MOND regime, the true acceleration of a test particle on a circular orbit is still the centripetal acceleration, so

$$\left(\frac{v_c^2}{r}\right)^2 = \frac{GMa_0}{r^2}$$

which is none other than the Baryonic Tully-Fisher relation

$$v_c^4 = GMa_0 \tag{5.50}$$

This would be great if (a) the BTFR scaled exactly with v^4 (obviously it doesn't), (b) didn't exist the bullet-cluster proof: It proved that during the collision of two clusters of galaxies, the gravitational lensing is centered on the gas-stripped clusters (where the DM would be) rather than on the stripped gas as MOND would predict.

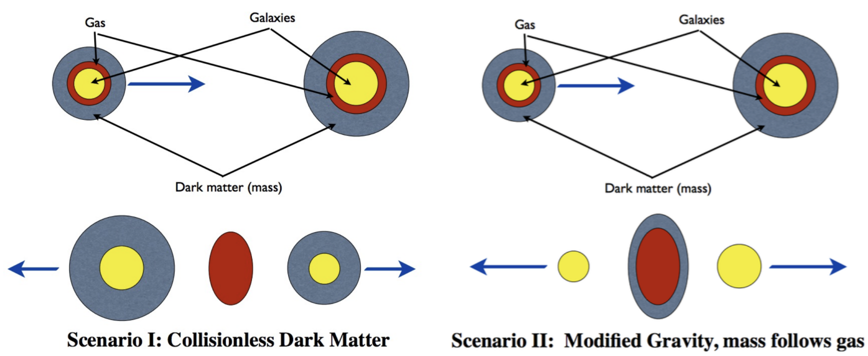

Figure 5.4: The two different scenarios that were surmised when the bullet cluster was first discovered.

Unfortunately for MOND, the scenario of collisionless DM prevailed, as shown by Fig.5.5.

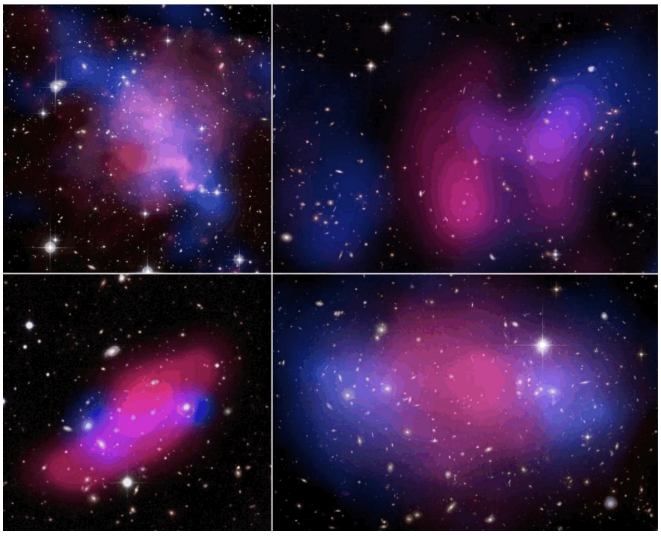

Figure 5.5: The so-called bullet cluster evidence. Three colliding galaxy clusters and one colliding group (at the lower-left), showing the separation between X-rays (pink) and gravitation (blue), indicative of dark matter.

5.4.1 Galaxy Clusters

Galaxy clusters, as the name suggests, are systems comprising of multiple galaxies. They were already object of studies back in the 1930s.

One cool thing about them is that they verify the *virial theorem*

$$2\langle T\rangle_{\text{trasl.}} = -\sum_k \langle \vec{F}_k \cdot \vec{r}_k\rangle_{\text{trasl.}} \tag{5.51}$$

which under the assumption of a conservative force associated to a central potential $U(r) \propto r^{-n}$ translates into the equation the theorem is most known with

$$2\langle T\rangle_{\text{trasl.}} = -n\langle U\rangle \tag{5.52}$$

Assuming gravitational potential, the virial theorem allows us to predict the total mass of a cluster by making a simple assumption about its symmetry (spherical, of course) and through measurement of the velocity dispersion of the galaxies within

$$M_{\text{tot}} = \frac{5R\langle v^2\rangle}{3G} \tag{5.53}$$

which, for example, for the Coma Cluster predicts a mass $M_{\text{tot}} \approx 2 \cdot 10^{15} M_\odot$. For such a cluster, the mass-to-light ratio M/L would be roughly equal to $250 M_\odot/L_\odot$. Even carrying out the due computations with some more care, you don't manage to bring it below $\sim 10 - 20 M_\odot/L_\odot$, which is not good since the expected mass-to-light ratio is of order $2 - 3 M_\odot/L_\odot$.

Cluster trivia. Did you know that the mass of a cluster is composed by:

- *1% stars;*
- *9% Intracluster Medium;*
- *90% Dark Matter.*

Moreover, as was ascertained from the aforementioned interaction between clusters, DM is also barely interacting with itself, let alone matter or radiation. One of the few viable ways we can detect its presence and its effects is gravitational lensing.

Research in the cosmological field mainly focuses on two major events: The BBN and the CMB, both of which allows us to measure Ω_m and Ω_b.

Clearly, if we assume only DM and baryonic matter to contribute to Ω_m, a simple subtraction makes us able to give an estimate to DM's density parameter. The current values set

$$\Omega_m \approx 0.3 \quad \Omega_b \approx 0.0486 \implies \Omega_{DM} = 0.2589 \tag{5.54}$$

so most of the matter contribution comes from DM.

What if we wanted to use the values of Ω_b and Ω_{DM} to put constrains on the mass our hypothetical DM particle should have? Would that be of any help? Present estimates constrain DM mass in the range

"PBH" stands for Primordial Black Holes, one of the exotic candidates for DM.

$$10^{-22}\,\text{eV} \lesssim m_{DM} \lesssim m_{PBH} \approx 10^{70}\,\text{eV} \approx 10^4 - 10^5 M_\odot \tag{5.55}$$

corresponding to scales of order

$$\lambda \sim (2\,\text{kpc}) \left(\frac{10^{-22}\,\text{eV}}{m}\right) \left(\frac{10\,\text{km s}}{v}\right) \tag{5.56}$$

With only this information at hand, the only candidate within the current Standard Model is the neutrino: It does not emit light, it's barely interacting with other stuff, it's non-baryonic, and it's stable on cosmological scales. However structure-formation theory has long discarded neutrinos as viable candidates for DM (they're not a major component, at least): To put it simply, neutrinos would be too "hot" (read: relativistic for far too long after their decoupling), and their free-streaming would smear out density fluctuations on scales of order

$$d_{\min} \approx ct_h \approx 13\,\text{kpc} \left(\frac{m_h c^2}{3\,\text{eV}}\right)^{-2} \tag{5.57}$$

$$r_{\min} = \frac{d_{\min}}{a(t_h)} \approx 55\,\text{Mpc} \left(\frac{m_h c^2}{3\,\text{eV}}\right)^{-1} \tag{5.58}$$

where the latter is the scale in comoving units (i.e., the size of the region *today*) below which fluctuations are erased.

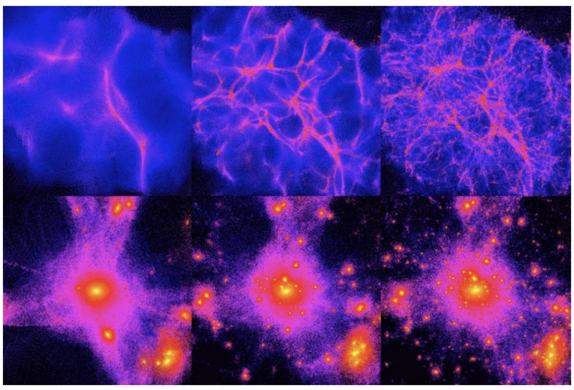

Figure 5.6: Top row: Simulations of what structure in Hot (left), Warm (middle), and Cold (Right) dark matter Universes would look like at high redshift (early times). Bottom row: same as top row, except now as they would look at the present time ($z = 0$). Credits: ITC University of Zurich.

Even considering the current estimated range for the mass of the neutrino, no mass in that range would allow us to observe *now* structures that have already collapsed and fragmented.

The only possibility for that to happen is that DM must be *cold* (i.e. non-relativistic) at decoupling. Technically speaking we cannot fully exclude the possibility of other candidates for DM like particles yet to be discovered or primordial black holes.

5.4.2 WIMPS

In the last section we've listed quite a few properties we're requiring DM to have. Let's briefly recall them:

1. Negligible light emission;
2. Weakly interacting with radiation, itself, and matter in general;
3. Non-baryonic;
4. Cold (non-relativistic at decoupling);
5. Either stable or with $\tau_{1/2} \gg \tau_{\text{Universe}}$.

A possible class of candidates that meets all these requirements is that of the WIMPS[4] (*Weakly Interacting Massive Particles*). The expected mass for the WIMPS would be in the range $1\,\text{GeV} \lesssim m_{\text{WIMPS}} \lesssim$ TeV.

Thermal Decoupling: Freeze-Out

The interested reader might refer also to [6] and [15] for a more comprehensive and extensive discussion on the subject.

As a matter of fact, the WIMPS decoupling is the non-relativistic version of neutrinos' decoupling. We start by calculating the freeze-out temperature

$$\Gamma \sim H \to n\sigma v \sim \frac{T_{FO}^2}{M_{Pl}} \tag{5.59}$$

We can define $x_{FO} = m_\chi / T_{FO}$ (with χ we're referring to WIMPS) so that

$$(m_\chi T_{FO})^{3/2} \exp\left(-m_\chi / T_{FO}\right) \sigma v \sim \frac{T_{FO}^2}{M_{Pl}} \tag{5.60}$$

which we can promptly rewrite as

$$\boxed{x_{FO}^{1/2} \exp(-x_{FO}) \sim \frac{1}{m_\chi M_{Pl} \sigma v} := C} \tag{5.61}$$

Let's throw in some numbers by setting a benchmark Model: $m_\chi \sim 100\,\text{GeV}$ (electroweak scale), $M_{Pl} \sim 10^{19}\,\text{GeV}$, $\sigma \sim 10^{-6}\,\text{GeV}^{-2}$, and $v \sim \text{o}(1)$, which all reflect into

$$C = 10^{-15} \quad x_{FO} \approx 36 \tag{5.62}$$

4 Currently, WIMPS are beginning to pop out in scenarios of beyond-SM physics, such as in supersymmetric theories.

Spanning through the whole WIMPS admitted range of masses bounds $x_{FO} \in [20, 40]$. We can also use the equipartition theorem

$$\frac{3}{2}kT_{FO} \sim \frac{1}{2}mv^2 \rightarrow v \sim \frac{1}{3} \tag{5.63}$$

WIMPS Miracle

It just remains to relate the freeze-out abundance of dark matter relics to the dark matter density today

$$\Omega_{\chi,0} = \frac{\rho_{\chi,0}}{\rho_c} = \frac{m_\chi n_{\chi,0}}{\rho_c} = \frac{m_\chi n_{\chi,0} T_0^3}{\rho_c T_0^3} \tag{5.64}$$

Since $n \propto a^{-3}$ and $a \propto T^{-1}$, $n/T^3 \sim$ const. This we can use to rewrite the expression for $\Omega_{\chi,0}$ as

$$\Omega_{\chi,0} = \frac{m_\chi T_0^3}{\rho_c}\left(\frac{n_{\chi,0}}{T_0^3}\right) = \frac{T_0^3}{\rho_c} x_{FO}\left(\frac{n_{\chi,0}}{T_0^2}\right) \tag{5.65}$$

but we also have the decoupling condition

$$n_{FO}\sigma v \sim \frac{T_{FO}^2}{M_{Pl}} \tag{5.66}$$

Putting everything together yields

$$\Omega_{\chi,0} = \left(\frac{T_0^3}{\rho_c M_{Pl}}\right) x_{FO} \frac{1}{\langle \sigma v \rangle} \tag{5.67}$$

or we can put some numbers in

$$\boxed{\left(\frac{\Omega_{\chi,0}h^2}{0.2}\right) \approx \left(\frac{x_{FO}}{20}\right)\left(\frac{10^{-8}\,\text{GeV}^{-2}}{\langle \sigma v \rangle}\right)} \tag{5.68}$$

This reproduces the observed dark matter density if

$$\sqrt{\langle \sigma v \rangle} \sim 10^{-4}\,\text{GeV}^{-1} \sim 0.1\sqrt{G_F} \tag{5.69}$$

The fact that a thermal relic with a cross section characteristic of the weak interaction gives the right dark matter abundance is called the WIMP miracle.

Boltmann Equation

Technically speaking, the formal tool to describe the evolution beyond equilibrium is the Boltzmann equation. Assume we want to study the interaction process reported below with cross section σ

$$1 + 2 \leftrightharpoons 3 + 4 \tag{5.70}$$

each species with a given numeric density n_i. In the absence of interactions, the number density of a particle species i evolves as

$$\dot{n}_i + 3\frac{\dot{a}}{a}n_i = 0 \iff \frac{1}{a^3}\frac{\mathrm{d}}{\mathrm{d}t}(n_i a^3) = 0 \tag{5.71}$$

This is simply a reflection of the fact that the number of particles in a fixed physical volume is conserved, so that the density dilutes with the expanding volume. To include the effects of interactions we add a collision term to the right-hand side

$$\frac{1}{a^3}\frac{\mathrm{d}}{\mathrm{d}t}(n_i a^3) = C_i[\{n_j\}] \tag{5.72}$$

This is the Boltzmann equation. The form of the collision term depends on the specific interactions under consideration. Suppose we are interested in tracking the number density n_1 of species 1. Obviously, the rate of change in the abundance of species 1 is given by the difference between the rates for producing and eliminating the species. The Boltzmann equation simply formalises this statement

$$\frac{1}{a^3}\frac{\mathrm{d}}{\mathrm{d}t}(n_1 a^3) = -\alpha n_1 n_2 + \beta n_3 n_4 \tag{5.73}$$

The first term, $-\alpha n_1 n_2$, describes the destruction of particles 1, while that second term, $\beta n_3 n_4$, describes the interaction of particles 3-4. The parameter $\alpha = \langle \sigma v \rangle$ is the *thermally averaged cross section*. The second parameter β can be related to α by noting that the collision term has to vanish in (chemical) equilibrium

$$\beta = \left(\frac{n_1 n_2}{n_3 n_4}\right)_{\mathrm{eq}} \alpha \tag{5.74}$$

where the densities n_i are the ones at equilibrium. We therefore find

$$\boxed{\frac{1}{a^3}\frac{\mathrm{d}}{\mathrm{d}t}(n_1 a^3) = -\langle \sigma v \rangle \left[n_1 n_2 - \left(\frac{n_1 n_2}{n_3 n_4}\right)_{\mathrm{eq}} n_3 n_4 \right]} \tag{5.75}$$

It is instructive to write this in terms of the number of particles in a comoving volume $N_i = n_i a^3$. Since

$$\mathrm{d}(\ln a) = \frac{\mathrm{d}a}{a} \implies \frac{\mathrm{d}(\ln N_i)}{\mathrm{d}(\ln a)} = \frac{a\,\mathrm{d}(\ln(n_i a^3))}{\mathrm{d}a} = \frac{1}{H}\frac{1}{n_i a^3}\frac{\mathrm{d}}{\mathrm{d}t}(n_i a^3) \tag{5.76}$$

we can rewrite (5.75) as

$$\frac{\mathrm{d}(\ln N_i)}{\mathrm{d}(\ln a)} = -\frac{n_2 \langle \sigma v \rangle}{H}\left[1 - \left(\frac{n_1 n_2}{n_3 n_4}\right)_{\mathrm{eq}} \frac{n_3 n_4}{n_1 n_2}\right] = -\frac{\Gamma}{H}\left[1 - \left(\frac{n_1 n_2}{n_3 n_4}\right)_{\mathrm{eq}} \frac{n_3 n_4}{n_1 n_2}\right] \tag{5.77}$$

The right-hand side of last equation contains a factor describing the interaction efficiency, Γ/H, and a factor characterizing the deviation from equilibrium, the one within parentheses. We can define two regimes:

1. $\Gamma/H \gg 1$: The system is always driven towards its equilibrium configuration. As long as the interaction rates are large, the system quickly relaxes to a steady state where the right hand side vanishes and the particles assume their equilibrium abundances;

2. $\Gamma/H \ll 1$: The right hand side gets suppressed and the comoving density of particles approaches a constant relic density (*freeze-out*).

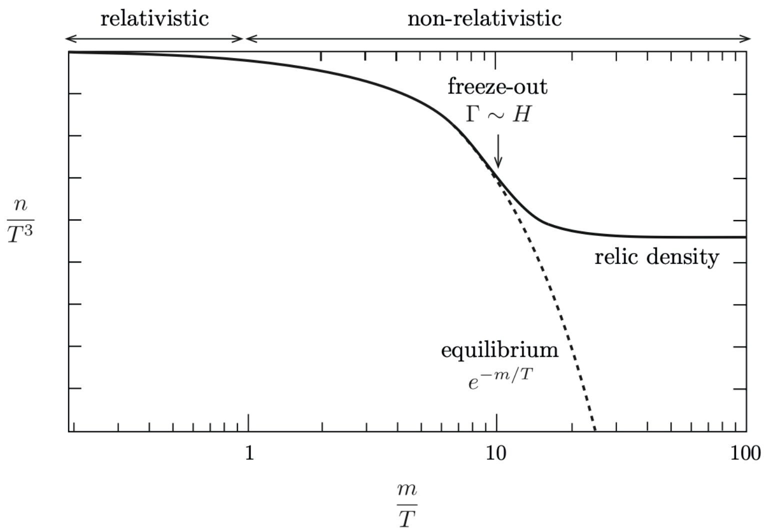

Figure 5.7: A schematic illustration of particle freeze-out. At high temperatures, $T \gg m$, the particle abundance tracks its equilibrium value. At low temperatures, $T \ll m$, the particles freeze out and maintain a density that is much larger than the Boltzmann-suppressed equilibrium abundance. Credits: [2].

Both scenarios are pictorially represented in Fig.5.7.

Another possibility is the "hot-decoupling," like that of neutrinos, that produces a hot relic. With reference to Fig.5.7, you might imagine it as though the curve never drops for a broad range of values for x.

In the integral, $t_{s.\,f.}$ stands for some time associated to structure formation.

For neutrinos we can define a free-streaming length

$$\lambda_{FS} = \int_{t_{\rm dec}}^{t_{\rm s.\,f.}} \frac{v(t)\,\mathrm{d}t}{a(t)} \sim 1\,\mathrm{Mpc}\left(\frac{1\,\mathrm{eV}}{m_\nu}\right) \tag{5.78}$$

which is what allows us to say that if DM were made of hot neutrinos, we wouldn't have structures below the Mpc-scale.

5.5 BIG BANG NUCLEOSYNTHESIS

Let us return to $T \sim 1\,\mathrm{MeV}$. Photons, electrons, and positrons are in equilibrium. Neutrinos are about to decouple. Baryons are non-relativistic and therefore much fewer in number than the relativistic species. Nevertheless, we now want to study what happened to these trace amounts of baryonic matter. The total number of nucleons stays constant due to baryon number conservation. This baryon number can be in the form of protons and neutrons or heavier nuclei. Weak nuclear reactions may convert neutrons and protons into each other and strong nuclear reactions may build nuclei from them.

The purpose of this section will be reaching the understanding of a single number

$$Y_p = \frac{4n_{\rm He}}{n_{\rm H}} \approx 0.25 \tag{5.79}$$

namely, the abundance of primordial Helium. The steps we're going to take to reach our final destination are pictorially summarized in Fig.5.8.

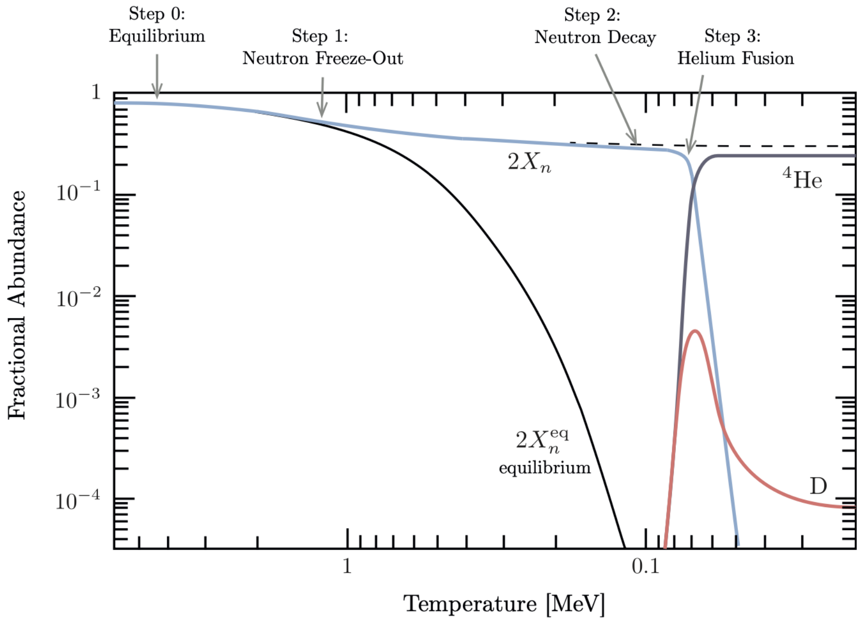

Figure 5.8: Numerical results for Helium production in the early Universe: Fractional abundance of neutrons as a function of temperature. Credits: [2].

So, we're starting from the MeV scale with free protons and free neutrons, which are in equilibrium through the following reactions

$$n + \nu_e \leftrightharpoons p^+ + e^-$$
$$n + e^+ \leftrightharpoons p^+ + \bar{\nu}_e$$

with a Q-value of order $Q \approx 1.3\,\text{MeV}$. As temperature changes, assuming the reactions don't drop out of equilibrium, the reactions will be leaning towards the production of protons

$$\left(\frac{n_n}{n_p}\right)_{\text{eq}} = \left(\frac{m_n}{m_p}\right)^{3/2} \exp\left(-\frac{m_n - m_p}{(k)T}\right) \tag{5.80}$$

We can define the neutronization ratio

$$X_n = \frac{n_n}{n_n + n_p} \tag{5.81}$$

so that, at equilibrium, the fractional abundance of neutrons is given by

$$X_n^{\text{eq}}(T) = \frac{\exp(-Q/T)}{1 + \exp(-Q/T)} \tag{5.82}$$

which is vanishing for either $T \to 0$ or $t \to \infty$. Neutrons would disappear on a timescale of about $\sim 1\,\text{s}$, when the Universe has cooled to about $\sim 1.3\,\text{MeV}$. This is somewhat...unhandy, considering that in a second there's not much time to do any nucleosynthesis.

What's the solution? Easy: The reactions quickly drop out of equilibrium, and neutrons freeze-out. This means that as soon as the Universe drops below the MeV scale, neutron freeze-out allows them to keep existing with a significant abundance.

To go any further than this, we have to recall Boltmann's equation (5.75). If we consider a reaction like

$$n + \text{lept}_1 \leftrightharpoons p + \text{lept}_2 \tag{5.83}$$

then the associated Boltmann equation is

$$\frac{1}{a^3}\frac{\mathrm{d}}{\mathrm{d}t}(n_n a^3) = -\Gamma\left(n_n - \left(\frac{n_n}{n_p}\right)_{\text{eq}} n_p\right) \qquad \Gamma = n_e^{\text{eq}}\langle\sigma v\rangle \tag{5.84}$$

Since $n_b = n_p + n_n$, the neutronization factor (5.81) is

$$X_n = \frac{n_n}{n_n + n_p} = \frac{X_n n_b}{X_n n_b + (1 - X_n) n_b} \implies n_n = X_n n_b \quad n_p = (1 - X_n) n_b \tag{5.85}$$

Note that we've used the expression for X_n to replace n_n and n_p with expressions dependent on n_b. Then, we exploited the fact that the product $n_b a^3$ is independent of time because of the scaling (4.91) for baryons.

which allows us to rewrite Boltmann's equation

$$\frac{1}{a^3}\frac{\mathrm{d}}{\mathrm{d}t}(X_n n_b a^3) = -\Gamma\left[X_n n_b - (1 - X_n) n_b \exp\left(-Q/T\right)\right] \tag{5.86}$$

hence

$$\frac{\mathrm{d}X_n}{\mathrm{d}t} = -\Gamma\left[X_n - (1 - X_n)\exp\left(-Q/T\right)\right] \tag{5.87}$$

We can define a new variable $x = Q/T$, so that

$$\boxed{\frac{\mathrm{d}X_n}{\mathrm{d}x} = \frac{\Gamma(x)}{xH(x)}\left(e^{-x} - X_n(1 - e^{-x})\right)} \tag{5.88}$$

where we have used

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -\frac{Q}{T^2}\frac{\mathrm{d}T}{\mathrm{d}t} = xH \tag{5.89}$$

since $T \propto a^{-1}$. During BBN we have

$$H = \sqrt{\frac{\rho}{3M_{Pl}^2}} = \frac{\pi}{3}\sqrt{\frac{g_\star}{10}}\frac{Q^2}{M_{Pl}}\frac{1}{x^2} := \frac{H_1}{x^2} \tag{5.90}$$

Solving (5.88) numerically, we find

$$X_n^\infty = X_n(\infty) = 0.15 \tag{5.91}$$

Finally, we need an expression for the neutron-proton conversion rate Γ, which we'll here pretend it has been revealed from God, your favorite deity, or [7]

$$\Gamma = \frac{255}{\tau_n}\frac{12 + 6x + x^2}{x^5}\,\mathrm{s}^{-1} \tag{5.92}$$

where $\tau_n = 878.4 \pm 0.5\,\mathrm{s}$ [14]. It is interesting to point out that the conversion time Γ^{-1} is comparable to the age of the Universe at the considered energy-scale, while at later times $T \propto t^{-1/2}$, and $\Gamma \propto T^3$, and thus the neutron-proton conversion time becomes longer than the age of the Universe. Hence the freeze-out.

For temperatures below 0.2 MeV the finite lifetime of the neutron becomes important. To include neutron decay in our computation we simply multiply the freeze-out abundance by an exponential decay factor

$$X_n(t) = X_n^\infty e^{-t/\tau_n} \tag{5.93}$$

which motivates the abrupt drop in the low temperature regions of Fig.5.8.

Thanks to the freeze-out, we've managed to extend a neutron's lifespan from less than a second to a little shy of 1000 s. However, we've yet not finished. In fact, if things were exactly as we've described so far, and nothing else occurred, all neutrons would vanish after some integer multiple of τ_n, which would kinda suck for the purpose of forming heavier elements.

How can we then prevent the natural and inevitable demise of free neutrons? You bound them into heavier nuclei

Note that during the BBN there's no formation of elements with $A = 5$ or $A = 8$ because of α-instability. You'll have to wait until the 1st generation of stars ignites Helium through the 3α reaction to breach through this gate and produce heavier elements.

$$
\begin{aligned}
{}^1_0\mathrm{n} + {}^1_1\mathrm{p} &\to {}^2_1\mathrm{D} + \gamma \\
{}^2_1\mathrm{D} + {}^1_1\mathrm{p} &\to {}^3_2\mathrm{He} + \gamma \\
{}^2_1\mathrm{D} + {}^2_1\mathrm{D} &\to {}^3_2\mathrm{He} + {}^1_1\mathrm{p} \\
&\to {}^3_2\mathrm{He} + {}^1_0\mathrm{n} \\
{}^3_2\mathrm{He} + {}^2_1\mathrm{D} &\to {}^4_2\mathrm{He} + {}^1_1\mathrm{p} \\
{}^3_2\mathrm{He} + {}^4_2\mathrm{He} &\to {}^7_4\mathrm{Be} + \gamma \\
{}^3_2\mathrm{He} + {}^4_2\mathrm{He} &\to {}^7_3\mathrm{Li} + \gamma
\end{aligned}
$$

but it's essentially the very first reaction that determines the timescale of this whole machination we have just penned. Incidentally, this is also a quite significant problem. Since Deuterium is formed directly from neutrons and protons, it can follow its equilibrium abundance as long as enough free neutrons are available. However, since Deuterium's binding energy is rather small, the Deuterium abundance becomes large rather late (at $T < 100\,\mathrm{keV}$) because it is very prone to being photodisintegrated.

So, although heavier nuclei have larger binding energies and hence would have larger equilibrium abundances, they cannot be formed until sufficient Deuterium has become available. This is the *Deuterium bottleneck*. Only when there is enough Deuterium, can Helium be produced.

Consider now the chemical balance equation at equilibrium

$$
\left(\frac{n_{\mathrm{D}}}{n_n n_p}\right)_{\mathrm{eq}} = \frac{3}{4}\left(\frac{m_{\mathrm{D}}}{m_p m_n}\frac{2\pi}{T}\right)^{3/2} \exp(B_{\mathrm{D}}/T) \tag{5.94}
$$

where B_{D} is Deuterium's binding energy $B_{\mathrm{D}} = m_n + m_p - m_{\mathrm{D}} = 2.22\,\mathrm{MeV}$.

We then use $n_n = X_n n_b$ which, thanks to the baryon to photon ratio, is also equal to

$$
n_n = X_n n_b = X_n \eta n_\gamma = X_n \eta \frac{2\zeta(3)}{\pi^2} T_\gamma^3 \tag{5.95}
$$

Hence, approximating $m_{\mathrm{D}} \approx 2m_p \approx 2m_n$, we conclude

$$
\left(\frac{n_{\mathrm{D}}}{n_p}\right)_{\mathrm{eq}} = \frac{3}{4} X_n^{\mathrm{eq}} \eta \frac{2\zeta(3)}{\pi^2} T_\gamma^3 \left(\frac{4\pi}{m_p T}\right)^{3/2} \exp(B_{\mathrm{D}}/T) \tag{5.96}
$$

The Deuterium bottleneck is overcome when $(n_{\mathrm{D}}/n_p)_{\mathrm{eq}} \sim 1$, which means that we have to wait until $T \sim 70\,\mathrm{keV}$, so that

$$
\eta \exp(B_{\mathrm{D}}/T) \sim 1 \tag{5.97}
$$

If we turn the temperature into a time, we find that the Deuterium bottleneck is breached past at $t_{\mathrm{nuc}} \approx 250\,\mathrm{s}$ which is less than the neutron half-life time!

At this temperature, the neutronization rate is

$$\frac{n_n}{n_b} = \frac{n_n}{n_p + n_n} = X_n \approx 0.11 \tag{5.98}$$

Since the binding energy of Helium is larger than that of Deuterium, the Boltzmann factor $\exp(B/T)$ favours Helium over Deuterium. Indeed, in Fig.5.8 we see that Helium is produced almost immediately after Deuterium. Virtually all remaining neutrons at $t \sim t_{\text{nuc}} \approx 250\,\text{s}$ are processed into ^{4}He. Since two neutrons go into one nucleus of ^{4}He, the final ^{4}He abundance is equal to half of the neutron abundance at t_{nuc}, so $n_{\text{He}} = n_n/2$. The fractional abundance of Helium will thus be

$$Y_p = \frac{4n_{\text{He}}}{n_{\text{H}}} = \frac{2n_n}{n_{\text{H}}} = \frac{2X_n(t_{\text{nuc}})}{1 - X_n(t_{\text{nuc}})} \approx 0.25 \tag{5.99}$$

This is of course a very simplified version of the much more complex calculation. From the primordial Helium abundance we can, in principle, infer a value for the baryon-to-photon ratio

$$Y_p = 0.2262 + 0.0135 \ln\left(\frac{\eta}{10^{-10}}\right) \tag{5.100}$$

Light Elements Synthesis

To determine the abundances of other light elements, the coupled Boltzmann equations have to be solved numerically, and actually theoretical predictions for the light element abundances as a function of η (or Ω_b) are found to match reasonably well with observations.

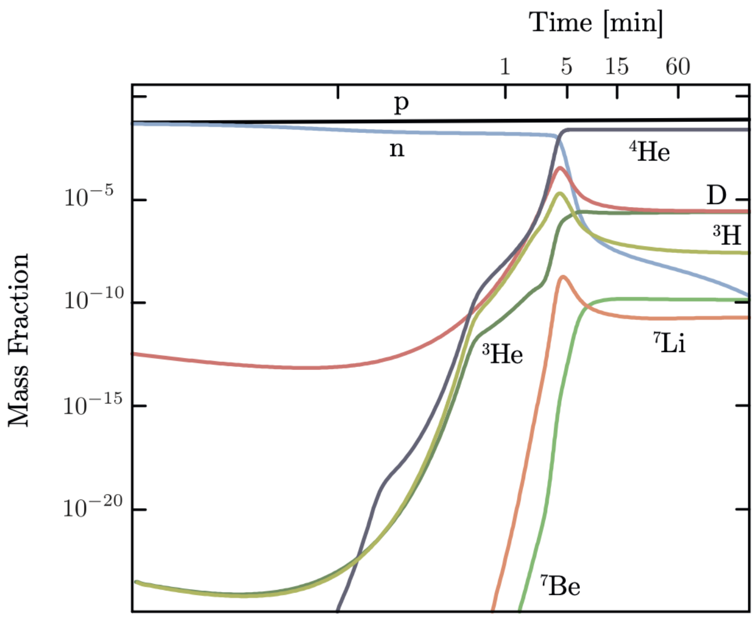

Figure 5.9: Numerical results for the evolution of light element abundances. Credits: [2].

The only element that has created a little trouble (and still does) is Lithium, which presents an anomalous behavior.

5.6 COSMIC MICROWAVE BACKGROUND

When we refer to the CMB (Cosmic Microwave Background), we're referring to a diffused cosmic background of photons, that is what remains of the photon fluid after the time of decoupling.

The CMB has quite a few interesting properties:

- *Almost perfect blackbody behavior (check Appendix B if you're not familiar);*
- *A temperature of $T \approx 2.72$ K and minimal fluctuations;*
- *$E \sim 10^{-4}$ eV (160 GHz).*

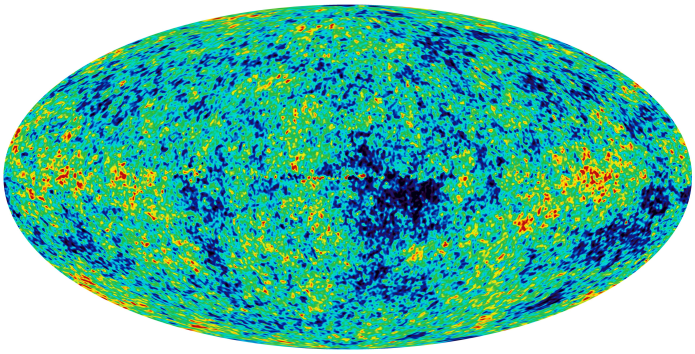

Figure 5.10: Behold! The Cosmic Microwave Background.

All the useful information about the CMB is hidden and preserved into its microscopic fluctuations

$$\frac{\Delta T}{T} \sim 10^{-5} \tag{5.101}$$

so all we have to do is to construct a formalism able to extract such information from the temperature contrast

$$\Theta(\hat{n}) = \frac{T(\hat{n}) - \langle T(\hat{n}) \rangle}{\langle T(\hat{n}) \rangle} \tag{5.102}$$

from which we can define the angular power spectrum and 2-point correlation function as such

$$C(\theta) = \langle \Theta(\hat{n}) \Theta(\hat{n}') \rangle \quad \hat{n} \cdot \hat{n}' = \cos\theta \tag{5.103}$$

At this point it's particularly useful to expand the temperature contrast with spherical harmonics, which means assigning a weight to angular patches that are able to capture finer and finer details. We'll use

$$\Theta(\hat{n}) = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} a_{lm} Y_{lm}(\theta, \phi) \tag{5.104}$$

where

$$Y_{lm} = (-1)^m \sqrt{\frac{(2l+1)(l-m)!}{4\pi(l+m)!}} P_l^m(\cos\theta) e^{im\phi} \tag{5.105}$$

$$P_l^m(x) = (1-x^2)^{m/2} \frac{\mathrm{d}^m P_l}{\mathrm{d}x^m} \tag{5.106}$$

where P_l are Legendre's polynomials. If the function to be expanded is real, then

$$Y_{lm}^* = (-1)^m Y_{l\,-m} \quad a_{lm}^* = (-1)^m a_{l\,-m} \tag{5.107}$$

where the former relation holds also for a complex-valued function. With these conventions, the 2-point correlation function becomes

$$C(\theta) = \sum_{l,m} \sum_{l',m'} \langle a_{lm} a_{l'm'} \rangle Y_{lm} Y_{l'm'} = \sum_{l,m} \sum_{l',m'} \langle a_{lm} a_{l'm'}^* \rangle Y_{lm} Y_{l'm'}^* \tag{5.108}$$

If we assume isotropy (hence invariance under angular traslations), patterns associated to different l are uncorrelated, hence

$$\langle a_{lm} a^*_{l'm'} \rangle = C_l \delta_{ll'} \delta_{mm'} \tag{5.109}$$

so

$$C(\theta) = \sum_l C_l \sum_{m=-l}^{+l} Y_{lm} Y^*_{lm} \tag{5.110}$$

Let's now use the property

$$P_l(\cos\theta) = \frac{4\pi}{2l+1} \sum_m Y_{lm} Y^*_{lm} \tag{5.111}$$

which allows us to conclude

Usually, you also end up exploiting Wiener-Chinčin theorem to define a power spectrum as the FT of the correlation function.

$$\boxed{C(\theta) = \sum_l \frac{(2l+1)}{4\pi} P_l(\cos\theta) C_l} \tag{5.112}$$

Here C_l is the factor that quantifies how much correlated is the CMB on the angular scale l. However, what you will usually find in plots is actually

$$\Delta_T^2 = \frac{l(l+1)}{2\pi} C_l \tag{5.113}$$

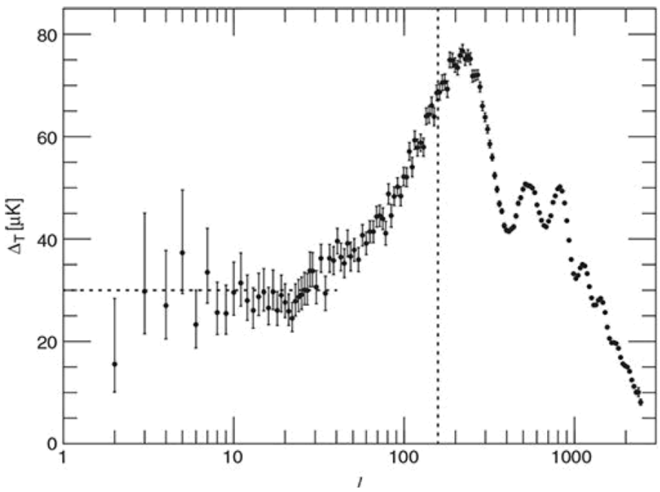

Figure 5.11: Temperature fluctuations Δ_T of the CMB as observed by the Planck collaboration, expressed as a function of the multipole l. The vertical dotted line shows l_{Hor}, the multipole corresponding to the horizon size at the epoch of last scattering.

The higher spread of the measurement on the larger scales (smaller l) is caused by the presence of only one CMB and progressively less patches to average on, reflecting in a higher variance (*cosmic variance*).

Our purpose now would be that to study the small perturbations of the cosmological fluid, to which is associated an initial power spectrum coming from the residuals of the inflationary period

Here $\mathcal{R}$ refers to the spatial curvature. The spectrum $P_{\mathcal{R}}(k)$ is sometimes known as Harrison-Zeldovič spectrum.

$$\Delta^2_{\mathcal{R}}(k) = \frac{k^3 P_{\mathcal{R}}(k)}{2\pi^2} \propto k^{n_s - 1} \tag{5.114}$$

from which a temporal evolution will be brought about. After the last scattering epoch, fluctuations will be free-streaming, even though they will be weakly interacting with structures on the larger scales. But regardless, the idea is to write

$$C_l = \int \mathrm{d}\ln(k)\Theta_l^2(k)\Delta_{\mathcal{R}}^2(k) \tag{5.115}$$

where $\Theta_l^2(k)$ is a transfer function containing the temporal details and projections on the power spectrum.

5.6.1 Perturbation Theory

Let's start from the conformal FLRW metric (4.38)

$$\mathrm{d}s^2 = a^2(\eta)\left[-\mathrm{d}\eta^2 + \delta_{ij}\,\mathrm{d}x^i\,\mathrm{d}x^j\right] \tag{5.116}$$

The more general way to add a perturbation to the metric is the following

$$\begin{aligned} \mathrm{d}s^2 = a^2(\eta)\Big[&-(1+2A(x^\mu))\,\mathrm{d}\eta^2 + 2B_i(x^\mu)\,\mathrm{d}\eta\,\mathrm{d}x^i + \\ &+ C\delta_{ij}\,\mathrm{d}x^i\,\mathrm{d}x^j + 2E_{ij}(x^\mu)\,\mathrm{d}x^i\,\mathrm{d}x^j\Big] \end{aligned}$$

It will be extremely useful to perform a scalar-vector-tensor (SVT) decomposition of the perturbations. For 3-vectors, this should be familiar. It simply means that we can split any 3-vector into the gradient of a scalar and a divergenceless vector

$$B_i = \partial_i B + \hat{B}_i \quad \partial^i \hat{B}_i = 0 \tag{5.117}$$

Similarly, any rank-2 symmetric tensor can be written as

$$E_{ij} = C\delta_{ij} + \partial_{\langle i}\partial_{j\rangle}E + \partial_{(i}\hat{E}_{j)} + 2\hat{E}_{ij} \tag{5.118}$$

where

$$\begin{aligned} \partial_{\langle i}\partial_{j\rangle}E &= \left(\partial_i\partial_j - \frac{\delta_{ij}}{3}\nabla^2\right)E \\ \partial_{(i}\hat{E}_{j)} &= \frac{1}{2}\left(\partial_i\hat{E}_j + \partial_j\hat{E}_i\right) \end{aligned}$$

As before, the hatted quantities are divergenceless, and the tensor perturbation is traceless, $E^i{}_i = 0$. The 10 degrees of freedom of the metric have thus been decomposed into 4 + 4 + 2 SVT degrees of freedom.

What makes the SVT-decomposition so powerful is the fact that the Einstein equations for scalars, vectors, and tensors don't mix at linear order and can therefore be treated separately. In the following, we'll only be interested in scalar fluctuations and the associated density perturbations (under the assumption of adiabatic processes). Vector perturbations aren't produced by inflation, but even if they were, they would decay quickly with the expansion of the Universe. Tensor perturbations are an important prediction of inflation but we won't deal with them[5].

5 We would if I had followed the 9 credits course rather than the 6 one. But since I didn't, there's no need to go that far.

Before we continue, we should, in principle, address an important subtlety. The metric perturbations we've introduced aren't uniquely defined, but are gauge dependent. It is possible to show (we won't) that one way to avoid the gauge problems is to define special combinations of metric perturbations that do not transform under a change of coordinates. Meet the *Bardeen variables* (or potentials)

You should recall from Chapter 4 that $\mathcal{H} = a'/a$ is the Hubble parameter in conformal time, and with the superscript we're referring to derivatives with respect to conformal time.

$$\psi(x^\mu) = A + \mathcal{H}(B - E') + (B - E')' \tag{5.119}$$

$$\phi(x^\mu) = -C + \frac{1}{3}\nabla^2 E - \mathcal{H}(B - E') \tag{5.120}$$

Technically speaking, also

$$\hat{\Phi}_i = \hat{E}_i - \hat{B}_i \quad \text{and} \quad \hat{E}_{ij} \tag{5.121}$$

are gauge invariant quantities. With significant effort, you might also find the three-dimensional Ricci scalar associated with the spatial part of the perturbed metric

$$a^2 R_{(3)} = -4\nabla^2\left(C - \frac{1}{3}\nabla^2 E\right) \tag{5.122}$$

where the quantity within parentheses is usually called *curvature perturbation*. On the other hand, the *comoving curvature perturbation* $\mathcal{R}$ is the curvature perturbation evaluated in the comoving gauge ($B_i = 0$). It will prove convenient to have a gauge-invariant expression for $\mathcal{R}$, so that we can evaluate it from the perturbations in any gauge (for example, in Newtonian gauge). Since B and v^6 vanish in the comoving gauge, we can always add linear combinations of these to curvature perturbation to form a gauge-invariant combination that equals $\mathcal{R}$. The correct gauge-invariant expression for the comoving curvature perturbation is

$$\mathcal{R} = C - \frac{1}{3}\nabla^2 E + \mathcal{H}(B + v) \tag{5.123}$$

At any rate, these gauge-invariant variables can be considered as the "real" spacetime perturbations since they cannot be removed by a gauge transformation.

We'll work in Newton's gauge $B = E = 0$, which implies $\psi = A$ and $\phi = -C$, thus producing the metric

$$\mathrm{d}s^2 = a^2(\eta)\left[-(1 + 2\psi)\,\mathrm{d}\eta^2 + (1 - 2\phi)\delta_{ij}\,\mathrm{d}x^i\,\mathrm{d}x^j\right] \tag{5.124}$$

For perturbations that decay at spatial infinity, the Newtonian gauge is unique (i.e. the gauge is fixed completely). In this gauge, the physics appears rather simple since the hypersurfaces of constant time are orthogonal to the worldlines of observers at rest in the coordinates (since $B = 0$) and the induced geometry of the constant-time hypersurfaces is isotropic (since $E = 0$).

In the absence of anisotropic stress, $\psi = \phi$. Note the similarity of the metric to the usual weak-field limit of GR about Minkowski space; we shall see that ψ plays the role of the gravitational potential.

Let's see how a photon moves in the perturbed metric, that is, let's check how its geodesics respond to the perturbation

$$\frac{\mathrm{d}p^\mu}{\mathrm{d}\lambda} = -\Gamma_{\alpha\ \beta}^{\ \mu} p^\alpha p^\beta \tag{5.125}$$

which gives

$$\frac{1}{E_\gamma}\frac{\mathrm{d}E_\gamma}{\mathrm{d}\eta} = -\mathcal{H} + \phi' - \hat{p}_i \partial_i \psi \tag{5.126}$$

Recall that the energy of a photon can be expressed through $E_\gamma = -u_\mu p^\mu$, where u_μ is completely determined by the request $u_\mu u^\mu = -1$. It directly follows that $E_\gamma \approx a(1+\psi)p^0$.

which is essentially a composition of the standard cosmological redshift (the Hubble term), a small perturbation on the redshift (ϕ'), and the gravitational redshift. Ultimately we find

$$\boxed{\frac{\mathrm{d}\ln(aE_\gamma)}{\mathrm{d}\eta} = -\frac{\mathrm{d}\psi}{\mathrm{d}\eta} + \phi' + \psi'} \tag{5.127}$$

Note that

$$\frac{d\psi}{d\eta} = \frac{\partial\psi}{\partial\eta} + \frac{\partial x^i}{\partial\eta}\partial_i\psi$$

where the derivative of x^i is essentially the unit vector tangent to the momentum.

Consider

$$\int_{\eta_{LS}}^{\eta_0} \frac{\mathrm{d}\ln(aE_\gamma)}{\mathrm{d}\eta}\,\mathrm{d}\eta = -\int_{\eta_{LS}}^{\eta_0}\left(\frac{\mathrm{d}\psi}{\mathrm{d}\eta} - \phi' - \psi'\right)\mathrm{d}\eta \tag{5.128}$$

which is immediately integrated

$$\ln\left(\frac{(aE_\gamma)_0}{(aE_\gamma)_{LS}}\right) = -(\psi_0 - \psi_{LS}) + \int_{LS}^{\eta_0} (\psi' + \phi')\,\mathrm{d}\eta \tag{5.129}$$

This equation essentially describes how the energy of a photon has changed along a geodesic from the last scattering up until today. We arbitrarily set the potential $\psi_0 = 0$, while ψ_{LS} represents the overdensities of the oscillating fluid at last scattering.

On the other hand, the integral term describes the behavior of fluctuations in the cosmological fluid based on how they're interacting with the photons. This is called *Integrated Sachs-Wolfe effect* [19]: It regulates how photons are acquiring/losing energy while passing through spatially and temporally changing gravitational potentials, thus introducing a secondary anisotropy in the CMB spectrum. Since

$$aE \propto aT \propto a(\bar{T} + \delta T) \propto a\bar{T}\left(1 + \frac{\delta T}{\bar{T}}\right) \tag{5.130}$$

and

$$\ln\left(1 + \frac{\delta T}{\bar{T}}\right) \sim \frac{\delta T}{\bar{T}} \tag{5.131}$$

we thus obtain an expression for the temperature constant in the CMB at present time

$$\left(\frac{\delta T}{T}\right)_0 (\hat{n}) = \left(\frac{\delta T}{T}\right)_{LS} (\hat{n}) + \psi_{LS} + \int_{LS}^{\eta_0} (\psi' + \phi')\,\mathrm{d}\eta - \hat{n}\cdot\vec{v} \tag{5.132}$$

where the last term $\hat{n}\cdot\vec{v}$ takes into account the small oscillations in the baryon-photon fluid. Alternatively, we can further expand the temperature contrast at LS

$$\boxed{\left(\frac{\delta T}{T}\right)_0 = \left[\frac{1}{4}\left(\frac{\delta\rho}{\rho}\right)_{LS} + \psi_{LS}\right] + \int_{LS}^{\eta_0} (\psi' + \phi')\,\mathrm{d}\eta - \hat{n}\cdot\vec{v}} \tag{5.133}$$

6 The scalar v is defined through the 3-dimensional velocity $v_i = \partial_i v$.

The term within square parentheses is the *Sachs-Wolfe effect*, caused by gravitational redshift occurring at the surface of last scattering[7]. The effect is not constant across the sky due to differences in the matter/energy density at the time of last scattering.

Sachs-Wolfe Effect

Note that necessarily

$$\left(\frac{\delta\rho}{\rho}\right)_{LS} \cdot \psi_{LS} < 0$$

and you can also prove that cold patches in the CMB are associated to the overdensities, producing the following relation

$$\frac{1}{4}\left(\frac{\delta\rho}{\rho}\right)_{LS} + \psi_{LS} = \frac{\psi_{LS}}{3}$$

We can switch to the FT[8] of $\Theta_0 = \delta T/T$

$$\left(\frac{\delta T}{T}\right)_{\eta=\eta_0} = \int \exp\left(i\vec{k}\cdot\hat{n}\chi_{LS}\right)\left[F_{LS}(k) - i(\hat{k}\cdot\hat{n})G_{LS}(k)\right] R_i(k)\,\frac{\mathrm{d}^3k}{(2\pi)^3} \tag{5.134}$$

Note that we're neglecting the integrated Sachs-Wolfe effect.

where we've defined

$$F_{LS} = \frac{\left[\frac{1}{4}\widehat{\left(\frac{\delta\rho}{\rho}\right)}(\eta_{LS},\vec{k}) + \hat{\psi}_{LS}(\eta_{LS},\vec{k})\right]}{R_i(k)} \qquad G_{LS} = \frac{\hat{v}_b}{R_i(k)} \tag{5.135}$$

which are (almost) the transfer functions we were talking about at the beginning of our discussion. The quantity $R_i(k)$ contains the initial conditions in terms of the scalar curvature, while χ_{LS} is the comoving distance of the last scattering horizon.

From this we want to compute the 2-point correlation function and project it on the Legendre polynomials to obtain the power spectrum C_l. It's useful to exploit one of the properties of the spherical Bessel functions j_l

$$\exp\left(i\hat{k}\cdot\hat{n}k\chi_{LS}\right) = \sum_l i^l(2l+1)j_l(k\chi_{LS})P_l(\hat{k}\cdot\hat{n}) \tag{5.136}$$

then we use

$$\frac{\partial\left[\exp\left(i\hat{k}\cdot\hat{n}k\chi_{LS}\right)\right]}{\partial(k\chi_{LS})} = i(\hat{k}\cdot\hat{n})\exp\left(i\hat{k}\cdot\hat{n}k\chi_{LS}\right) \tag{5.137}$$

hence

$$-k\frac{\partial\left[\exp\left(i\hat{k}\cdot\hat{n}k\chi_{LS}\right)\right]}{\partial(k\chi_{LS})} = -ik(\hat{k}\cdot\hat{n})\exp\left(i\hat{k}\cdot\hat{n}k\chi_{LS}\right) \tag{5.138}$$

Applying the derivative to the spherical-Bessel expansion gives

$$-i(\hat{k}\cdot\hat{n})\exp\left(i\hat{k}\cdot\hat{n}k\chi_{LS}\right) = -\sum_l i^l(2l+1)j_l'(k\chi_{LS})P_l(\vec{k}\cdot\hat{n}) \tag{5.139}$$

7 Which is also the main difference from the ISW, which captures the effects stemming from photons falling into *all* potentials from all structures from LS and beyond.
8 For an alternative derivation, the interested reader can check either this website, or [4].

so that putting it in the FT of Θ_0 yields

$$\Theta(\hat{n}) = \sum_l i^l(2l+1) \int \Theta_l(k) R_i P_l(\hat{k}\cdot\hat{n}) \frac{\mathrm{d}^3k}{(2\pi)^3} \tag{5.140}$$

where

$$\Theta_l(k) = F_{LS}(k) j_l(k\chi_{LS}) - G_{LS}(k) j_l'(k\chi_{LS}) \tag{5.141}$$

Recall the definition of $C(\hat{n}\cdot\hat{n}')$

$$C(\hat{n}\cdot\hat{n}') = \langle\Theta(\hat{n})\Theta(\hat{n}')\rangle \tag{5.142}$$

where the average is over the ensemble. Now substitute the value for $\Theta(\hat{n})$ we've just found

$$C(\hat{n}\cdot\hat{n}') = \sum_{ll'} i^l i^{l'}(2l+1)(2l'+1) \int\int \frac{\mathrm{d}^3k}{(2\pi)^3}\frac{\mathrm{d}^3k'}{(2\pi)^3}\mathcal{F} \tag{5.143}$$

where the function $\mathcal{F}$ is defined as[9]

$$\mathcal{F} = \langle R_i(\vec{k}) R_i(\vec{k}')\rangle \left[\Theta_l(k)\Theta_l(k')\right] \cdot \left[P_l(\hat{k}\cdot\hat{n}) P_{l'}(\hat{k}'\cdot\hat{n}')\right] \tag{5.144}$$

Now we exploit a useful property/definition

$$\langle R_i(\vec{k}) R_i(\vec{k}')\rangle = (2\pi)^3\delta^{(3)}(\vec{k}+\vec{k}')P_{\mathcal{R}}(k) \tag{5.145}$$

which allows us to considerably simplify the integral above, especially once we recall that for real-valued functions $\Theta_{l'}(-k) = \Theta^*_{l'}(k)$ and that Legendre polynomials have the following properties

$$\int P_l(\hat{k}\cdot\hat{n}) P_{l'}(\hat{k}\cdot\hat{n}')\,\mathrm{d}\Omega_k = \frac{4\pi}{2l+1} P_l(\hat{n}\cdot\hat{n}')\delta_{ll'} \tag{5.146}$$

$$P_{l'}(-\hat{k}\cdot\hat{n}') = (-1)^{l'} P_l(\hat{k}\cdot\hat{n}') \tag{5.147}$$

which allows us to write

$$C(\hat{n}\cdot\hat{n}') = \sum_l (2l+1)\frac{4\pi}{(2\pi)^3} P_l(\hat{n}\cdot\hat{n}') \int |\Theta_l(k)|^2 P_{\mathcal{R}} \cdot k^2\,\mathrm{d}k \tag{5.148}$$

We can now introduce the post-inflation dimensionless power spectrum

$$\Delta^2_{\mathcal{R}} = \frac{k^3}{2\pi^2} P_{\mathcal{R}}(k) \tag{5.149}$$

which ultimately yields for the 2-point correlation function

$$C(\hat{n}\cdot\hat{n}') = \sum_l \frac{2l+1}{4\pi}\left\{4\pi\int \mathrm{d}\ln(k)\Theta_l^2(k)\Delta^2_{\mathcal{R}}(k)\right\} P_l(\hat{n}\cdot\hat{n}') \tag{5.150}$$

where we have added the identity $4\pi/4\pi$ so that it's easier to recognize the power spectrum

$$\boxed{C_l = 4\pi\int \mathrm{d}\ln(k)\Theta_l^2(k)\Delta^2_{\mathcal{R}}(k)} \tag{5.151}$$

9 Please note that nobody ever defines this function. I only defined it because the whole expression for $C(\hat{n}\cdot\hat{n}')$ couldn't be written in a single line.

Before moving over, let's consider the transmission probability for photons

$$P_t = \exp\left(-\tau(z_0, z_1)\right) \tag{5.152}$$

where the optical depth τ is defined as

$$\tau = \int \Gamma \, \mathrm{d}t = \int \sigma_T n_e \, \mathrm{d}t = \int \frac{\mathrm{d}z}{H(z)(1+z)} \sigma_T n_b X_e(z) \tag{5.153}$$

Sometimes we might also define a *visibility function* $g(z) = \mathrm{d}\tau/\mathrm{d}z$, but at the end of the day what both are telling us is that there exists a residual probability of scattering also at small redshift.

Note that there is also a dipole anisotropy ($l = 1$) of significant amplitude $\delta T/T \sim 10^{-3}$ which is usually interpreted as a Doppler effect caused by the relative motion of the observer (us) with respect to the frame in which the CMB is isotropic[10]. Such velocity turns out to be of order $v \sim 396\,\mathrm{km\,s^{-1}}$, and the value is corrected annually.

Well now, recall the definition of $\Delta^2_{\mathcal{R}}$

$$\Delta^2_{\mathcal{R}}(k) = \frac{k^3 P_{\mathcal{R}}(k)}{2\pi^2} \propto k^{n_s - 1} \tag{5.154}$$

with $n_s \approx 0.965$ for a Harrison-Zeldovič spectrum, which means that the initial (adimensional) spectrum is (almost) scale invariant. If you recall, it enters in the expression for the FT of the temperature contrast through R_i

For the sake of simplicity, we're working in the approximation of ***instantaneous decoupling*** *$g(z) = \delta(z - z_{LS})$. A much more elegant discussion would consider how the solutions to the Boltmann equation are distributed in l.*

$$\int \exp\left(i\vec{k}\cdot\hat{n}\chi_{LS}\right)\left[F_{LS}(k) - i(\hat{k}\cdot\hat{n})G_{LS}(k)\right] R_i(k) \frac{\mathrm{d}^3k}{(2\pi)^3} \tag{5.155}$$

when calculating the correlation function and power spectrum (5.151). If we approximate the Bessel function to a δ-function

$$j_l(k\chi_{LS}) \approx \delta(l - k\chi_{LS}) \tag{5.156}$$

the power spectrum is significantly simplified

$$C_l^{SW} \propto F^2_{LS}(k)\Delta^2_{\mathcal{R}}(k)|_{k=l/\chi_{LS}} \tag{5.157}$$

5.6.2 Photon-Baryon Fluid Oscillations

At high values of l, so smaller angular scales, we observe an effect called silk damping, which is caused by diffusive effects of photons from hot to cold regions

$$r^2_{diff} = \int_0^{\eta_{LS}} l_\gamma \, d\eta$$

where l_γ is the mean free path of photons. The integral above evaluates to $49\,Mpc^2$.

The oscillations we're going to investigate shortly in this section exist only on the smaller l scales, which are those within the sound horizon at last scattering

$$d < r_{\text{sound}}(t_{LS}) = \int_0^{\eta_{LS}} \mathrm{d}\eta c_s(\eta) \approx 145\,\mathrm{Mpc} \quad (\text{comoving}) \tag{5.158}$$

which is the maximum scale at which coherent oscillations can exist. We thus expect to see these oscillations on scales $\theta \sim 1°$ or, equivalently, $l \approx 100$.

10 Essentially, the dipole anisotropy is a combination of peculiar velocities: Earth wrt Sun, Sun wrt MW, MW wrt M31, Local Group wrt Virgo cluster etc.

Temporal evolution

In the non-relativistic limit, we can consider small perturbations on the continuity and Euler equations

$$\begin{cases} \partial_t \rho + \nabla(\rho\vec{u}) = 0 \\ \rho\left[\partial_t + \vec{u}\cdot\nabla\right]\vec{u} = -\nabla p \end{cases} \tag{5.159}$$

We thus linearize the equations and assume a barotropic equation of state $p(\rho)$, which leads us to the following equation

$$\partial_t^2 \delta\rho = \nabla^2 \delta p \to (\partial_t^2 - c_s^2 \nabla^2)\delta\rho = 0 \tag{5.160}$$

This is all nice and dandy, but we're not considering two major points: (a) we should add the contribution of (self)gravity in Euler equation and thus introduce Poisson's equation

$$\nabla^2 \phi = 4\pi G\rho \tag{5.161}$$

and (b) we should also take into account the expansion of the Universe. This can be easily done switching to comoving coordinates $\vec{r}_{phy.} = a(t)\vec{x}_c$ and defining the velocity $\vec{v} = H\vec{r} + a\dot{\vec{x}}$, where the latter contribution is the comoving (or peculiar) velocity. In the Newtonian limit, the cosmological equation for the evolution of the density contrast $\delta = \delta\rho/\rho$ is

$$\ddot{\delta} + 2H\dot{\delta} - \left(\frac{c_s^2\nabla^2}{a^2} + 4\pi G\bar{\rho}\right)\delta = 0 \tag{5.162}$$

which represents a damped (Hubble friction) harmonic oscillator with a driving (gravity and pressure). We can also produce a more relativistically-acceptable equation

$$\boxed{\delta'' + \frac{R'}{1+R}\delta' - \frac{1}{3(1+R)}\nabla^2\delta = \frac{4}{3}\nabla^2\psi + 4\phi'' + \frac{4R^2}{1+R}\phi'} \tag{5.163}$$

where ϕ, ψ are the Bardeen potentials we've introduced earlier and R is the *baryon loading*

$$R = \frac{3}{4}\frac{\rho_b}{\rho_\gamma} = 0.6\left(\frac{\Omega_b}{0.02}\right)\left(\frac{a}{10^{-3}}\right) \tag{5.164}$$

Note that the sound speed is

$$c_s = \sqrt{\frac{1}{3(1+R)}}(c) \tag{5.165}$$

which tends to $c/\sqrt{3}$ in RD ($R \to 0$) and much smaller in MD.

The objective here is to find some solution to the relativistic equation above and then investigate the transfer function that descends from there, namely $F_{LS}(k)$. If you were to do so (we won't), you'd find

$$F_{LS}(k) = \frac{1}{5}\left[\frac{S(k)}{(1+R_{LS})^{1/4}}\cos(kr_S^{LS} + \theta(k)) - 3R_{LS}T(k)\right] \tag{5.166}$$

with S, θ, T fitting functions popping out the numerical solution. The important part is all in the phase of the cosine

$$kr_S^{LS} + \theta(k) \tag{5.167}$$

which tells us that peaks in the oscillations are found for values of k

$$k_n \approx \frac{n\pi}{r_s(\eta_{LS})} \tag{5.168}$$

where the presence of the fitting function introduces only a small displacement from the reported value. We can also calculate

Remarkably, the position of the 1st peak is influenced by the curvature of the Universe, while the height-ratio between different peaks depends on the density parameter of the various components involved via Bardeen potentials.

$$l_n = k_n \chi_{LS} \approx \frac{n\pi\chi_{LS}}{r_S(\eta_{LS})} \approx 302n \tag{5.169}$$

In reality the first peak in the oscillations is found at $l \approx 220$ and not at $l \approx 302$ because of the shift introduced by θ.

5.7 EXTRA: INDIRECT OBSERVATIONS OF DM

How can we do so? For example we can assume that DM is weakly coupled with particles of the Standard Model and there exist interactions like

$$\begin{aligned} \chi + \chi &\to \mathrm{SM} + \mathrm{SM} \\ \chi &\to \mathrm{SM} + \mathrm{SM} \end{aligned}$$

where SM refers to some particle in the Standard Model. In particular the latter reaction, a decay, would eventually produce a small flux of photons ϕ_γ which we can hope to detect. In principle we can expect to measure

$$\frac{\#\text{ events}}{TmcA} = \frac{M}{m_\chi}\Gamma\frac{1}{4\pi r^2} \tag{5.170}$$

Are there any numbers we can throw in there to even begin to hope we could observe something?

Take for example the Virgo cluster $R \approx 16\,\mathrm{Mpc}$, $M \approx 10^{19} M_\odot$, and suppose someone has given you for your birthday a detector that is most sensitive in the GeV-TeV range. With these numbers, if you wanted to measure (say) 100 events in a whole year, the interaction rate would be $\Gamma \approx 10^{-29} - 10^{-26}\,\mathrm{s}^{-1}$, which would correspond to timescales $\tau = \Gamma^{-1}$ much larger than the age of the Universe!

If DM were to annihilate somehow, to produce the correct Ω_{CDM} we would need $\langle \sigma v \rangle \approx 3 \cdot 10^{-26}\,\mathrm{cm}^3\,\mathrm{s}^{-1}$, and so

$$\Gamma = n_{\mathrm{DM}} \langle \sigma v \rangle = \frac{\rho_{\mathrm{DM}} \langle \sigma v \rangle}{m_{\mathrm{DM}}} \tag{5.171}$$

The inferred density of DM in the Milky Way is $\rho_{\mathrm{DM}} \approx 0.3\,\mathrm{GeV\,cm^{-3}}$, producing an interaction rate

$$\Gamma \approx \frac{10^{-26}\,\mathrm{s}^{-1}}{(m_{\mathrm{DM}}/\,\mathrm{GeV})} \tag{5.172}$$

Even in this case there would be little hope for detection. But the morale behind all this is that if you're able to aim an appropriate experiment towards a significantly large astronomical object, then you might hope to observe anomalous fluxes in the GeV-TeV range. This has the nice effect to produce bounds for $\langle \sigma v \rangle$.

5.7.1 Indirect Cosmological DM Detection

Consider now the number of annihilations per comoving volume that might have occurred in a Hubble time

$$\frac{\#\text{ events}}{V_c t_H} \approx \frac{n_{\text{phy}}^2 a^3 \langle \sigma v \rangle}{2H} \tag{5.173}$$

which implies that the annihilation rate is

$$\mathcal{A} \propto a^{-3} H^{-1} \langle \sigma v \rangle \tag{5.174}$$

In RD it would be $\mathcal{A} \propto a^{-1}$, in MD $\mathcal{A} \propto a^{-3/2}$, so in any case it is decreasing. Let's compute it at freeze-out and at the eV scale.

At freeze-out $\Gamma = \mathcal{A} \sim H$, so the fraction of annihilating particles would be of order 1! At the intermediate scales of the CMB, on the other hand, that fraction would have dropped below $f \sim 10^{-9}$. That is good, because we don't want DM to disappear in bulk.

What happens is that DM would be injecting about $1\,\text{eV} - 1\,\text{GeV}$ per annihilation, which is not that far away from what's needed to reionize the Universe. This, however, would produce an increase in the ionization fraction and would change significantly the visibility function, introducing an asymmetry. This would create a deviation from the approximation of instantaneous scattering which would suppress the peaks in the CMB spectrum and move the 1st peak as well. As a consequence, recombination and last scattering would also have to be postponed.

But worry not. Such deviation from expectations has never been observed (yet), further constraining the interaction cross section of DM.

6 INFLATION

The first few sections of this chapter aren't directly related to inflation, but are useful tools that will allow us to address some of the problems brought about by the standard (non-inflationary) cosmological model.

6.1 LUMINOSITY DISTANCE

If the redshift is not very small, we have to think more carefully about what we mean by "distance" in cosmology. The instantaneous physical distance (proper distance) is a convenient construct, but not itself observable, since observation always refer to our past light cone. In Euclidean space, there are a number of different ways to infer the distance of an object, for example comparing its apparent brightness to its intrinsic luminosity.

In a Euclidean space, we could define the *luminosity distance* as

$$d_L = \left(\frac{L}{4\pi F} \right)^{1/2}$$

with intrisic luminosity L and measured flux F. But how does this change in a curved and expanding Universe, and why?

For two reasons mainly:

1. *Geometry*: The space is no longer Euclidean, and the surface of the 2-sphere onto which the luminosity is diluted is $4\pi a_0^2 \tilde{r}_s^2 = 4\pi a_0^2 f_k(\chi)^2$, where $f_k(\chi)$ is

$$\chi = \int_0^{\tilde{r}_s} \frac{\mathrm{d}r}{\sqrt{1-kr^2}} = f_k^{-1}(r_s) = \begin{cases} \tilde{r}_s & k=0 \\ \arcsin(\tilde{r}_s) & k=1 \\ \operatorname{arcsinh}(\tilde{r}_s) & k=-1 \end{cases} \tag{6.1}$$

2. *Correction to the concept of luminosity*: In an expanding Universe, we have to be a little more cautious with what we refer to as "intrinsic luminosity." The necessary steps to be taken are explained in detail in the following.

Because of the expansion, the wavelength of these photons will change according to (4.51). Therefore, the energy of a photon will decrease

$$E_0 \sim \frac{c}{\lambda_0} = \frac{c}{(1+z)\lambda_e} \sim \frac{E_e}{1+z}$$

On top of that, two wavefronts separated by δt_e will arrive with a delay $\delta t_0 = \delta t_e(1+z)$, so the arrival rate will decrease of another $1+z$

$$L_e \sim \frac{E_e}{\delta t_e} = \frac{(1+z)^2 E_0}{\delta t_0} = (1+z)^2 L_0$$

Since the flux is equal to energy per unit time per unit area, we can write

$$F = \frac{L_e}{4\pi a_0^2 \tilde{r}_s^2 (1+z)^2} = \frac{L_e}{4\pi a_0^2 f_k(\chi)^2 (1+z)^2}$$

so that by direct comparison with our initial equation, we find that the *luminosity distance* is

$$d_L = a_0 \tilde{r}_s (1+z) = (1+z) a_0 f_k(\chi) \tag{6.2}$$

The luminosity distance is something we might hope to measure, since there are some astrophysical sources whose absolute luminosity are known, but χ (and, for extension, the comoving distance $\tilde{r}_s$), is not directly observable, so we have to remove it from our equation. On a null radial geodesic (for convenience), we have

$$0 = \mathrm{d}s^2 = -\mathrm{d}t^2 + a^2 \mathrm{d}\chi^2$$

or more simply

$$\chi = \int \frac{\mathrm{d}t}{a} = \int \frac{\mathrm{d}a}{a^2 H(a)}$$

It is conventional to convert the scale factor to redshift using (4.52) $a = a_0(1+z)^{-1}$, so we have

$$\chi(z) = \int_0^z \frac{\mathrm{d}z'}{a_0 H(z')}$$

In order to evaluate the Hubble parameter in this integral we can use Friedmann equation (4.123), thus finding

$$d_L(z) = (1+z) a_0 f_k \left[a_0^{-1} H_0^{-1} \int_0^z \frac{\mathrm{d}z'}{E(z')} \right]$$

where we've defined $E(z) = H(z)/H_0$. Still, it is more common to speak in terms of $\Omega_{c,0} = -k/a_0^2 H_0^2$, which can be measured. We can therefore write the luminosity in terms of measurable cosmological parameters as

$$\boxed{d_L(z) = (1+z) \frac{H_0^{-1}}{\sqrt{|\Omega_{c,0}|}} f_k \left[\sqrt{|\Omega_{c,0}|} \int \frac{\mathrm{d}z'}{E(z')} \right]} \tag{6.3}$$

Recall that the spatial density parameter $\Omega_{c,0}$ must abide the consistency relation

$$1 = \sum_i \Omega_{i,0} + \Omega_{c,0}$$

Although it appears unwieldy (and it kinda is), this equation is of central importance in cosmology. Given the observables H_0 and $\Omega_{i,0}$, we can straightforwardly calculate the luminosity distance to an object at any redshift z.

Experimentally, it's more common to use a quantity known as *distance module* to quantify distances. It is related to the luminosity distance through the definition of absolute magnitude

$$\mu = 5 \log_{10} \left(\frac{d_L}{10\,\mathrm{pc}} \right) \tag{6.4}$$

Remarkably, going back in time (so at higher redshift) we observe a change in the slope of the luminosity distance curve (or, equivalently, of the distance module) at roughly $z = 0.5$ caused by the change from an accelerating Universe (ΛDU) to a decelerating Universe (MDU).

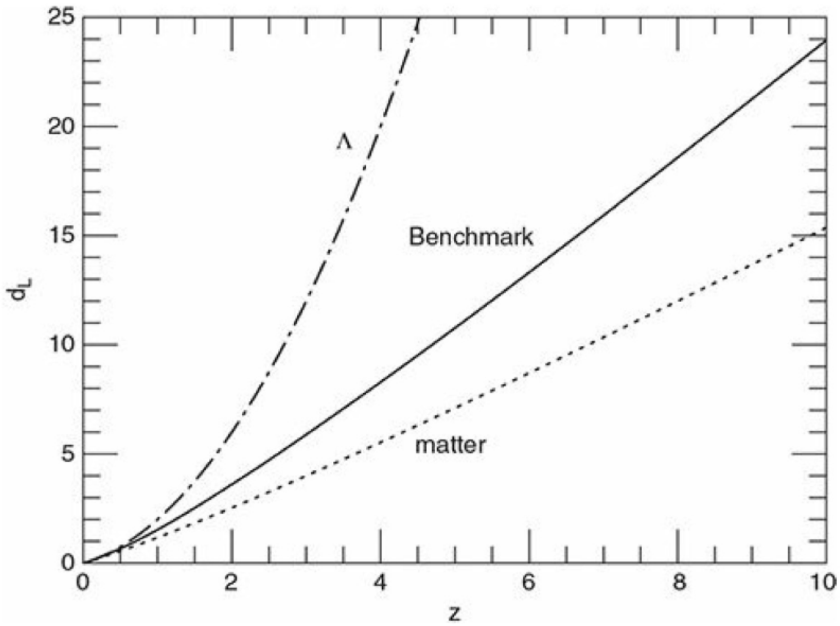

Figure 6.1: Luminosity distance of a standard candle with observed redshift z, in units of the Hubble distance, c/H_0. The bold solid line gives the result for the Benchmark Model. For comparison, the dot-dash line indicates a flat, Λ-dominated Universe, and the dotted line a flat, matter-dominated Universe.

If we define z_Λ the redshift at which there is equivalence between the Λ and matter density parameters

$$\Omega_{\Lambda,0} = \Omega_{m,0}(1+z_\Lambda)^3 \implies z_\Lambda = \left(\frac{\Omega_{\Lambda,0}}{\Omega_{m,0}}\right)^{1/3} - 1 \tag{6.5}$$

we find it to be $z_\Lambda \approx 0.44$, which is the exact cause of the *cosmic coincidence problem*, since it appears rather odd that the Cosmological Constant is dominating at a redshift so "close" to present time.

6.2 SECOND ORDER HUBBLE LAW

If you recall, earlier we've defined Hubble-Le Maître's law (4.54) at the linear order. It is the purpose of this section to stretch our approximation a bit further, including larger redshifts and, by extension, a second-order correction to the aforementioned law.

Let us now approximate $a(t)$ in a neighborhood of today

$$a(t) = a(t_0) + \left(\frac{\partial a}{\partial t}\right)_{t_0}(t-t_0) + \frac{1}{2}\left(\frac{\partial^2 a}{\partial t^2}\right)_{t_0}(t-t_0)^2 + o\left((t-t_0)^3\right)$$

so that

$$a(t) = a_0\left[1 + H_0(t-t_0) - \frac{1}{2}q_0H_0^2(t-t_0)^2 + o\left((t-t_0)^3\right)\right] \tag{6.6}$$

where we've defined

$$H_0 = \left(\frac{\dot{a}(t)}{a(t)}\right)_{t_0} \qquad q_0 = -\left(\frac{\ddot{a}(t)a(t)}{\dot{a}^2(t)}\right)_{t_0} \tag{6.7}$$

We also know that

$$z = \frac{a(t_0)}{a(t)} - 1 \approx -1 + \frac{1}{1 - H_0(t_0 - t) - \frac{1}{2}H_0^2 q_0 (t_0 - t)^2}$$

We can find an approximate relation for the redshift Taylor expanding its definition

$$z \approx H_0(t_0 - t) + \left(1 + \frac{q_0}{2}\right) H_0^2 (t_0 - t)^2$$

Remember that at first order we found the relation $z(t) = H_0(t_0 - t)$, which we can plug into last equation introducing corrections only to the third order

$$H_0(t_0 - t) \approx z(t) - \frac{1}{2}(q_0 + 2)z^2 + \mathrm{o}(z^3) \tag{6.8}$$

Now we'll turn to investigate how all of this enters into the definition of proper distance

$$\tilde{r}_s = \int_0^{r_s} \frac{\mathrm{d}r}{\sqrt{1 - kr^2}} = \int_{t_e}^{t_0} \frac{\mathrm{d}t}{a} \tag{6.9}$$

where we're considering a null, radial geodesic for the FLRW metric (4.35). We can plug in the second order expansion for $a(t)$

$$\tilde{r}_s = \frac{1}{a_0} \int_{t_e}^{t_0} \mathrm{d}t \left[1 + H_0(t_0 - t) + \frac{1}{2}H_0^2 q_0 (t_0 - t)^2\right]^{-1} \tag{6.10}$$

We can neglect the third-order contribution brought by the third term in the right-hand side of the equation and expand once more

$$\tilde{r}_s \approx \frac{1}{a_0} \int_{t_e}^{t_0} \mathrm{d}t \left[1 - H_0(t_0 - t)\right] = \frac{(t_0 - t_e)}{a_0} - \frac{H_0(t_0 - t_e)^2}{2a_0} \tag{6.11}$$

which means that our proper distance is ($c = 1$)

$$\tilde{r}_s = \frac{1}{a_0}\left[(t_0 - t_e) + \frac{1}{2}H_0(t_0 - t_e)^2\right] \tag{6.12}$$

or, in terms of the redshift and reintroducing the missing c factors

$$\tilde{r}_s(t_0) = \frac{c}{H_0} z \left[1 - \frac{(1 + q_0)}{2} z\right] \tag{6.13}$$

We can substitute the expression at second order for the proper distance in the definition of distance luminosity (6.2) to find

$$d_L^2 = \frac{1}{H_0^2}\left(z - \frac{1}{2}(q_0 + 2)z^2 + \frac{z^2}{2}\right)^2 (1 + z)^2 \tag{6.14}$$

which we can expand to quadratic order

$$H_0 d_L = \left[z - \frac{1}{2}(q_0 + 1)z^2\right](1 + z) = z + z^2 - \frac{1}{2}(q_0 + 1)z^2 + \mathrm{o}(z^3) \tag{6.15}$$

which yields

$$\boxed{H_0 d_L = z + \frac{1}{2}(1 - q_0)z^2 + \mathrm{o}(z^3)} \tag{6.16}$$

6.3 ANGULAR DIAMETER DISTANCE

Let's consider a curved, expanding Universe described by the FLRW metric and choose your comoving coordinate system so that the observer is at its origin.

Consider a yardstick of constant proper length l perpendicular to the line of sight and at a comoving coordinate distance r_e. At a time t_e, the yardstick emitted the light that you observe at time t_0. The comoving coordinates of the two ends of the yardstick, at the time the light was emitted, were (r_e, θ_1, φ) and (r_e, θ_2, φ).

The distance $\mathrm{d}s$ between the two ends of the yardstick, measured at the time t_e when the light was emitted, can be found from the FLRW metric

$$\mathrm{d}s = a(t_e) f_k(\chi)\, \mathrm{d}\Omega = a(t_e)\tilde{r}_s\, \mathrm{d}\Omega$$

where $a(t_e)\tilde{r}_s$ is the physical (proper) distance of the emitter at the time of emission. We can set $\mathrm{d}s = l$ and define

$$d_A := \frac{\mathrm{d}s}{\mathrm{d}\Omega} = \tilde{r}_s \frac{a(t_e)}{a_0} a_0 = \frac{\tilde{r}_s a_0}{(1+z)} \tag{6.17}$$

from which we conclude

$$d_A = \frac{a_0 f_k(\chi)}{1+z} = \frac{d_L}{(1+z)^2} \tag{6.18}$$

You can also show that

$$\begin{aligned} H_0 d_A &= \left[z - \frac{1}{2}(1-q_0)z^2 + \mathrm{o}(z^3)\right](1 - 2z + \mathrm{o}(z^2)) \\ &= z - \frac{1}{2}(3+q_0)z^2 + \mathrm{o}(z^3) \end{aligned}$$

which is the Hubble law as a function of the angular distance.

We can define the apparent brightness B as

$$B = \frac{l}{\Omega} = \frac{L}{4\pi d_L^2}\frac{d_A^2}{S} \tag{6.19}$$

where l is the apparent luminosity, Ω is the angular extension, and S is the physical extension of the source. We can further manipulate the definition introducing the absolute luminosity per unit area, which depends only on the intrinsic characteristic of the source

$$B = \frac{\mathcal{L}}{4\pi}\left(\frac{d_A}{d_L}\right)^2 = \frac{\mathcal{L}}{4\pi}(1+z)^{-4} \tag{6.20}$$

Suppose that the redshift we observe were produced by some absorbing medium between us and the object. In an Euclidean Universe

$$d_L = d(1+z)^{1/2} \tag{6.21}$$

while $d_A = d$, and so

$$\frac{d_A}{d_L} = \frac{1}{(1+z)^{1/2}} \implies B \propto (1+z)^{-1} \tag{6.22}$$

but (un)fortunately observations are telling us that $B \propto (1+z)^{-4}$, thus allowing us to discard the scenario in which the redshift is wholly produced by absorbing media (rather than an expanding Universe).

6.4 PARTICLE HORIZON

Consider a photon emitted by some source at time t_e, for a light path $\mathrm{d}s = 0$. The maximum comoving distance the photon might have travelled to present day is

$$a^{-1}(t)\,\mathrm{d}t = \frac{\mathrm{d}r}{\sqrt{1-kr^2}} \implies \int_0^{t_0} \frac{\mathrm{d}t}{a(t)} = \int_0^{d_H} \frac{\mathrm{d}r}{\sqrt{1-kr^2}} \tag{6.23}$$

For a single component (flat) Universe, the expression for $a(t)$ is simple

$$a(t) \propto t^n \quad n = \frac{2}{3(1+w)} \tag{6.24}$$

and the integral is well defined provided that $n < 1$, which implies $w > -1/3$ in the EoS of the Cosmological fluid (accelerating Universe). The maximum (physical) distance travelled by a photon is then defined as

$$d_{\max} = R_H(t_0) = a(t_0) \int_0^{t_0} \frac{\mathrm{d}t}{a(t)} = a(t_0)\tilde{r}_{\max}(t_0) \tag{6.25}$$

Such distance is commonly known as *particle horizon* and represents the maximum distance from which we might have possibly received information from the past via electromagnetic waves. For a flat Universe ($k = 0$)

Recall the definition of x

$$x = \frac{a(t)}{a_0} = \frac{1}{1+z(t)}$$

$$\mathrm{d}t = \frac{\mathrm{d}x}{xH_0E(x)} \implies d_{\max}(t_0) = \int_0^1 \frac{\mathrm{d}x}{H_0xE(x)} \tag{6.26}$$

while for a "standard" single-component Universe

$$R_H(t) = a(t) \int_{t_i}^{t} \frac{\mathrm{d}t'}{a(t')} \quad a(t) \propto t^n \tag{6.27}$$

which is also equal to ($t \gg t_i$)

$$R_H(t) = t^n \left[\frac{1}{1-n} t^{1-n}\right]_{t_i}^{t} \approx \frac{t}{1-n} \tag{6.28}$$

We might relate this last result to the Hubble radius H^{-1}, obtaining

$$\frac{H^{-1}}{R_H} = \frac{1-n}{n} \tag{6.29}$$

but please note that this last equation holds only for a simple cosmological model with a single dominating component.

Event Horizon

We define the *event horizon* as the maximum distance at which an object can be so that a signal emitted by that same object reaches us at a future infinite. Fixed an emission time $t_\star$

$$\int_{t_\star}^{\infty} \frac{\mathrm{d}t}{a(t)} = \int_0^{r_\star} \frac{\mathrm{d}r}{\sqrt{1-kr^2}} \tag{6.30}$$

If $r_s < r_\star$, the event will never be observed, meaning there could be regions of spacetime that are impossible to "communicate" with. To see if the integral can actually be finite, you will usually check

$$r_\star = -\frac{1}{AH} e^{-Ht}\Big|_{t_\star}^{\infty} < \infty \tag{6.31}$$

For example, if the EoS has $w > -1/3$, $r_\star \to \infty$ and the horizon is not properly defined.

We're assuming a flat Universe $k = 0$ for which

$$a(t) = Ae^{Ht}$$

6.5 FLATNESS PROBLEM

Consider a Universe with vanishing Cosmological Constant $\Lambda = 0$ which is also such that Einstein's equation is *always* valid, also at $t = 0$. If we define

$$\Omega_k = 1 - \Omega = \frac{\rho_c - \rho}{\rho_c} = -\frac{k}{a^2 H^2} \tag{6.32}$$

Now assume that $k \neq 0$ and Ω_k can evolve independently from the dominant contribution to the cosmological fluid. If

- MDU: $a(t) \propto t^{2/3}$ and $H \propto a^{-3/2} \implies \Omega_k \sim a \sim t^{2/3} \sim \Gamma^{-1}$;
- RDU: $a(t) \propto t^{1/2}$ and $H \propto a^{-2} \implies \Omega_k \sim a^2 \sim t \sim \Gamma^{-2}$.

Going back in time has the effect of decreasing the value of Ω_k. Through studies of the CMB (see Chapter 5 for more), the Planck collaboration managed to put bounds to the value of Ω_k, which was found to be

$$|\Omega_k(t_0)| < 0.05 \tag{6.33}$$

At the time of matter-radiation equality, the spatial density parameter was

$$\left|\frac{\Omega_k(t_{eq})}{\Omega_k(t_0)}\right|^{-1} = \frac{T_{eq}}{T_0} \sim 10^4 \implies |\Omega_k(t_{eq})| < 10^{-4} \tag{6.34}$$

Similarly, if we extrapolate backward to the time of Big Bang nucleosynthesis, at $a_{BBN} \approx 3.6 \cdot 10^{-9}$

$$|1 - \Omega_{BBN}| < 7 \cdot 10^{-16}$$

and the distance from unity would only decrease if we went even further back in time. Setting then those orders of magnitude as initial conditions imply a *huge fine-tuning problem*.

There is no known reason for the density of the Universe to be so close to the critical density, and this appears to be an unacceptably strange coincidence in the view of most astronomers and cosmologists. This is the so called *Flatness problem*.

Let us define

$$x = \log\left(\frac{a}{a_0}\right) \implies H = \frac{\mathrm{d}\ln(a)}{\mathrm{d}t} \implies \mathrm{d}t = \frac{1}{H}\mathrm{d}\ln(a) = \frac{1}{H}\mathrm{d}x \tag{6.35}$$

and recall Friedmann equation (4.84) for a perfect fluid

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\rho(1+3w) \implies 2q = (1+3w)\Omega(t) \tag{6.36}$$

and by direct substitution you can check that

$$\frac{\dot{H}}{H^2} = -(1+q) \tag{6.37}$$

Putting all of this to work, we gather

$$\frac{\mathrm{d}\Omega_k}{\mathrm{d}t} = \frac{k}{a^3}\frac{2\dot{H}}{H^3} + \frac{k}{H^2}\frac{2\dot{a}}{a^3} = -2\Omega_k H\left(1+\frac{\dot{H}}{H^2}\right) = -2\Omega_k H(1-(1+q)) \tag{6.38}$$

If we want the final result to be independent from the particular cosmology that we've chosen

Here the prime derivative is with respect to x, so

$$dt = \frac{1}{H}dx$$

and not conformal time.

$$\Omega_k' = 2\Omega_k q \tag{6.39}$$

$$2q = (1+3w)\Omega = (1+3w)(1-\Omega_k) \tag{6.40}$$

which means

$$\boxed{\Omega_k' = (1+3w)(1-\Omega_k)\Omega_k} \tag{6.41}$$

This tells us that if $\Omega_k \approx 0$, the stationary point is unstable if $1+3w > 0$! Similarly, $\Omega_k \approx 0$ is a stable attractor if $1+3w < 0$, which means $w < -1/3$.

The flatness problem is thus resolved if we postulate a Universe that accelerates for enough time. Such period of time is called *inflationary epoch* or just *inflation*. If during the inflation $H \sim$ const, then

$$\Omega_k \approx -\frac{k}{a^2H^2} \tag{6.42}$$

Let t_i be inflation's starting time and t_f the ending time

$$\frac{\Omega_k(t_f)}{\Omega_k(t_i)} = \left(\frac{a_f}{a_i}\right)^2 = e^{-2N} \tag{6.43}$$

where we defined $N = \ln(a_f/a_i)$ the number of e-folds of the inflationary epoch. We want that at the end of inflation $|\Omega_k(t_f)| \sim 10^{-61}$, while we assume a strongly non-flat Universe at the beginning of inflation $|\Omega_k(t_i)| \sim 1$. For our requirements to be met, the Universe should expand a number of e-folds such that

$$e^{-2N} \approx 10^{-60} \implies N \geq 60 \tag{6.44}$$

During inflation, then, the comoving Hubble radius is decreasing

$$\boxed{\frac{1}{aH}\downarrow \implies |\Omega_k| = \frac{|k|}{a^2H^2}\downarrow} \tag{6.45}$$

thus resulting in a quasi-de Sitter Universe with $p = -\rho$ and $H \sim$ const.

6.6 HORIZON PROBLEM

Another problem remains to be addressed. We know that photons decouple from ordinary matter at $z \approx 1100$ (decoupling/last scattering). Currently, within our particle horizon, we observe a roughly homogeneous temperature distribution, which is that of the CMB. So far so good.

From last scattering until present day, the Universe was dominated by non-relativistic matter, hence

$$R_H(t_{LS}) \approx H_{LS}^{-1} = H_0^{-1}\left(\frac{a_{LS}}{a_0}\right)^{3/2} \approx R_H(t_0)\left(\frac{a_{LS}}{a_0}\right)^{3/2} \tag{6.46}$$

But here comes the problem.

Since

"*Scales scale with the scale factor*"

meaning

$$\lambda_{H_0}(t_{LS}) = \frac{R_H(t_0)}{1+z_{LS}} \tag{6.47}$$

This is essentially the corresponding particle wavelength at last scattering.

we get a gargantuan problem $R_H(t_{LS}) \ll \lambda_{H_0}(t_{LS})$, which can be rewritten in a perhaps clearer form

$$\left(\frac{\lambda_{H_0}(t_{LS})}{R_H(t_{LS})}\right)^3 = \left(\frac{a_{LS}}{a_0}\right)^{-3/2} \approx 10^5 \tag{6.48}$$

which means that 10^5 volumes are *not* in causal connection!

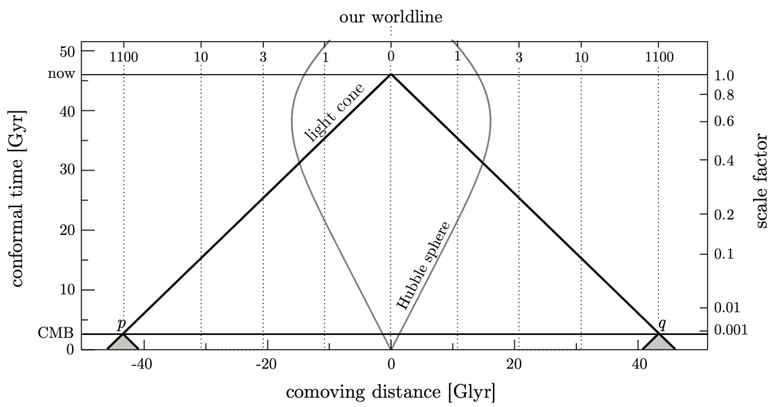

Figure 6.2: The horizon problem in the conventional Big Bang model. All events that we currently observe are on our past light cone. The intersection of our past light cone with the spacelike slice labelled CMB corresponds to two opposite points in the observed CMB. Their past light cones don't overlap before they hit the singularity, $a = 0$, so the points appear never to have been in causal contact. The same applies to any two points in the CMB that are separated by more than 1 degree on the sky. Credits: [2].

This is a huge problem in order to find a sensible explanation for the impressive homogeneity of the CMB (since after decoupling photons are no longer interacting with matter).

Perhaps you won't be surprised when I say that inflation comes to the rescue this time as well.

During inflation $H_i^{-1} \approx$ const, while structures continue to grow until they eventually cross the horizon. If we assume a large enough inflationary period, the scales within which we observe the current homogeneity could be able to enter once more into the horizon because they're exponentially suppressed.

At the end of an à la de Sitter inflation (but French-sounding) the particle horizon was

$$R_H = a_f \int_{t_i}^{t_f} \frac{dt}{a(t)} \tag{6.49}$$

with

$$a(t) = a(t_i) \exp(H_i(t - t_i)) \tag{6.50}$$

meaning that

$$\frac{a_f}{a_i} = \frac{a(t_f)}{a(t_i)} = \exp(H_i(t_f - t_i)) \tag{6.51}$$

We can also invert the expression

$$a(t_i) = a(t_f) \exp(-H_i(t_f - t_i)) \tag{6.52}$$

and define the number of e-folds like we've done earlier $N = H_i(t_f - t_i)$. This allows us to write

$$R_H(t_f) = a(t_f) \int_{t_i}^{t_f} \frac{dt}{a(t_f)} \exp(H_i(t_f - t)) \tag{6.53}$$

If we assume that the particle horizon at last scattering is such that

$$R_H(t_{LS}) = a(t_{LS}) \int_{t_i}^{t_{LS}} \frac{dt}{a(t)} \approx a(t_{LS}) \int_{t_i}^{t_f} \frac{dt}{a(t)} \exp(H_i(t_f - t)) \tag{6.54}$$

we conclude

$$\boxed{R_H(t_{LS}) = \frac{a_{LS}}{a(t_f)} \frac{1}{H_i} \left[\exp(H_i(t_f - t_i)) - 1\right] = \frac{a_{LS}}{a(t_f)} \frac{1}{H_i} (e^N - 1)} \tag{6.55}$$

Our problem is solved if $R_H(t_{LS}) > \lambda_{H_0}(t_{LS})$

$$\frac{a_{LS}}{a(t_f)} \frac{1}{H_i} (e^N - 1) > \frac{1}{H_0(1 + z_{LS})} = \frac{a_{LS}}{a_0 H_0} \tag{6.56}$$

hence if

$$\boxed{e^N \gtrsim \frac{a(t_f) H_i}{a(t_0) H_0}} \tag{6.57}$$

How do we interpret this last result? All scales we observe today had always been causally connected during the whole expansion of the Universe.

For the sake of completeness, let's prove that the conditions to solve both the flatness and horizon problems are the same.

$$|\Omega_k(t_0)| = \frac{|k|}{a_0^2 H_0^2} \qquad |\Omega_k(t_f)| = \frac{|k|}{a_f^2 H^2(t_f)} \tag{6.58}$$

Recall that during inflation the Hubble constant is not changing.

This implies

$$|\Omega_k(t_0)| a_0^2 H_0^2 = |\Omega_k(t_f)| a_f^2 H^2(t_f) = |\Omega_k(t_f)| a_f^2 H_i^2 \tag{6.59}$$

but

$$\left|\frac{\Omega_k(t_f)}{\Omega_k(t_i)}\right| = e^{-2N} \iff |\Omega_k(t_0)| = e^{-2N} |\Omega_k(t_i)| \left(\frac{a_f^2 H_i^2}{a_0^2 H_0^2}\right) \tag{6.60}$$

The request $|\Omega_k(t_0)| < 1$ purely relies on observational evidence and on (4.120).

If we set $|\Omega_k(t_i)| \approx 1$

$$|\Omega_k(t_0)| \approx e^{-2N} \left(\frac{a_f^2 H_i^2}{a_0^2 H_0^2}\right) < 1 \iff e^N > \frac{a(t_f) H_i}{a_0 H_0} \tag{6.61}$$

which is the condition that solves the horizon problem.

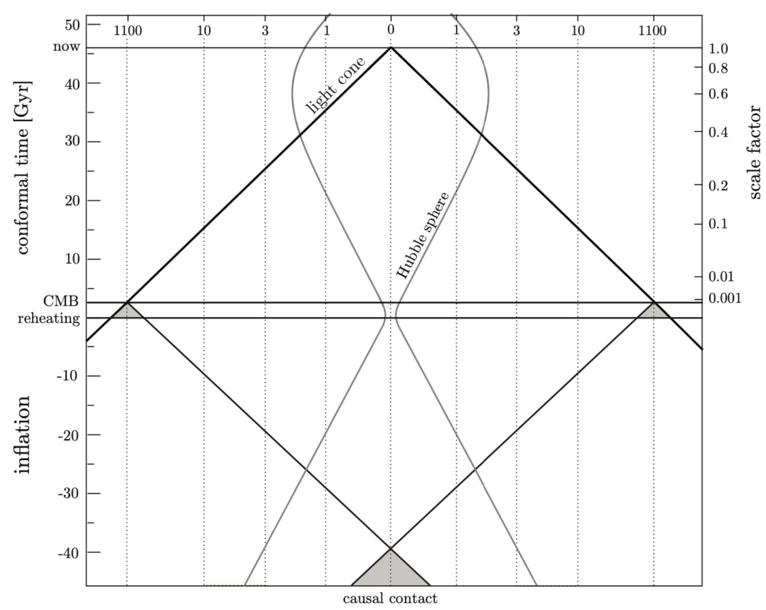

Figure 6.3: Inflationary solution to the horizon problem. The comoving Hubble sphere shrinks during inflation and expands during the conventional Big Bang evolution (at least until dark energy takes over at $a \approx 0.5$). Conformal time during inflation is negative. The spacelike singularity of the standard Big Bang is replaced by the reheating surface, i.e. rather than marking the beginning of time it now corresponds simply to the transition from inflation to the standard Big Bang evolution. All points in the CMB have overlapping past light cones and therefore originated from a causally connected region of space. Credits: [2].

6.7 HUBBLE RADIUS

The Hubble radius we've defined as $R_H \sim H^{-1}$ for both MDU and RDU is different from that we have in an accelerated Universe. The Hubble radius at a given time t is roughly equal to the distance travelled by a photon during an "expansion time," (the time needed for the scale parameter to double) starting from said time t.

As a consequence, two particles that are separated by $\lambda > H^{-1}(t_0)$ will also be causally connected if the distance that separated them at some earlier time was $\lambda(t) < H^{-1}(t)$.

Say that $\tilde{R}_H(t_0)$ is the distance crossed by a photon in $t \to t_\star$ so that $a(t_\star) = 2a(t)$ (so, an expansion time). By virtue of its own definition

$$\tilde{R}_H(t) = a(t_\star) \int_t^{t_\star} \frac{\mathrm{d}t}{a(t)} = a(t_\star) \int_{a(t)}^{a(t_\star)} \frac{\mathrm{d}a}{a^2 H} = a(t_\star) \int_{a(t)}^{2a(t)} \frac{\mathrm{d}a}{a^2 H} \tag{6.62}$$

Consider a small expansion time (respect to the age of the Universe) so that approximately $H(t) \sim \text{const}$. Then

$$\tilde{R}_H(t) = \frac{1}{H(t)} \tag{6.63}$$

but if now $\lambda < H_0^{-1}$, while $\lambda > H(t_{LS})^{-1}$ at an earlier time, then there must necessarily exist an epoch during which λ is growing faster than H^{-1}

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\lambda}{H^{-1}}\right) > 0 \tag{6.64}$$

but if $\lambda \sim a$, this implies $\ddot{a} > 0$, hence the Universe must be accelerating, hence the necessity of inflation.

What we surmise, then, is that the comoving Hubble radius $(aH)^{-1}$ must be increasing right after the BBN, but for earlier times it must be decreasing, as pictorially shown in Fig.6.4.

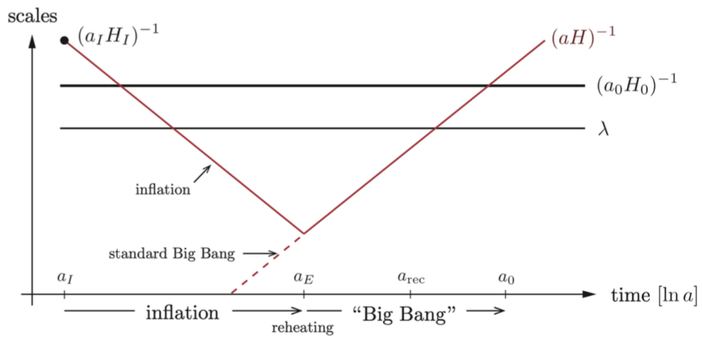

Figure 6.4: Scales of cosmological interest were larger than the Hubble radius until $a \approx 10^{-5}$ (where today is at $a(t_0) = 1$). However, at very early times, before inflation operated, all scales of interest were smaller than the Hubble radius and therefore susceptible to microphysical processing. Notice the symmetry of the inflationary solution. Scales just entering the horizon today, 60 e-folds after the end of inflation, left the horizon 60 e-folds before the end of inflation. Credits: [2].

The existence of an epoch of accelerated expansion corresponds to having $p < -\rho/3$ and violates the *strong energy condition*.

Several cosmologists began to postulate the necessity of an inflationary period starting from the end of 1970s. Remarkably, Starobinskiǐ [20] noted that quantum corrections to GR should be important for the early Universe.

The solution to the Einstein field equations in the presence of curvature squared terms, when the curvatures are large, leads to an effective Cosmological Constant. Therefore, he proposed that the early Universe went through an inflationary de Sitter era, but never mentioned that such inflationary epoch could solve the problems intrinsic of the standard hBBN model.

Only two years later, Guth [9], and the following month still Guth and Weinberg [10], showed that the horizon and flatness problems could be solved if we allowed the existence of an extraordinary supercooling in the early history of the Universe to temperatures 28 or more orders of magnitude below the critical temperature for some phase transition. But

"Unfortunately, the scenario seems to lead to some unacceptable consequences, so modifications must be sought."

In short, they were postulating the existesnce of a scalar field φ. The associated potential $U(\varphi)$ had to present a local maximum where the phase

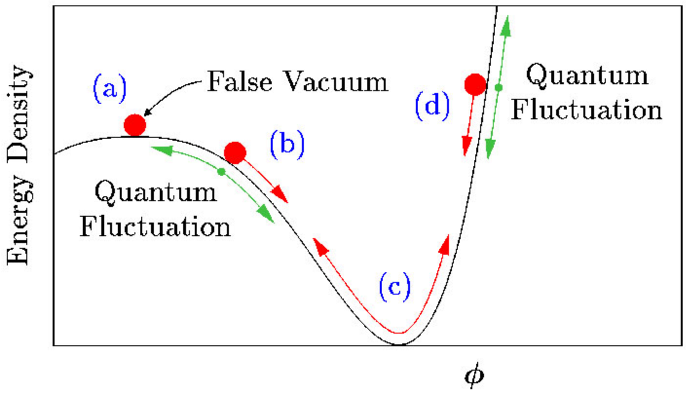

Figure 6.5: When the scalar field φ is near region (a), the energy density will remain nearly constant, $\rho \approx \rho_f$, even as the Universe expands. Moreover, cosmic expansion acts like a frictional drag, slowing the motion of φ: Even near regions (b) and (d), φ behaves more like a marble moving in a bowl of molasses, slowly creeping down the side of its potential, rather than like a marble sliding down the inside of a polished bowl. During this period of "slow roll," ρ remains nearly constant. Only after φ has slid most of the way down its potential will it begin to oscillate around its minimum, as in region (c), ending inflation. Credits: this website.

transition occurred (so at the beginning of inflation), and a local minimum where $U(\varphi) \approx \text{const.} = \Lambda$, hence a metastable de Sitter phase during which tunnel effect occurred towards true vacuum. This was the so-called *old inflation*.

What had gone wrong with Guth's and Weinberg's model was solved only one year later by Andrej Dmitrievič Linde [12], [13] and independently by Andreas Albrecht and Paul Steinhardt [1] in a model named *new inflation* or *slow-roll inflation*.

In this model, instead of tunneling out of a false vacuum state, inflation occurred by a scalar field rolling down a potential energy hill. When the field rolls very slowly compared to the expansion of the Universe, inflation occurs. However, when the hill becomes steeper, inflation ends and reheating can occur.

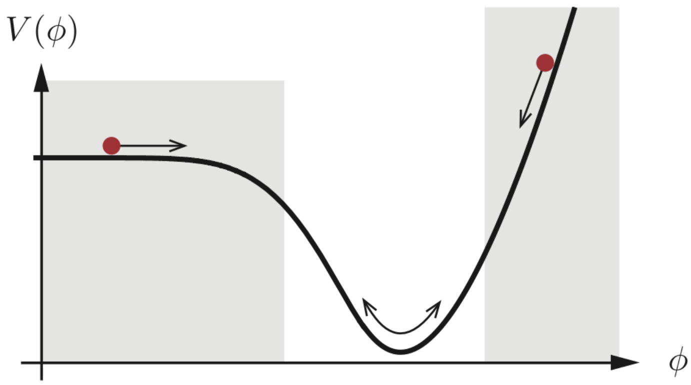

Figure 6.6: Example of a slow-roll potential as postulated by Linde, Albrecht, and Steinhardt. Inflation occurs in the shaded parts of the potential. Credits: [2].

6.8 SLOW-ROLL INFLATION

Let's consider an inflationary model as brought about by a scalar field φ we will call, in a very witty fashion, *inflaton*. To the scalar field φ is associated an action S

$$S = \int d^4x \sqrt{-g}\mathcal{L} = \int d^4x \sqrt{-g} \left[\frac{1}{2} g^{\mu\nu} \partial_\mu \varphi \partial_\nu \varphi - V(\varphi) \right] \tag{6.65}$$

where $\sqrt{-g} = a^3$ for the FLRW metric (4.35). We can calculate Euler-Lagrange equations

$$\partial^\mu \frac{\delta(\sqrt{-g}\mathcal{L})}{\delta(\partial^\mu \varphi)} - \frac{\delta(\sqrt{-g}\mathcal{L})}{\delta \varphi} = 0 \tag{6.66}$$

which yield

Note the presence of a Hubble-friction term $3H\dot{\varphi}$ that we wouldn't have in a static Universe.

$$\ddot{\varphi} + 3H\dot{\varphi} - \frac{\nabla^2 \varphi}{a^2} + V'(\varphi) = 0 \quad V'(\varphi) = \frac{dV(\varphi)}{d\varphi} \tag{6.67}$$

Consider also the energy-momentum tensor associated to the action S (6.65), which we can obtain simply variating the action with respect to the metric

$$T_{\mu\nu} = \frac{2}{\sqrt{-g}} \frac{\delta(\sqrt{-g}\mathcal{L})}{\delta g_{\mu\nu}} \tag{6.68}$$

obtaining

$$T_{\mu\nu} = \partial_\mu \varphi \partial_\nu \varphi - g_{\mu\nu} \mathcal{L} \tag{6.69}$$

We can give a little twist to this last expression so that it looks like the energy-momentum tensor of a perfect fluid

$$T^{\mu\nu} = p_\varphi g^{\mu\nu} + (p_\varphi + \rho_\varphi) u^\mu u^\nu \tag{6.70}$$

where ρ_φ and p_φ are respectively

$$\rho_\varphi = T^{00} = \frac{\dot{\varphi}^2}{2} + V(\varphi) + \frac{(\nabla \varphi)^2}{2a^2} \tag{6.71}$$

$$p_\varphi = \frac{T^{ii}}{3} = \frac{\dot{\varphi}^2}{2} - V(\varphi) - \frac{(\nabla \varphi)^2}{6a^2} \tag{6.72}$$

If the field is strongly dishomogeneous and, as a consequence, $|\nabla \varphi|$ is large

$$\rho_\varphi \approx + \frac{(\nabla \varphi)^2}{2a^2} \tag{6.73}$$

$$p_\varphi \approx - \frac{(\nabla \varphi)^2}{6a^2} \tag{6.74}$$

which means

$$p_\varphi \approx -\frac{1}{3} \rho_\varphi \tag{6.75}$$

and hence $w = -1/3$. However, a Universe so dishomogeneous cannot give raise to an inflationary epoch. So let's consider instead a scalar field that is decomposed as

$$\varphi(\vec{x}, t) = \varphi_0(t) + \delta\varphi(\vec{x}, t) \tag{6.76}$$

where the former term is that associated to a "classical" field ($\lambda \to \infty$), while the latter is associated to quantum fluctuations. If we assume the quantum fluctuations to be negligible respect to the classical field $\delta\varphi \ll \varphi_0$, we can neglect the contribution brought by $\delta\varphi$ to the evolution (*back-reaction*). Hence

$$T^{00} = \rho_\varphi = \frac{\dot\varphi_0^2}{2} + V(\varphi_0) \tag{6.77}$$

$$\frac{T^{ii}}{3} = p_\varphi = \frac{\dot\varphi_0^2}{2} - V(\varphi_0) \tag{6.78}$$

which grants $p_\varphi \approx -\rho_\varphi$, thus producing a pseudo-de Sitter evolution. The equation of motion then turns into

$$\boxed{\ddot\varphi_0 + 3H\dot\varphi_0 + V'(\varphi_0) = 0} \tag{6.79}$$

which is the Klein-Gordon equation for the inflaton. In the limit $V(\varphi_0) \gg \dot\varphi_0^2/2$, the scalar field is "slowly-rolling" in the potential. And this is all thanks to the friction term!

The 2nd Friedmann equation (4.89) for the background density is

$$H^2 = \frac{8\pi G}{3}\left(\frac{1}{2}\dot\varphi^2 + V(\varphi)\right) \tag{6.80}$$

which we can differentiate with respect to time to yield

$$2H\dot H = \frac{8\pi G}{3}\left(\dot\varphi\ddot\varphi + \frac{\mathrm{d}V}{\mathrm{d}\varphi}\dot\varphi\right) = -8\pi G H\dot\varphi^2 \tag{6.81}$$

ultimately leading to

$$\boxed{\dot H = -4\pi G\dot\varphi^2} \tag{6.82}$$

From the first Friedmann equation, on the other hand, we have

$$\begin{aligned}\frac{\ddot a}{a} &= -\frac{4\pi G}{3}(\rho + 3p) = -\frac{4\pi G}{3}(2\dot\varphi^2 - 2V(\varphi)) \\ &= -\frac{4\pi G}{3}\left[3\dot\varphi^2 - 2\left(\frac{1}{2}\dot\varphi^2 + V(\varphi)\right)\right]\end{aligned}$$

so that recalling (6.80), (6.82), the 1st Friedmann equation is rewritten as

$$\frac{\ddot a}{a} = \dot H + H^2 \tag{6.83}$$

and we can define the 1st slow-roll parameter ε

$$\boxed{\varepsilon = -\frac{\dot H}{H^2}} \tag{6.84}$$

so that

$$\frac{\ddot a}{a} = (1-\varepsilon)H^2 \tag{6.85}$$

Note that more than one slow-roll parameter exist, but only the first, ε, determines the inflation.

and the Universe is accelerating only if $\varepsilon < 1$. If we now request the Universe to be (ideally) a pseudo-de Sitter with $H \approx \text{const}$, we can consider the relative variation $\dot H/H$ over an Hubble time (H^{-1}) and require

$$\left|\frac{\dot H}{H}\frac{1}{H}\right| \ll 1 \implies |\varepsilon| \ll 1 \tag{6.86}$$

In terms of the number of e-folds from the beginning of inflation up to time t we can also write

$$\varepsilon = -\frac{\mathrm{d}\ln(H)}{\mathrm{d}N} \qquad N = \log\left(\frac{a(t)}{a_i}\right) \tag{6.87}$$

which also means $\mathrm{d}N = H\,\mathrm{d}t$ and

$$-\frac{\dot{H}^2}{H^2} = -\frac{\mathrm{d}\ln(H)}{\mathrm{d}N} \tag{6.88}$$

The 1st slow-roll parameter ε thus describes, in some sense, the displacement from a purely de Sitter Universe.

Let's now show that $\varepsilon \ll 1 \iff V(\varphi) \gg \dot{\varphi}^2/2$

$$(\implies) \qquad -\dot{H} = 4\pi G\dot{\varphi}^2 \ll H^2 = \frac{8\pi G}{3}\left(\frac{1}{2}\dot{\varphi}^2 + V(\varphi)\right)$$
$$\implies \dot{\varphi}^2 \ll V(\varphi)$$

which is the first direction of the implication. Let's now prove that the reverse holds as well

$$(\impliedby) \quad p \approx -\rho \to \begin{cases} \dot{\rho} + 3H(p+\rho) = 0 & \implies \dot{\rho} \approx 0 \\ H^2 \approx \frac{8\pi G}{3}\rho & \implies H^2 \approx \text{const.} \end{cases} \tag{6.89}$$

and at this point concluding is trivial $H^2 \approx$ const. $\implies$ $H \approx$ const $\implies$ $\varepsilon \ll 1$.

Let's define the *second slow-roll parameter* δ (dimensionless acceleration per Hubble time)

$$\boxed{\delta = -\frac{\ddot{\varphi}}{H\dot{\varphi}}} \tag{6.90}$$

and require it to be $\delta \ll 1$. This is because the variation of ε during an e-fold must be small, and so

$$\frac{\mathrm{d}\varepsilon}{\mathrm{d}N} = \frac{1}{H}\frac{\mathrm{d}\varepsilon}{\mathrm{d}t} = \frac{1}{H}\left(-\frac{\ddot{H}}{H^2} + \frac{2\dot{H}^2}{H^3}\right) = -\frac{\dot{H}}{H^2}\left(\frac{\ddot{H}}{H\dot{H}} - 2\frac{\dot{H}}{H^2}\right) \tag{6.91}$$

which implies

You can prove the last equality directly substituting from $-\dot{H} = 4\pi G\dot{\varphi}^2$

$$\frac{1}{\varepsilon}\frac{\mathrm{d}\varepsilon}{\mathrm{d}N} = \left(\frac{\ddot{H}}{H\dot{H}} - 2\frac{\dot{H}}{H^2}\right) \implies \frac{\ddot{\varphi}}{H\dot{\varphi}} = |\delta| \ll 1 \tag{6.92}$$

So far, no approximations have been made. We simply noted that in a regime where $\{\varepsilon, |\delta|\} \ll 1$, inflation occurs and persists. We now use these conditions to simplify the equations of motion. This is called the slow-roll approximation. The condition $\varepsilon \ll 1$ implies $V(\varphi) \gg \dot{\varphi}^2/2$ and hence leads to the following simplification of the Friedmann equation

$$\boxed{H^2 \approx \frac{8\pi G}{3}V(\varphi)} \tag{6.93}$$

In the slow-roll approximation, the Hubble expansion is determined completely by the potential energy. The condition $|\delta| \ll 1$ then simplifies the Klein-Gordon equation (6.79) to

$$\dot{\varphi} \approx -\frac{V'(\varphi)}{3H} \overset{\varepsilon \ll 1}{=} -V'\left[24\pi G V(\varphi)\right]^{-1/2} \tag{6.94}$$

This provides a simple relationship between the gradient of the potential and the "speed" of the inflaton. This is the so-called *α-relation*

$$\boxed{\dot{\varphi} \approx -\frac{V'}{\sqrt{24\pi G V(\varphi)}}} \tag{6.95}$$

Moreover, if $\epsilon, \delta \ll 1$

$$\epsilon = -\frac{\dot{H}}{H^2} = \frac{4\pi G \dot{\varphi}^2}{\frac{8\pi G}{3} V(\varphi)} \approx \frac{1}{16\pi G}\left(\frac{V'(\varphi)}{V(\varphi)}\right)^2 \tag{6.96}$$

where we've introduced the α-relation in the last equality. This leads to the *first flatness condition*

$$\boxed{\left|\frac{V'(\varphi)}{V(\varphi)}\right| \ll \sqrt{16\pi G}} \tag{6.97}$$

Consider now

$$\ddot{\varphi} = -\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{V'}{3H}\right) \implies \ddot{\varphi} = -\frac{V''\dot{\varphi}}{H} + \frac{V'\dot{H}}{3H^2} \tag{6.98}$$

which, requiring $\epsilon, \delta \ll 1$, yields

$$\ddot{\varphi} = -\frac{V''}{3H}\left(-\frac{V'}{3H}\right) + \frac{V'}{3}\left(\frac{1}{16\pi G}\frac{V'^2}{V^2}\right) \quad \frac{V'}{3}\left(\frac{1}{16\pi G}\frac{V'^2}{V^2}\right) \ll \frac{V'}{3} \tag{6.99}$$

Hence, since

$$|\ddot{\varphi}| \ll |H\dot{\varphi}| = \frac{1}{3}|V'(\varphi)| \tag{6.100}$$

we conclude

$$\left|\frac{V''}{3H}\right| \cdot \left|\frac{V'}{3H}\right| \ll \frac{V'}{3} \tag{6.101}$$

which also means $|V''| \ll 3H^2$, thus leading us to the *second flatness condition*

$$\boxed{\left|\frac{V''(\varphi)}{V(\varphi)}\right| \ll 8\pi G} \tag{6.102}$$

When both flatness conditions hold, the expansion of the Universe is exponential (pseudo-de Sitter) and is also exponentially long. The inflation epoch will end only when, at a certain time t_f, the first slow-roll parameter is $\varepsilon = 1$

$$N(\varphi) = \ln\left(\frac{a(t_f)}{a(t)}\right) = \int_t^{t_f} H\,\mathrm{d}t = \int_{\varphi}^{\varphi_f} \frac{H}{\dot{\varphi}}\,\mathrm{d}\varphi \tag{6.103}$$

which, after requiring $\epsilon, \delta \ll 1$, turns into

$$N(\varphi) \approx \int_{\varphi}^{\varphi_f} \left[\frac{8\pi G V}{3}\frac{24\pi G V}{V'^2}\right]^{1/2} \mathrm{d}\varphi \tag{6.104}$$

The number of *e*-folds from the end of inflation starting from an arbitrary time t is then

$$\boxed{N(\varphi) = 8\pi G \int_{\varphi}^{\varphi_f} \frac{V}{V'}\,\mathrm{d}\varphi} \tag{6.105}$$

The first flatness condition told us that

$$\left|\frac{V}{V'}\right| \gg (16\pi G)^{-1/2} \tag{6.106}$$

which allows us to constrain (6.105)

$$N(\varphi) = 8\pi G \int_{\varphi}^{\varphi_f} \frac{V}{V'}\, \mathrm{d}\varphi \gg \sqrt{4\pi G}|\varphi_f - \varphi| \tag{6.107}$$

Hence the number of e-folds for our inflationary model will always be

$$N \gg \sqrt{4\pi G}|\varphi_f - \varphi_i| \tag{6.108}$$

which, in principle, we can invert to find the range spanned by the scalar field φ

$$\Delta\varphi \ll \frac{N}{\sqrt{4\pi G}} \tag{6.109}$$

with the minimum number of e-folds necessary to replicate the current Universe ($N_{\text{min}} \approx 60$), the estimated upper bound for the field variation is $\Delta\varphi \ll 10^{20}\,\text{GeV}$.

The End.

Part I

APPENDIX

7 APPENDIX

APPENDIX B: BLACKBODY FUNCTION

Even at an undergraduate level, we're all fairly familiar with *blackbody radiation*. The easiest way to deduce the expression for the energy density of photons in *thermal equilibrium* (STE) inside a cavity is by the means of statistical mechanics.

Remember the Bose-Einstein distribution ($\mu = 0$)

$$n = \frac{1}{\exp(h\nu/kT) - 1}$$

and the phase space density of states (per unit volume)

$$\rho(\nu)\,\mathrm{d}\nu = \frac{4\pi h g \nu^3}{c^3}\,\mathrm{d}\nu$$

from which deducing the expression from internal energy is straightforward. Remembering $g = 2$ is the quantum degeneracy of photons, a simple moltiplication of the previous expressions yields

$$U_\nu\,\mathrm{d}\nu = \frac{8\pi h}{c^3}\frac{\nu^3}{\exp(h\nu/kT) - 1}\,\mathrm{d}\nu \tag{7.1}$$

The energy of a photon is $E_\gamma = h\nu$, meaning the number of photons in the frequency range $\nu \to \nu + \mathrm{d}\nu$ is

$$n_\nu\,\mathrm{d}\nu = \frac{8\pi}{c^3}\frac{\nu^2}{\exp(h\nu/kT) - 1}\,\mathrm{d}\nu \tag{7.2}$$

which can be integrated over all frequencies, thus obtaining the number density of photons in blackbody radiation

$$n_\gamma = \beta T^3 \quad \beta = \frac{2.4041 k^3}{\pi^2 \hbar^3 c^3} \tag{7.3}$$

Since blackbody radiation is isotropic (it depends only on the absolute temperature T), the definition of mean monochromatic intensity yields

$$\boxed{B_\nu(T) = \frac{2h\nu^3}{c^2}\frac{1}{\exp(h\nu/kT) - 1}} \tag{7.4}$$

It's important to notice that, in principle, such a fundamental result holds only in *strict thermodynamic equilibrium* (STE), but it's possible to generalize this formulation for less "restrictive" environments.

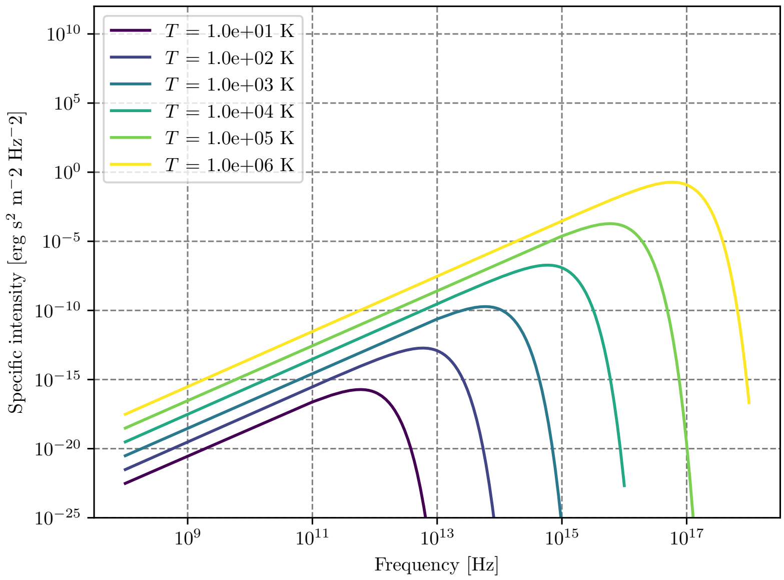

Figure 7.1: Blackbody frequency spectrum.

An incredible number of important results descends from (7.4), and it may be worthwhile to cite at least some of them, starting from Stefan-Boltzmann law. We'll use the following result without proving it

$$\int_0^{+\infty} B_\nu(T)\,\mathrm{d}\nu = \frac{2h}{c^2}\frac{\pi^4}{15}\left(\frac{kT}{h}\right)^4$$

Computing the bolometric flux and the bolometric energy density by integrating over all frequencies using what we've just written down, you find the following

$$U(T) = aT^4 \quad F(T) = \sigma_{SB}T^4$$

Clearly the two constants a and σ_{SB} cannot be independent, and are actually related by the integral we've previously calculated. Using for example[1]

$$F(T) = \pi \int_0^{+\infty} B_\nu(T)\,\mathrm{d}\nu$$

you can easily find out that the *Stefan-Boltzmann constant* is equal to

$$\sigma_{SB} = \frac{2\pi^5 k^4}{15c^2h^3}$$

and the relation with a is simply $\sigma_{SB} = ac/4$.

The equation

$$F(T) = \frac{2\pi^5 k^4}{15c^2h^3}T^4 \tag{7.5}$$

is what is usually known as the *Stefan-Boltzmann law*.

1 The emergent flux from an isotropically emitting surface (such as a blackbody) is $\pi \cdot$ brightness which is none other than the specific intensity.

Let us now consider two different regimes for eq.7.4: $h\nu/kT \ll 1$ and $h\nu/kT \gg 1$. The first yields what is commonly known as the Rayleigh-Jeans Law which is, sadly, pretty much relevant only for radioastronomy.

Since

$$\exp\left(\frac{h\nu}{kT}\right) = 1 + \frac{h\nu}{kT} + o\left(\frac{h\nu}{kT}\right)^2$$

the blackbody radiation assumes the much simpler form of

$$B_\nu^{RJ} = \frac{2\nu^2}{c^2}kT$$

Another important results is achieved in the opposite regime, when the exponential term is rather larger than unity

$$B_\nu^W = \frac{2h\nu^3}{c^2}\exp\left(-\frac{h\nu}{kT}\right)$$

This expression is known as Wien's Law.

Characteristic Temperatures

Starting from the blackbody spectrum we can give various definitions of temperature, that are going to be more and less useful when dealing with stars. The definitions are those of [18].

Brightness Temperature

One way of characterizing brightness (specific intensity) at a certain frequency is to give the temperature of the blackbody having the same brightness at that frequency. That is, for any value I_ν we define $T_b(\nu)$ by the relation

$$I_\nu = B_\nu(T_b)$$

this way of specifying brightness has the advantage of being closely connected with the physical properties of the emitter, and has the simple unit (K).

This procedure is used especially in radio astronomy, where the Rayleigh-Jeans law is usually applicable.

$$T_b = \frac{c^2}{2\nu k}I_\nu \quad h\nu/kT \ll 1 \tag{7.6}$$

Note that the uniqueness of the definition of brightness temperature relies on the monotonicity property of Planck's law. Also note that, in general, the brightness temperature is a function of ν.

Color Temperature

Often a spectrum is measured to have a shape more or less of blackbody form, but not necessarily of the proper absolute value. For example, by measuring F_ν, from an unresolved source we cannot find I_ν, unless we know the distance to the source and its physical size.

By fitting the data to a blackbody curve without regard to vertical scale, a color temperature T_c, is obtained. Often the "fitting" procedure is nothing more than estimating the peak of the spectrum and applying Wien's displacement law to find a temperature.

Effective Temperature

The effective temperature of a source T_{eff} is derived from the total amount of flux, integrated over all frequencies, radiated at the source. We obtain T_{eff} by equating the actual flux F to the flux of a blackbody at temperature T_{eff}

$$F = \int I_\nu \cos\theta \, \mathrm{d}\nu \, \mathrm{d}\Omega := \sigma T_{eff}^4 \tag{7.7}$$

APPENDIX C: CHRISTOFFEL SYMBOLS

For a symmetric, metric-compatible covariant derivative ∇_μ, the Christoffel symbols are given by

$$\Gamma_{\mu\ \nu}^{\ \alpha} = \frac{1}{2}g^{\lambda\alpha}(\partial_\mu g_{\lambda\nu} + \partial_\nu g_{\mu\lambda} - \partial_\lambda g_{\mu\nu}) \tag{7.8}$$

For the FLWR metric (4.35), the Christoffel symbols are given by

$$\begin{aligned}
\Gamma_{1\ 1}^{\ 0} &= \frac{a\dot{a}}{1-kr^2} & \Gamma_{1\ 1}^{\ 1} &= \frac{kr}{1-kr^2} \\
\Gamma_{2\ 2}^{\ 0} &= a\dot{a}r^2 & \Gamma_{3\ 3}^{\ 0} &= a\dot{a}r^2\sin^2\theta \\
\Gamma_{0\ j}^{\ i} &= \frac{\dot{a}}{a}\delta_j^i = H\delta_j^i & \Gamma_{0\ 1}^{\ 1} &= \frac{\dot{a}}{a} = H \\
\Gamma_{2\ 2}^{\ 1} &= -r(1-kr^2) & \Gamma_{3\ 3}^{\ 1} &= -r(1-kr^2)\sin^2\theta \\
\Gamma_{1\ 2}^{\ 2} &= \Gamma_{1\ 3}^{\ 3} = \frac{1}{r} & \Gamma_{3\ 3}^{\ 2} &= -\sin\theta\cos\theta \\
\Gamma_{2\ 3}^{\ 3} &= \cot\theta & \Gamma_{0\ 0}^{\ i} &= 0 \\
\Gamma_{0\ 0}^{\ 0} &= 0 & \Gamma_{i\ 0}^{\ 0} &= 0
\end{aligned}$$

or related to this by symmetry. The non-zero components of the Ricci tensor are

$$\begin{aligned}
R_{00} &= -3\frac{\ddot{a}}{a} \\
R_{11} &= \frac{a\ddot{a} + 2\dot{a}^2 + 2k}{1-kr^2} \\
R_{22} &= r^2(a\ddot{a} + 2\dot{a}^2 + 2k) \\
R_{33} &= r^2(a\ddot{a} + 2\dot{a}^2 + 2k)\sin^2\theta
\end{aligned}$$

and the Ricci scalar is then

$$R = 6\left[\frac{\ddot{a}}{a}\left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2}\right]$$

APPENDIX D: WHERE THE D STANDS FOR DEMONSTRATION

The FLRW Killing Tensor

In Chapter 4 we've defined at a certain point a Killing tensor (4.49) but I never showed that it actually satisfies Killing's equation. Our goal is to show that the tensor

$$K_{\mu\nu} = a^2(g_{\mu\nu} + u_\mu u_\nu) \tag{7.9}$$

satisfies $\nabla_{(\sigma}K_{\mu\nu)} = 0$.

Let's start from observing that $K_{\mu\nu}$ is a spatial projector

$$u^\nu K_{\mu\nu} = a^2(u^\nu g_{\mu\nu} + u_\mu u_\nu u^\nu) = 0 \tag{7.10}$$

since $u_\nu u^\nu = -1$. This means that we can define $h_{\mu\nu} = g_{\mu\nu} + u_{\mu\nu}$ so that $K_{\mu\nu} = a^2 h_{\mu\nu}$. Note that $h_{\mu\nu}$ coincides with the spatial sections of the FLRW metric.

Before calculating the covariant derivative of the alleged Killing tensor, let's first check

$$\begin{aligned}
\nabla_\mu u_\nu &= \partial_\mu u_\nu - \Gamma_{\mu\ \nu}^{\ \lambda} u_\lambda \\
&= +\Gamma_{\mu\ \nu}^{\ 0} = \frac{1}{2} g^{00}\partial_0 g_{\mu\nu} \\
&= -\frac{1}{2}\partial_0(h_{\mu\nu} - u_\mu u_\nu) \\
&= -\frac{1}{2}\partial_0 h_{\mu\nu} = -\frac{1}{2}\partial_0(a^2\tilde{h}_{\mu\nu}) = -a\dot{a}\tilde{h}_{\mu\nu} = -Hh_{\mu\nu}
\end{aligned}$$

We can now expand the covariant derivative of $K_{\mu\nu}$

$$\begin{aligned}
\nabla_\lambda K_{\mu\nu} &= h_{\mu\nu}\nabla_\lambda a^2 + a^2\nabla_\lambda h_{\mu\nu} \\
&= 2a\dot{a}u_\lambda h_{\mu\nu} + a^2\nabla_\lambda h_{\mu\nu} \\
&= 2a\dot{a}u_\lambda h_{\mu\nu} + a^2(u_{\mu\nabla_\lambda u_\nu} + u_\nu\nabla_\lambda u_\mu)
\end{aligned}$$

where we can now substitute $\nabla_\mu u_\nu = -Hh_{\mu\nu}$ to obtain

$$\begin{aligned}
\nabla_\lambda K_{\mu\nu} &= 2a\dot{a}u_\lambda h_{\mu\nu} - a\dot{a}(u_\mu h_{\nu\lambda} + u_\nu h_{\mu\lambda}) \\
&= a\dot{a}\left[(u_\lambda h_{\mu\nu} - u_\mu h_{\lambda\nu}) + (u_\lambda h_{\mu\nu} - u_\nu h_{\lambda\mu})\right] \\
&= 2a\dot{a}\left[u_{[\lambda}h_{\mu]\nu} + u_{[\lambda}h_{\nu]\mu}\right]
\end{aligned}$$

At this point, we can easily see that the symmetrization $\nabla_{(\sigma}K_{\mu\nu)}$ of the previous expression yields zero, thus proving that $K_{\mu\nu}$ is indeed a Killing tensor.

ABOUT ME

Hi there, I'm Giacomo, sometimes known as Cael by a very small niche of people, the author of the notes you just finished reading.

I'm a Batchelor-graduate student at the University of Pisa now in the process of undertaking a major (Master degree) in Astronomy and Astrophysics at the same university. In my (meager) free time I'm an avid reader, gamer, anime-enthusiast as well as an independent author in the very process of publishing his first novel.

For those of you that may be interested I'll leave both the link to the novel's page as well to my instagram profile in the QRcodes right below.

Figure 7.2: On the left is the link to my instagram profile, on the right the one to my novel's page (Python really has a library for anything, huh?).

Come say hi if you fancy action-fantasy stories with magic, swords, some comedy to garnish and a lot of other stuff :)

At last but not the least, I want to thank all my collegues and fellow students that helped (and hopefully will help me again) during the drafting of this text, pointing out errors and occasions for further expansion.

If you'd like your names to be included and take your fair share of credits, feel free to contact me and I'll add your name right away in the few lines that are left of this page.

I hope my understading of the complex interplayings going on in our Universe has managed to facilitate your own understading.

If not, well... oopsie.

See you, space cowboys...

Giacomo

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